

Prepared for:

Chris Misenis
U.S. Environmental Protection Agency
OAR/OAQPS
109 T.W. Alexander Drive
Research Triangle Park, NC 27711

Prepared by:

Prakash Karamchandani, Katie Tuite, Pradeepa Vennam and Greg Yarwood
Ramboll Americas Engineering Solutions, Inc.
7250 Redwood Blvd., Suite 105
Novato, California 94945

Under a subcontract from:

Institute for the Environment
The University of North Carolina at Chapel Hill
123 W. Franklin St., Suite 330B
Chapel Hill, NC 27516

March 4, 2025

Implementation of the SAPRC-22 Mechanism in CMAQv5.4

Task Order 68HERD21A0002

U.S. EPA Contract No. 68HERH24F0296

Draft Report



Contents

1.0 INTRODUCTION	1
2.0 SAPRC-22 IMPLEMENTATION	2
2.1 SAPRC-22 Mechanism	2
2.2 Approach	2
2.3 SOA Treatment	3
3.0 TESTING THE SAPRC-22 IMPLEMENTATION	5
3.1 Test Case Description	5
3.2 Test Simulations	5
3.3 Results	6
3.3.1 SAPRC-07T vs. SAPRC-22	6
3.3.2 SAPRC-22 EBI vs. SAPRC-22 Rosenbrock	16
3.3.3 Model Runtimes	20
4.0 GUIDANCE FOR SPECIATING TOTAL ORGANIC GAS (TOG) EMISSIONS FOR THE SAPRC-22 CHEMICAL MECHANISM IN CMAQ	21
5.0 SUMMARY	27
6.0 REFERENCES	28
Appendix A: SAPRC-22 Reactions and Species	1

TABLES

Table 3-1. Surface layer concentrations of key species averaged over the domain and over 10 days (July 6 to July 15, 2016). Concentrations of gas species are in ppb and PM species are in $\mu\text{g}/\text{m}^3$	16
Table 3-2. Model runtimes for SAPRC-07T/EBI, SAPRC-22/EBI and SAPRC-22/Rosenbrock.	20
Table 4-1. Assignments of TOG compounds to SAPRC-22 and SAPRC-07T model species.	21

FIGURES

Figure 3-1. Test case modeling domain.	5
Figure 3-2. Comparison of hourly surface concentrations (all grid cells and 10 simulation days) of O_3 , NO, NO_2 and OH from CMAQ simulations with the SAPRC-22 mechanism with those from the SAPRC-07T mechanism.	7
Figure 3-3. Comparison of hourly surface concentrations (all grid cells and 10 simulation days) of HO_2 , HNO_3 , $\text{PM}_{2.5}$ and $\text{PM}_{2.5}$ -Nitrate from CMAQ simulations with the SAPRC-22 mechanism with those from the SAPRC-07T mechanism.	8
Figure 3-4. Comparison of hourly surface concentrations (all grid cells and 10 simulation days) of $\text{PM}_{2.5}$ -Sulfate, $\text{PM}_{2.5}$ -Ammonium, $\text{PM}_{2.5}$ -Organic Carbon and SO_2 from CMAQ simulations with the SAPRC-22 mechanism with those from the SAPRC-07T mechanism.	9
Figure 3-5. MDA8 ozone concentrations on July 7: SAPRC-07T chemistry (top); SAPRC-22 chemistry (middle); and differences in concentrations (bottom).	10
Figure 3-6. 24-hr average $\text{PM}_{2.5}$ concentrations on July 7: SAPRC-07T chemistry (top); SAPRC-22 chemistry (middle); and differences in concentrations (bottom).	11

Figure 3-7.	24-hr average PM _{2.5} nitrate concentrations on July 7: SAPRC-07T chemistry (top); SAPRC-22 chemistry (middle); and differences in concentrations (bottom).	12
Figure 3-8.	24-hr average PM _{2.5} sulfate concentrations on July 7: SAPRC-07T chemistry (top); SAPRC-22 chemistry (middle); and differences in concentrations (bottom).	13
Figure 3-9.	24-hr average PM _{2.5} ammonium concentrations on July 7: SAPRC-07T chemistry (top); SAPRC-22 chemistry (middle); and differences in concentrations (bottom).	14
Figure 3-10.	24-hr average PM _{2.5} organic carbon concentrations on July 7: SAPRC-07T chemistry (top); SAPRC-22 chemistry (middle); and differences in concentrations (bottom).	15
Figure 3-11.	Comparison of hourly surface concentrations (all grid cells and 10 simulation days) of O ₃ , NO, NO ₂ and OH from CMAQ simulations with the SAPRC-22 mechanism using the EBI solver with those using the Rosenbrock solver.	17
Figure 3-12.	Comparison of hourly surface concentrations (all grid cells and 10 simulation days) of HO ₂ , HNO ₃ , PM _{2.5} and PM _{2.5} -Nitrate from CMAQ simulations with the SAPRC-22 mechanism using the EBI solver with those using the Rosenbrock solver.	18
Figure 3-13.	Comparison of hourly surface concentrations (all grid cells and 10 simulation days) of PM _{2.5} -Sulfate, PM _{2.5} -Ammonium, PM _{2.5} -Organic Carbon and SO ₂ from CMAQ simulations with the SAPRC-22 mechanism using the EBI solver with those using the Rosenbrock solver.	19

1.0 INTRODUCTION

The Community Multiscale Air Quality (CMAQ) model is a state-of-the-science photochemical grid model developed and maintained by the EPA's Office of Research and Development, which the Air Quality Modeling Group (AQMG) uses for a variety of regulatory and research functions. Model users have historically had the option to choose among several chemical mechanisms, including that of the California Statewide Air Pollution Research Center (SAPRC). The SAPRC version in CMAQ version 5.4 (release date October 14, 2022) is based on the 2007 version (SAPRC-07: Carter, 2010) revised to explicitly represent additional toxic compounds in SAPRC-07T (Hutzell et al., 2012). The core SAPRC mechanism has been updated twice since the release of SAPRC-07. The first update, SAPRC-18, was not widely used and was not implemented in CMAQ. The latest version of the mechanism is SAPRC-22, which was initially released in September 2023 (Carter, 2023), followed by subsequent updates in 2024. In comparison to SAPRC-07T, SAPRC-22 incorporates some potentially significant updates to the organic chemistry and more tracking of toxic species.

In this project, Ramboll implemented the version of SAPRC-22 available in June 2024 into CMAQv5.4. This report describes the implementation and testing of the updated model to ensure that the model is robust, and results are reasonable. Some differences in model results between SAPRC-22 and SAPRC-07T in CMAQ are expected because of the differences in the two chemistry mechanisms. However, the general spatial patterns in predicted concentrations between the two versions are expected to be qualitatively similar.

2.0 SAPRC-22 IMPLEMENTATION

2.1 SAPRC-22 Mechanism

The implementation was conducted using a version of the SAPRC-22 gas-phase chemical mechanism provided directly to Ramboll by the mechanism developer in June 2024 and also made publicly available at <https://intra.engr.ucr.edu/~carter/SAPRC/22/>. The mechanism has a total of 751 gas phase reactions and 261 gas phase species. As advised by the U.S. EPA, we used the base CMAQv5.4 version for the implementation although there have been subsequent updates to CMAQ during the course of the project.

2.2 Approach

The following approach was used for implementing the chemistry updates in CMAQ:

- The base CMAQv5.4 repository was downloaded from EPA's CMAQ GitHub website (<https://github.com/USEPA/CMAQ>) and installed on Ramboll's computers.
- A separate branch ("saprc22") was created for making the updates.
- The SAPRC-22 gas-phase reaction list was reformatted in a mechanism definition (mech.def) input file that was compatible with CMAQ's chemical mechanism processor, CHEMMECH.
- Additional reactions were added to the mechanism definition file to link gas-phase chemistry with aerosol and heterogeneous chemistry. These included reactions of organic aerosol precursors (e.g., anthropogenic VOCs) to form low volatility and semi-volatile organic gases; oligomerization reactions for secondary organic aerosols (SOA); aging reactions for primary organic carbon including functionalization and fragmentation reactions; heterogeneous formation of nitric acid (HNO₃); and heterogeneous uptake of glyoxal (GLY) and methylglyoxal (MGLY) to form SOA. More details on the SOA treatment in CMAQ with the standard SAPRC-22 mechanism are provided below in Section 2.3.
- Reactions to track primary formaldehyde (HCHO), acetaldehyde (CCHO), and acrolein (ACRO) as reactive tracers were also added consistent with the current SAPRC-07T implementation in CMAQ.
- CMAQ's gas-phase namelist file (GC_mechname.nml) was updated for consistency with the new mechanism definition file. The gas-phase species namelist file includes species name; molecular weight; the surrogate species to use for initial concentration, background concentration, dry deposition, wet deposition, gas to aerosol linkage, and gas to aqueous linkage, if applicable; any scaling factors that should be applied to the surrogate; flags to control species transport (advection and diffusion); and flags to control species information written to output files. We retained the SAPRC-22 species names in the new gas-phase species namelist. These names sometimes differ from those in the SAPRC-07T implementation in CMAQ for species that are identical (e.g., ethene is ETEN in SAPRC-22 and ETHE in SAPRC-07T). To determine if wet and dry deposition should be applied to each chemical species, we began with assignments used in the SAPRC-07T GC namelist file and made updates considering Henry's Law constants (H), assuming that all species with H greater than 100 M/atm are deposited. We reviewed all surrogate species and made updates where needed by considering H, chemical structure, and reactivity of the mechanism species.

- Similarly, CMAQ's particle-phase species namelist (AE_mechname.nml) was updated for consistency with the new mechanism definition file. For the most part, the AE namelist file is similar to the current file for SAPRC-07T, with the exceptions noted in Section 2.3.
- The CMAQ non-reactive species namelist (NR_mechname.nml) was also updated to include chlorine (CL2). The standard SAPRC-22 mechanism does not have halogen chemistry, but chlorine is required as a model species. If the SAPRC-22 mechanism is updated in the future to include halogen chemistry, then CL2 should be moved from the NR namelist to the GC namelist.
- The CHEMMECH processor was built and run with the updated mechanism definition and namelist files for SAPRC-22. CHEMMECH outputs include reaction rates at standard temperature and pressure (298.15 K and 1 atm), which agreed with calculations performed by Ramboll using the rate equations provided in the SAPRC-22 mechanism documentation.
- Cross section (CS) and quantum yield (QY) data for the SAPRC-22 photolysis reactions were also updated in CMAQ, using data provided from the SAPRC mechanism development website (<https://intra.engr.ucr.edu/~carter/SAPRC/22/>). The SAPRC-22 mechanism includes 6 new photolysis reactions compared to SAPRC-07T, and an additional 11 reactions use updated CS and QY data compared to SAPRC-07T. CMAQ's in-line photolysis module was therefore run for SAPRC-22 using these new and updated datasets.
- To create the Euler Backward Iterative (EBI) chemical solver for SAPRC-22, we used CMAQ's create_ebi utility and encountered a difficulty. Since SAPRC22 does not include an acetyl peroxy radical self-reaction, which is a requirement to use create_ebi, we patched the create_ebi utility so that it can create a functional EBI solver when the acetyl peroxy radical self-reaction is absent. The updated create_ebi program now gives a warning when the self-reaction is missing, rather than aborting with an error, and it assigns a small, negligible value to the self-reaction rate. With this update, Ramboll successfully created the EBI solver for SAPRC-22.

Appendix A provides the final set of reactions in the mechanism as defined in the mechanism definition file as well as tables of the gas and aerosol species in the updated SAPRC-22 mechanism implemented in CMAQv5.4.

2.3 SOA Treatment

The current version of SAPRC-07T in CMAQ is linked to both the Aero6 (AE6) and Aero7i (AE7i) aerosol modules. The AE7i treatment is the most current version in CMAQ. However, there are some important differences between SAPRC-07T and SAPRC-22 that influence the linkages, making it necessary to use a hybrid version of the two aerosol modules for SAPRC-22. We used the approach described below to keep most of the AE7i linkages in the SAPRC-22 implementation with some exceptions. We label the SAPRC-22 version of SOA treatment as AE65 since it includes features of both the AE6 and AE7 treatments.

The CMAQ AE7i implementation of SAPRC-07T extensively modified the standard SAPRC-07 gas-phase reactions of isoprene (ISOP) and lumped terpenes other than α -pinene (TERP) to interface with CMAQ SOA schemes. Modifications for ISOP add production of isoprene epoxydiols (IEPOX), hydroxymethyl-methyl- α -lactone (HMML) and methacrylic acid epoxide (MAE) that can then form SOA via heterogeneous uptake to aerosols (note that the AE6 implementation has a more condensed

treatment of IEPOX and SOA formation without HMML or MAE formation). AE7i modifications for TERP add production of a terpene-specific organic nitrate (MTNO3) that can then form SOA via heterogeneous uptake and hydrolysis. We did not modify SAPRC-22 to link with these AE7i features because making the needed changes to SAPRC-22 would create a new mechanism which is beyond the scope of this project. Updating SAPRC-22 ISOP chemistry to produce IEPOX (even for the simple treatment in AE6), HMML and MAE would substantially alter the mechanism and could change oxidant production in ways that should be evaluated. Updating TERP to produce MTNO3 appears less complex than updating ISOP chemistry, but we are uncertain what changes should be made because the SAPRC-07T updates to TERP and α -pinene (APIN) differ for reasons that aren't fully apparent. Instead, we utilize the CMAQ AE6 SOA schemes for ISOP, TERP and APIN (APINE in SAPRC-22) that remain available. Specifically, ISOP oxidation reactions produce species ISOPRXN that then links to SOA production, and TERP/APINE oxidation reactions produce species TERPRXN except for the APINE reaction with NO3 which doesn't produce TERPRXN (consistent with AE7i). SAPRC-22 adds β -pinene (BPINE) as an explicit species and we represent SOA formation from BPINE in the same way as TERP by adding TERPRXN as a product of the reactions of BPINE with OH, O3, NO3 and O(3P).

All other SOA formation pathways follow the AE7i approach. These include: the formation of low volatility (SVAVB1) and semivolatile (SVAVB2, SVAVB3, SVAVB4) organic gases from anthropogenic VOCs; oligomerization reactions; aging reactions for primary organic carbon including functionalization and fragmentation reactions; and heterogeneous uptake of glyoxal (GLY) and methylglyoxal (MGLY) to form SOA.

3.0 TESTING THE SAPRC-22 IMPLEMENTATION

3.1 Test Case Description

EPA's 2016 CMAQ platform was used to perform the testing of SAPRC-22 implementation in CMAQv5.4. The national-scale modeling consists of the standard 12-kilometer CMAQ continental modeling domain (12US2) shown in Figure 3-1 below. The horizontal resolution of the grid for this modeling domain is 12 kilometers with 396 grid cells in the east-west direction and 246 cells in the north-south direction.

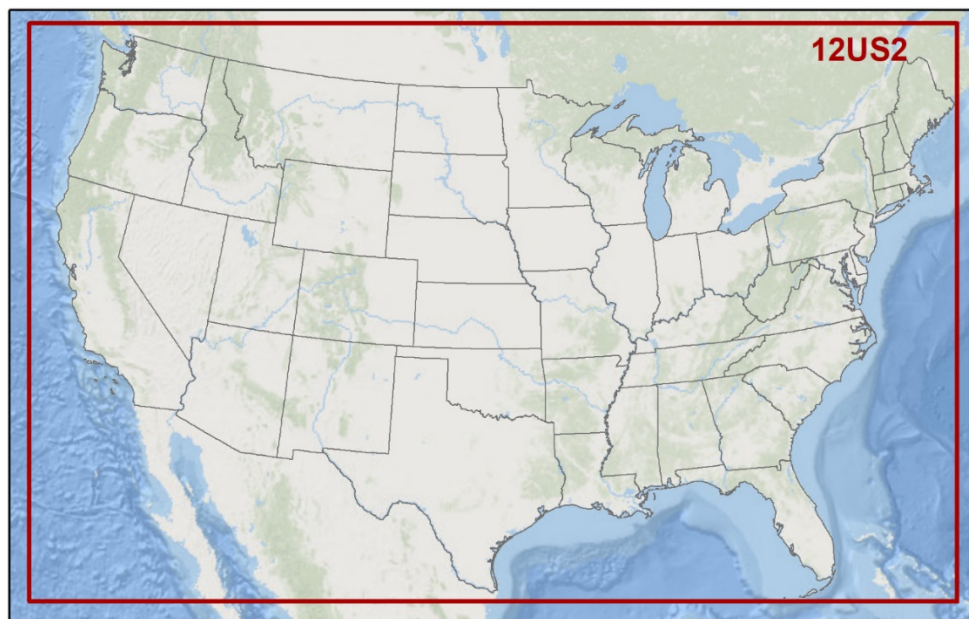


Figure 3-1. Test case modeling domain.

3.2 Test Simulations

All modeling inputs, including emissions (2016v3), meteorology, and initial boundary conditions, were sourced directly from EPA's 2016 CMAQ platform (EPA, 2023). CMAQv5.4 was used to conduct a 15-day modeling test (July 1–July 15) using both the SAPRC-07T and SAPRC-22 chemical mechanisms. Since the emissions and IC/BC inputs in the 2016 platform were designed for the Carbon Bond Mechanism (CB6r5), we mapped CB6r5 species to SAPRC-07T species and converted emissions and IC/BC inputs to be SAPRC-compatible using EPA's Combine tool (CMAQ/POST/combine/README.md at main · USEPA/CMAQ · GitHub). We next converted the SAPRC-07T inputs generated with the Combine tool to SAPRC-22 inputs, again with the Combine tool. Note that this was done for testing purposes. As discussed in Section 4, we do not recommend using model ready emissions that were prepared for SAPRC-07T with SAPRC-22 because species definitions have changed between the two mechanisms.

The following simulations were conducted for the 15-day test case:

1. CMAQv5.4 with SAPRC-07T chemistry and the EBI solver
2. CMAQv5.4 with SAPRC-22 chemistry and the EBI solver
3. CMAQv5.4 with SAPRC-22 chemistry and the Rosenbrock solver

The first 2 simulations provide a measure of the differences in results between the two chemistry mechanisms. The third simulation with a generalized solver provides a benchmark for the EBI solver, which is a mechanism-specific solver that combines analytical solutions for specific mechanism species and a numerical method for the remaining mechanism species. The EBI solver subroutines have to be created for a specific mechanism, as described in Section 2.2.

3.3 Results

To compare results from the CMAQ simulations, we discarded the first 5 days (July 1 to July 5) as a spin-up period. The results shown here are for the July 6 to July 15 period. We first discuss differences in results with the two chemistry mechanisms (SAPRC-07T and SAPRC-22), both using the EBI solver followed by a comparison of results with the new SAPRC-22 mechanism using the EBI solver and the Rosenbrock solver.

3.3.1 SAPRC-07T vs. SAPRC-22

Figures 3-2 through 3-4 show scatterplots (paired in space and time) comparing hourly values of surface concentrations of key species over all grid cells for the 10-day period from July 6 to July 15. In Figure 3-2, we see that O_3 , NO and NO_2 concentrations compare well, but there is more scatter in OH concentrations, showing that differences in the mechanisms have some impacts on radical species. Figure 3-3 shows that the scatter in HO_2 radical concentrations is similar to the scatter in OH concentrations. Total $PM_{2.5}$ concentrations are in good agreement, but HNO_3 and aerosol nitrate concentrations show some differences. This may be due to the sensitivity of ISORROPIA to small changes in gas-phase HNO_3 concentrations occasionally leading to a disproportionate change in partitioning between $HNO_3(g)$ and $NO_3(p)$. This is a well-known artifact of the numerical methods applied within ISORROPIA (Koo et al., 2020). This also explains the differences in particulate ammonium, as shown in Figure 3-4. SO_2 and particulate organic carbon concentrations show very little scatter between the two mechanisms.

Figures 3-5 through 3-10 show the spatial patterns of MDA8 ozone, 24-hour average $PM_{2.5}$, and 24-hour average $PM_{2.5}$ components (nitrate, ammonium, organic carbon, and sulfate) for July 7, the day with the largest differences in O_3 and $PM_{2.5}$ concentrations. Each figure shows the results from CMAQ simulations with the two mechanisms, as well as the difference between the two. The spatial patterns are qualitatively similar, but there are some noticeable differences in MDA8 ozone, total $PM_{2.5}$ and organic carbon concentrations. In particular, SAPRC-22 tends to predict lower total $PM_{2.5}$ and organic carbon than SAPRC-07T, especially in the eastern U.S. and northern California regions. This may be related to the more detailed SOA treatment in the CMAQ version of SAPRC-07T. As discussed in Section 2.3, this SAPRC-22 implementation does not include the detailed SOA formation from isoprene and terpene-specific organic nitrate that is available in the CMAQ SAPRC-07T version. Incorporating these features in the current project would have required changes to the SAPRC-22 mechanism that were beyond the scope of the current project.

Table 3-1 shows surface concentrations of key species averaged over the domain for all hours and days of the 10-day period. MDA8 ozone concentrations are averaged over the domain for the 10 days. As shown in the table, the average differences are small. The largest differences are for $PM_{2.5}$ and organic carbon, with the SAPRC-22 mechanism predicting lower values than the SAPRC-07T, consistent with the discussion above.

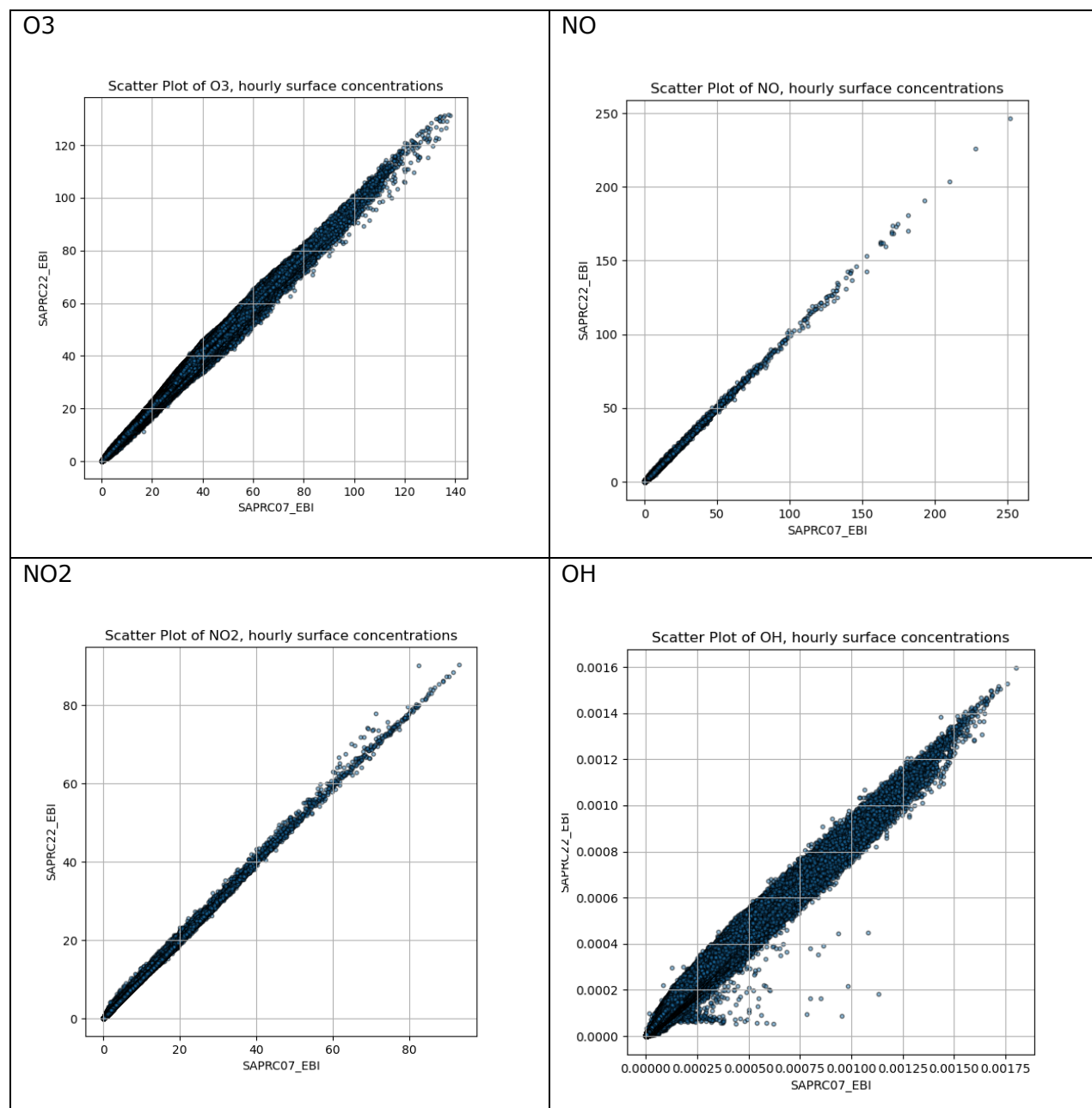


Figure 3-2. Comparison of hourly surface concentrations (all grid cells and 10 simulation days) of O₃, NO, NO₂ and OH from CMAQ simulations with the SAPRC-22 mechanism with those from the SAPRC-07T mechanism.

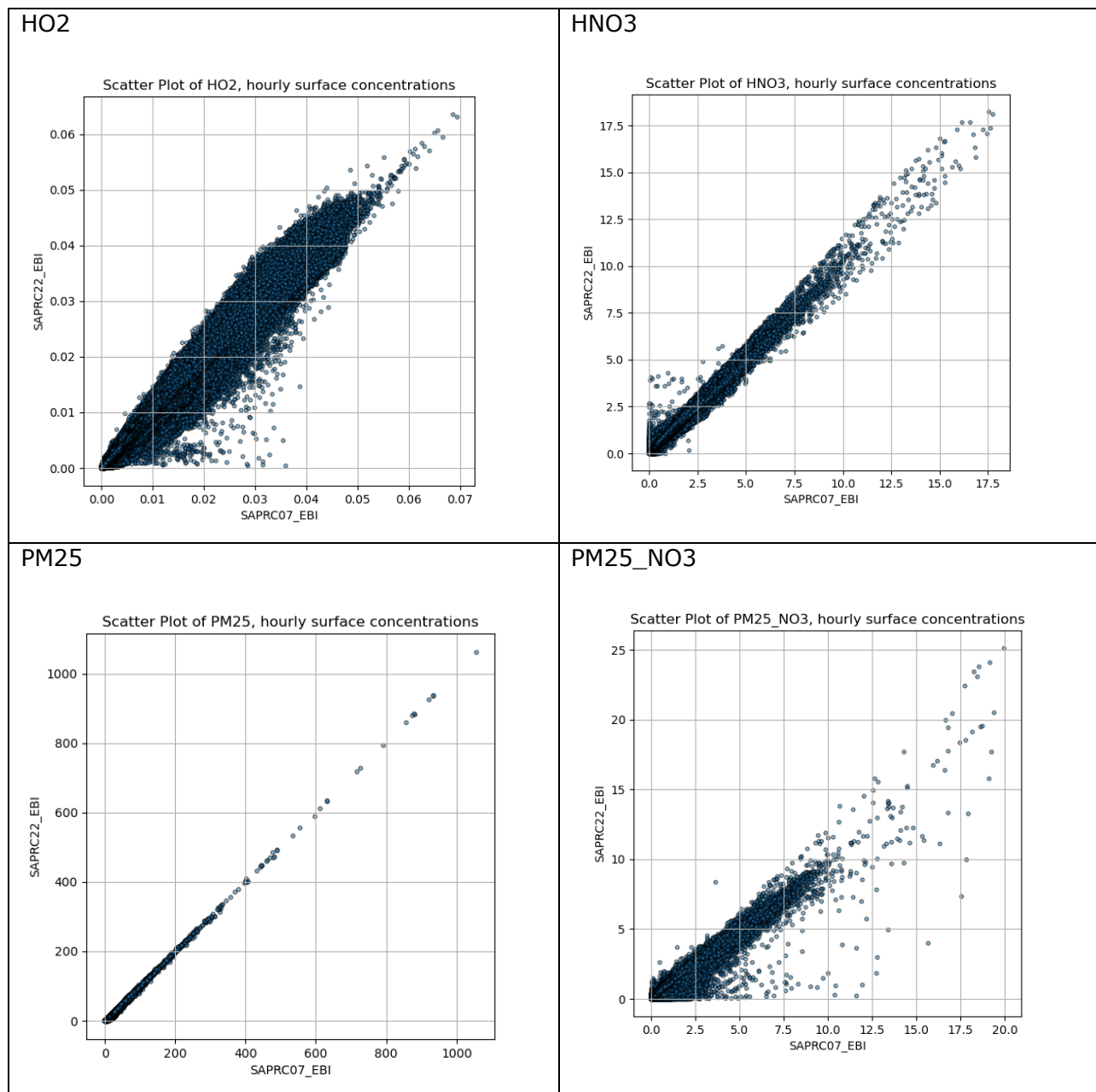


Figure 3-3. Comparison of hourly surface concentrations (all grid cells and 10 simulation days) of HO₂, HNO₃, PM_{2.5} and PM_{2.5}-Nitrate from CMAQ simulations with the SAPRC-22 mechanism with those from the SAPRC-07T mechanism.

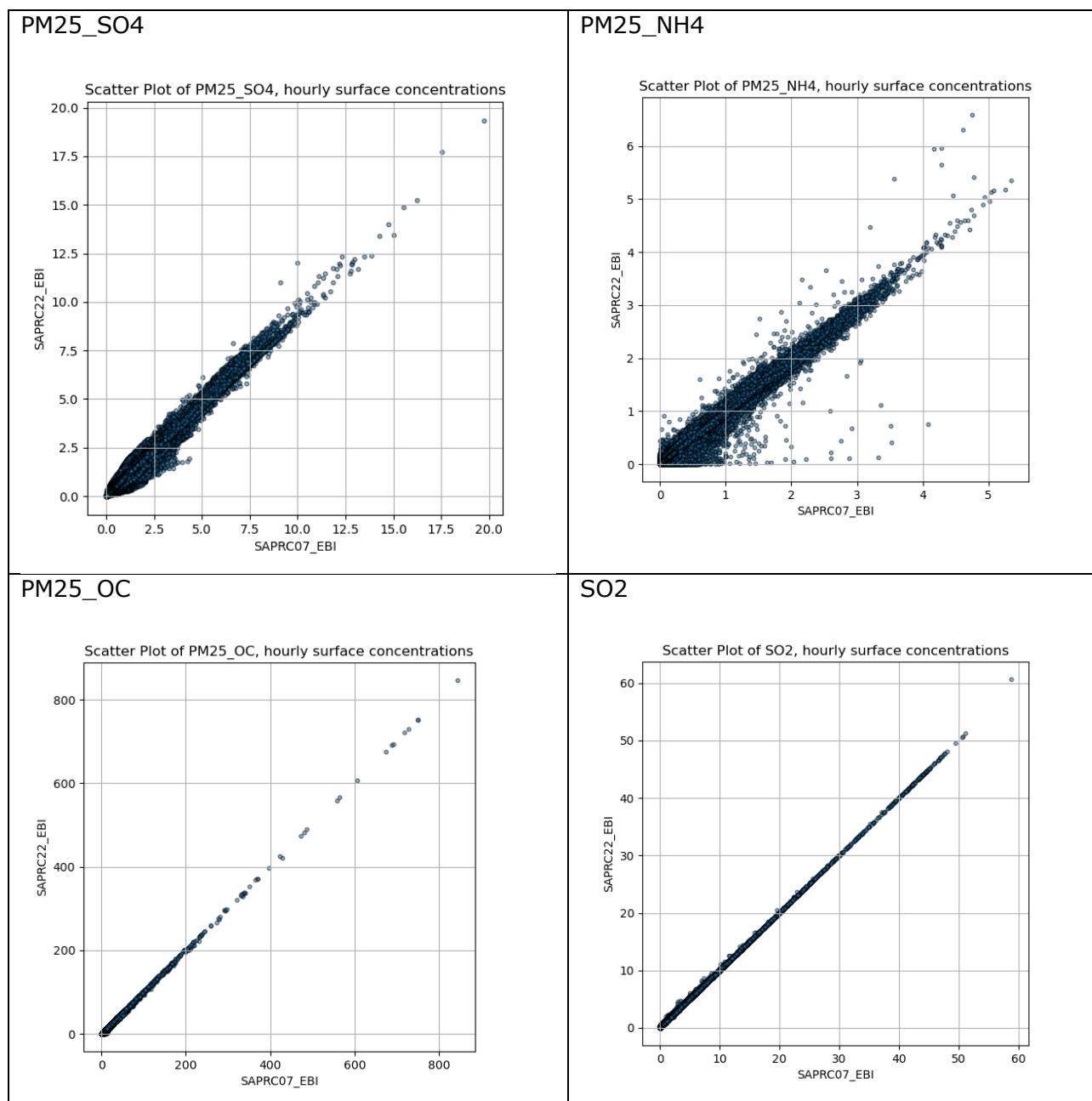


Figure 3-4. Comparison of hourly surface concentrations (all grid cells and 10 simulation days) of PM_{2.5}-Sulfate, PM_{2.5}-Ammonium, PM_{2.5}-Organic Carbon and SO₂ from CMAQ simulations with the SAPRC-22 mechanism with those from the SAPRC-07T mechanism.

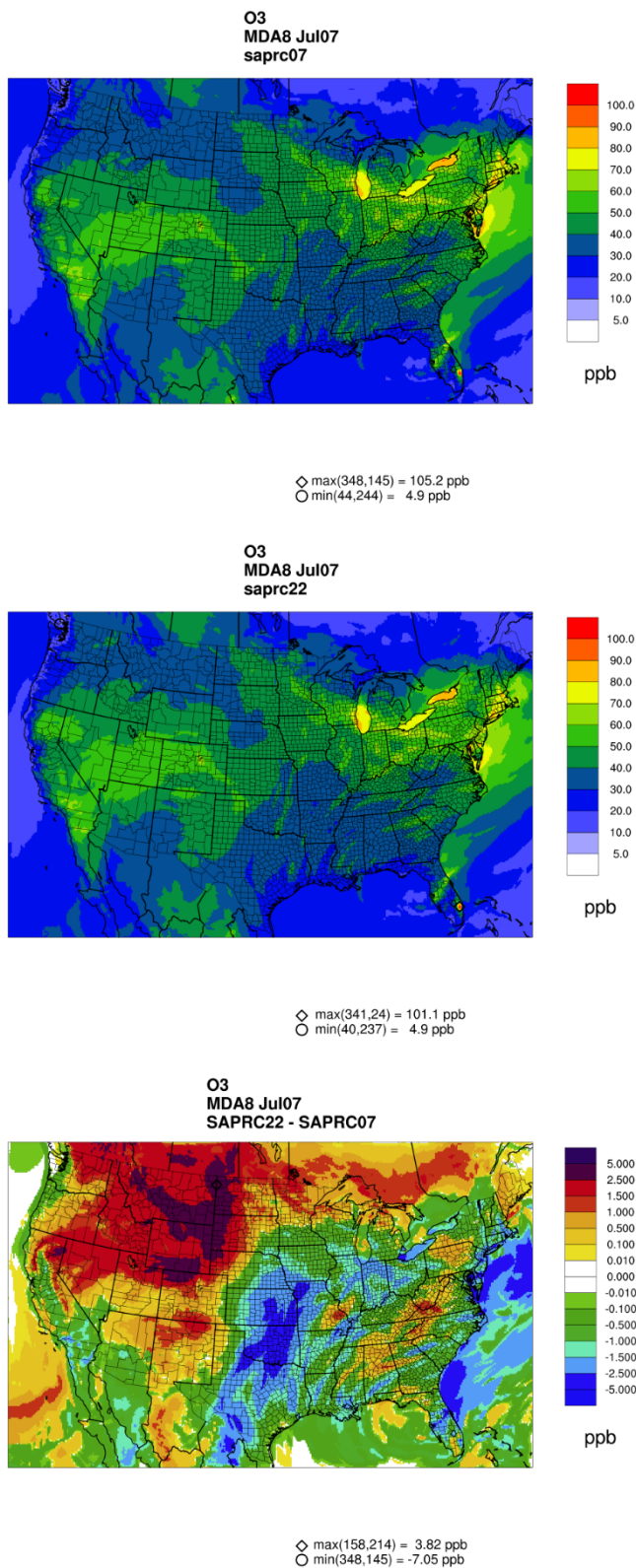


Figure 3-5. MDA8 ozone concentrations on July 7: SAPRC-07T chemistry (top); SAPRC-22 chemistry (middle); and differences in concentrations (bottom).

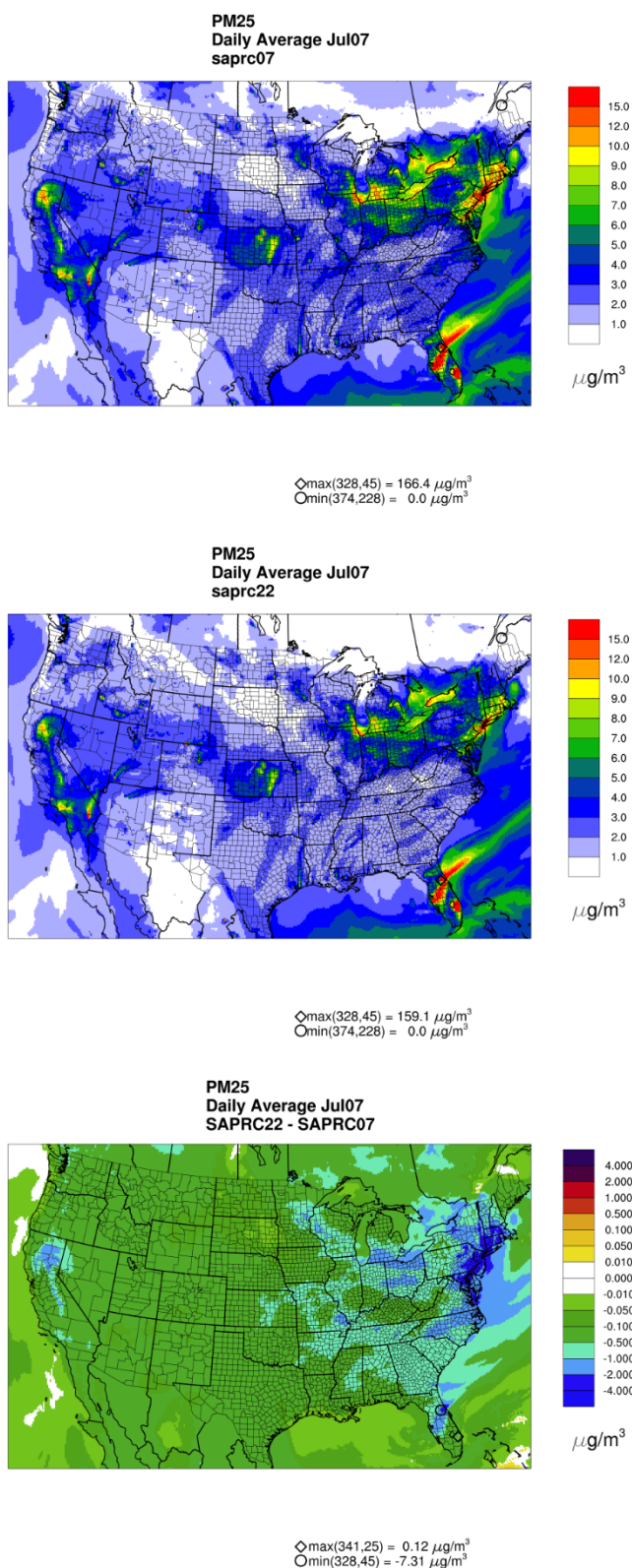
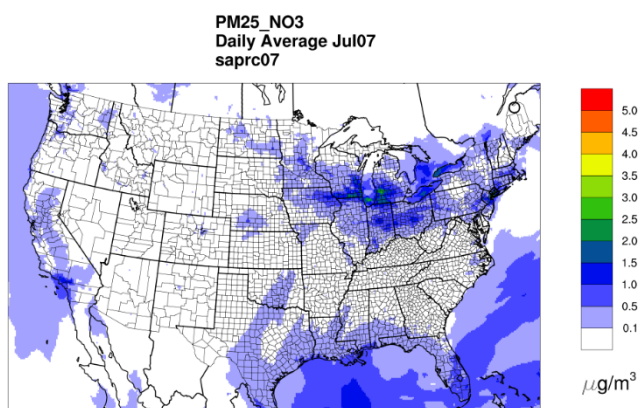
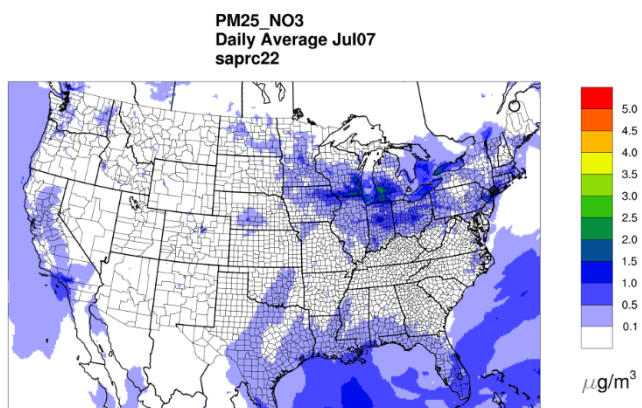


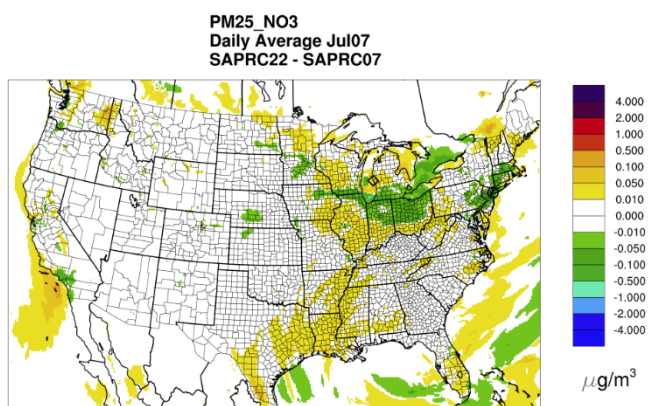
Figure 3-6. 24-hr average PM_{2.5} concentrations on July 7: SAPRC-07T chemistry (top); SAPRC-22 chemistry (middle); and differences in concentrations (bottom).



$\diamond\text{max}(362,165) = 3.9 \mu\text{g}/\text{m}^3$
 $\diamond\text{min}(377,228) = 0.0 \mu\text{g}/\text{m}^3$



$\diamond\text{max}(362,165) = 3.5 \mu\text{g}/\text{m}^3$
 $\diamond\text{min}(377,228) = 0.0 \mu\text{g}/\text{m}^3$



$\diamond\text{max}(272,171) = 0.17 \mu\text{g}/\text{m}^3$
 $\diamond\text{min}(361,162) = -0.62 \mu\text{g}/\text{m}^3$

Figure 3-7. 24-hr average PM_{2.5} nitrate concentrations on July 7: SAPRC-07T chemistry (top); SAPRC-22 chemistry (middle); and differences in concentrations (bottom).

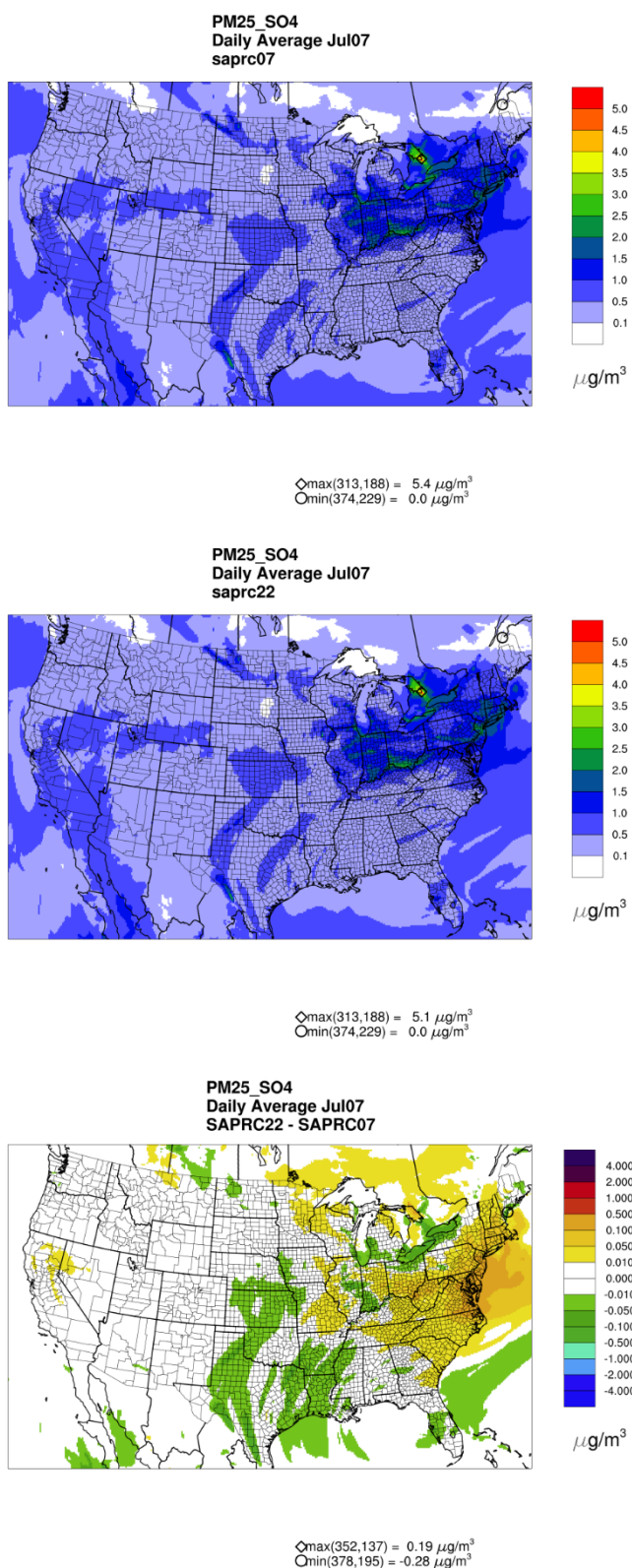


Figure 3-8. 24-hr average PM_{2.5} sulfate concentrations on July 7: SAPRC-07T chemistry (top); SAPRC-22 chemistry (middle); and differences in concentrations (bottom).

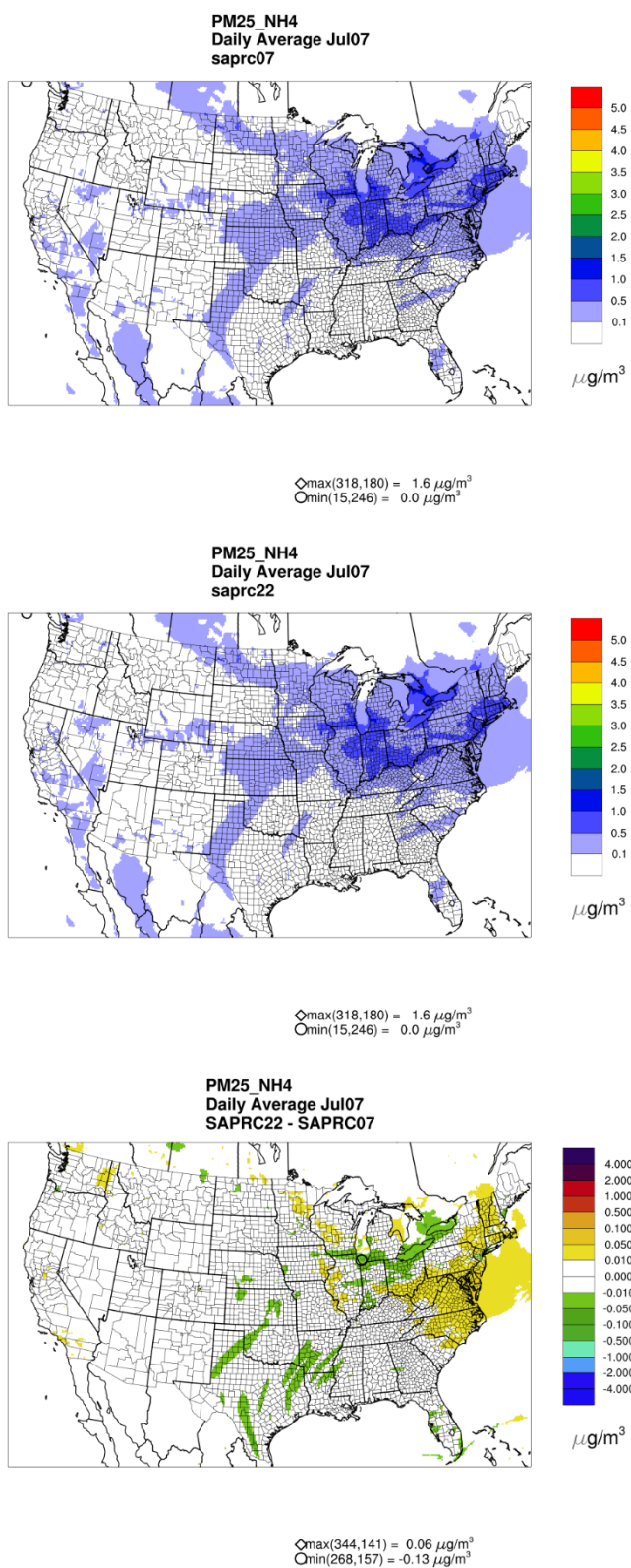


Figure 3-9. 24-hr average PM_{2.5} ammonium concentrations on July 7: SAPRC-07T chemistry (top); SAPRC-22 chemistry (middle); and differences in concentrations (bottom).

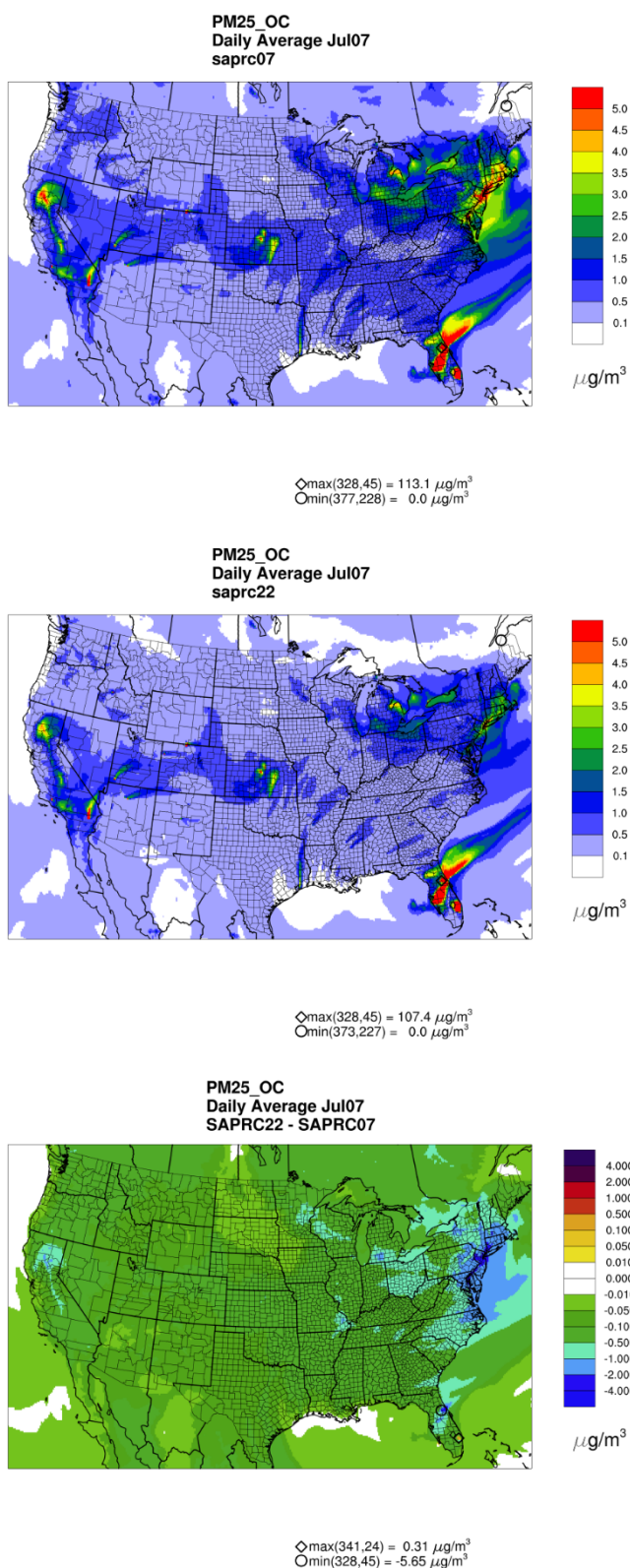


Figure 3-10. 24-hr average PM_{2.5} organic carbon concentrations on July 7: SAPRC-07T chemistry (top); SAPRC-22 chemistry (middle); and differences in concentrations (bottom).

Table 3-1. Surface layer concentrations of key species averaged over the domain and over 10 days (July 6 to July 15, 2016). Concentrations of gas species are in ppb and PM species are in $\mu\text{g}/\text{m}^3$.

Species	SAPRC-07T	SAPRC-22	SAPRC-22 – SAPRC-07T
NO	0.093	0.096	0.004
NO ₂	0.842	0.850	0.008
O ₃	29.817	29.700	-0.117
MDA8 O ₃	37.07	36.95	-0.116
HNO ₃	0.178	0.198	0.020
OH	0.00011	0.00010	-9.9×10^{-6}
HO ₂	0.0094	0.0090	-3.4×10^{-4}
SO ₂	0.129	0.131	0.002
PM ₂₅	2.909	2.566	-0.343
PM ₂₅ _SO ₄	0.470	0.474	0.004
PM ₂₅ _NO ₃	0.184	0.190	0.006
PM ₂₅ _NH ₄	0.088	0.090	0.002
PM ₂₅ _OC	0.675	0.460	-0.214

3.3.2 SAPRC-22 EBI vs. SAPRC-22 Rosenbrock

Figures 3-11 through 3-13 show scatterplots comparing hourly values of surface concentrations of key species from the two solvers over all grid cells for the 10-day period from July 6 to July 15. The comparisons demonstrate excellent agreement for the species involved in oxidant chemistry including NO, NO₂, O₃ and OH with minimal scatter away from the expected 1:1 agreement. However, more scatter is apparent for HO₂ even though the vast majority of points have near 1:1 agreement. SAPRC-22 accounts for complex dependencies of HO₂ production on other species, notably NO, using a system of operators. Apparently, HO₂ production in SAPRC22 is more sensitive to the numerical integration procedure than other species which is consistent with the HO₂ operator system behaving stiffly. We have not seen similar comparisons of the two solvers for other mechanisms and are unable to comment on whether HO₂ shows more scatter for SAPRC-22 than other mechanisms. Overall, the minor differences seen for HO₂ do not propagate to related important species (OH, O₃, PM_{2.5}) and therefore are not concerning. There are a few outliers for particulate nitrate. These are likely due to ISORROPIA sensitivity to small changes in HNO₃ concentrations, as discussed previously in Section 3.3.1.

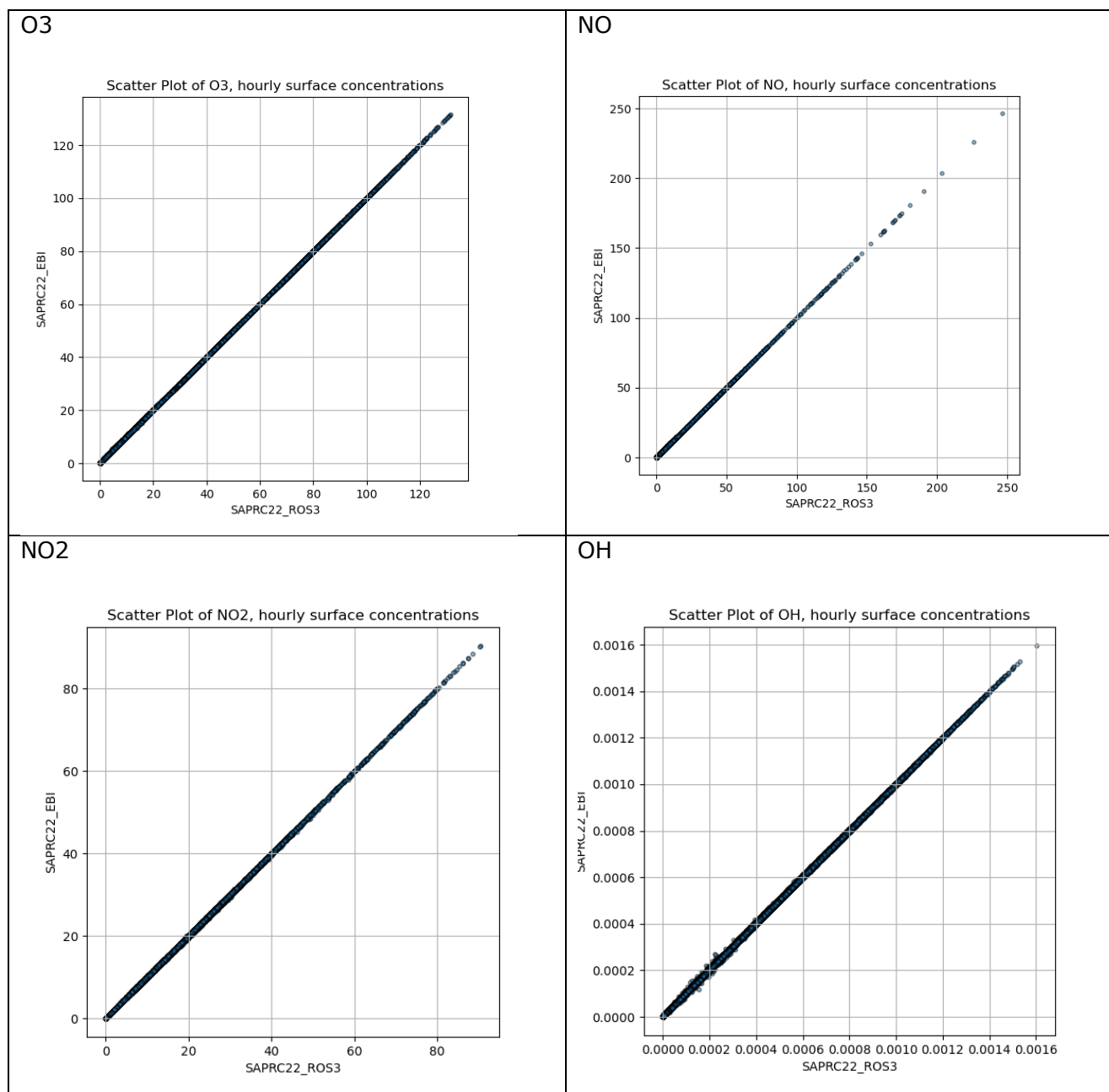


Figure 3-11. Comparison of hourly surface concentrations (all grid cells and 10 simulation days) of O₃, NO, NO₂ and OH from CMAQ simulations with the SAPRC-22 mechanism using the EBI solver with those using the Rosenbrock solver.

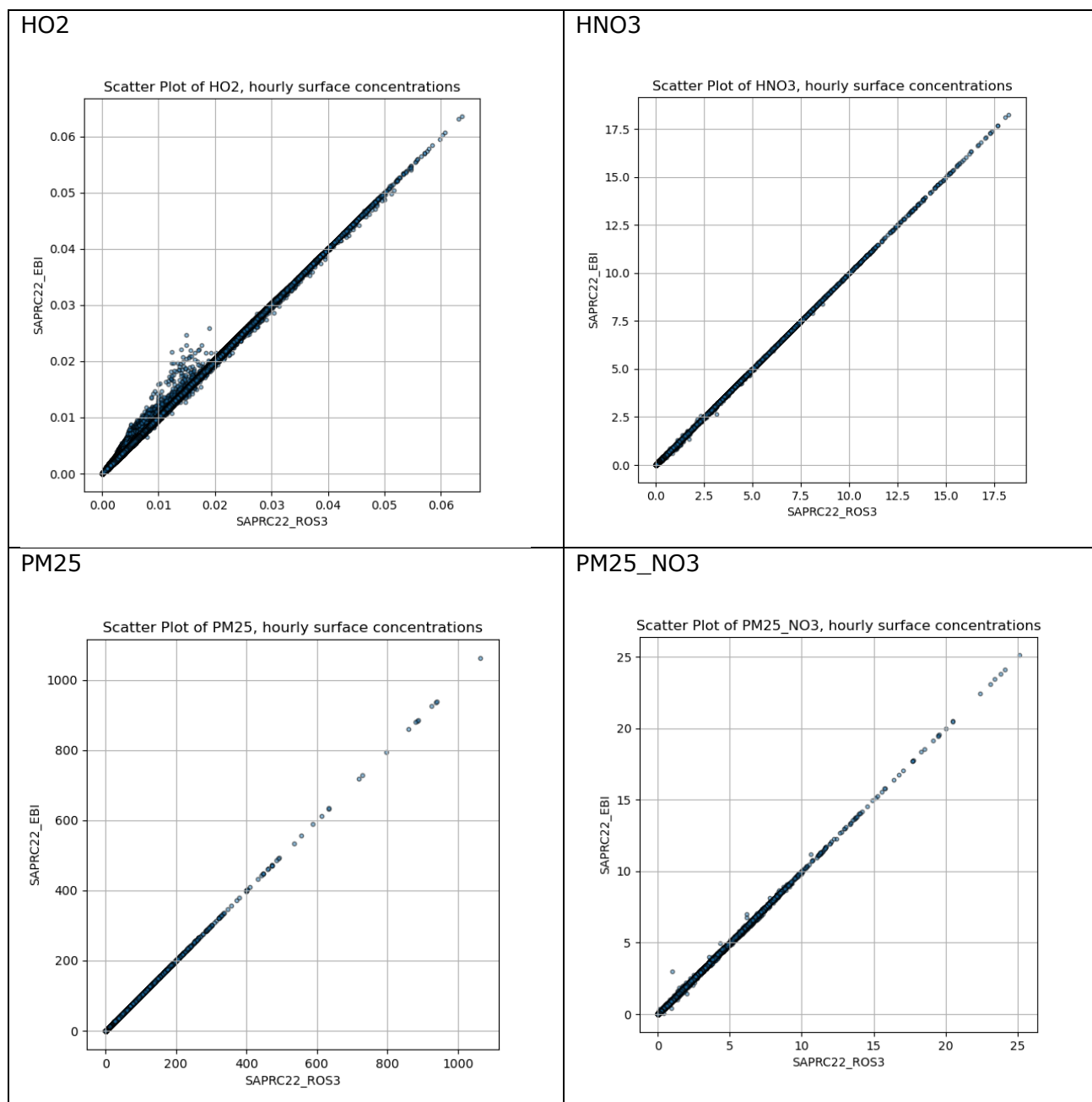


Figure 3-12. Comparison of hourly surface concentrations (all grid cells and 10 simulation days) of HO₂, HNO₃, PM_{2.5} and PM_{2.5}-Nitrate from CMAQ simulations with the SAPRC-22 mechanism using the EBI solver with those using the Rosenbrock solver.

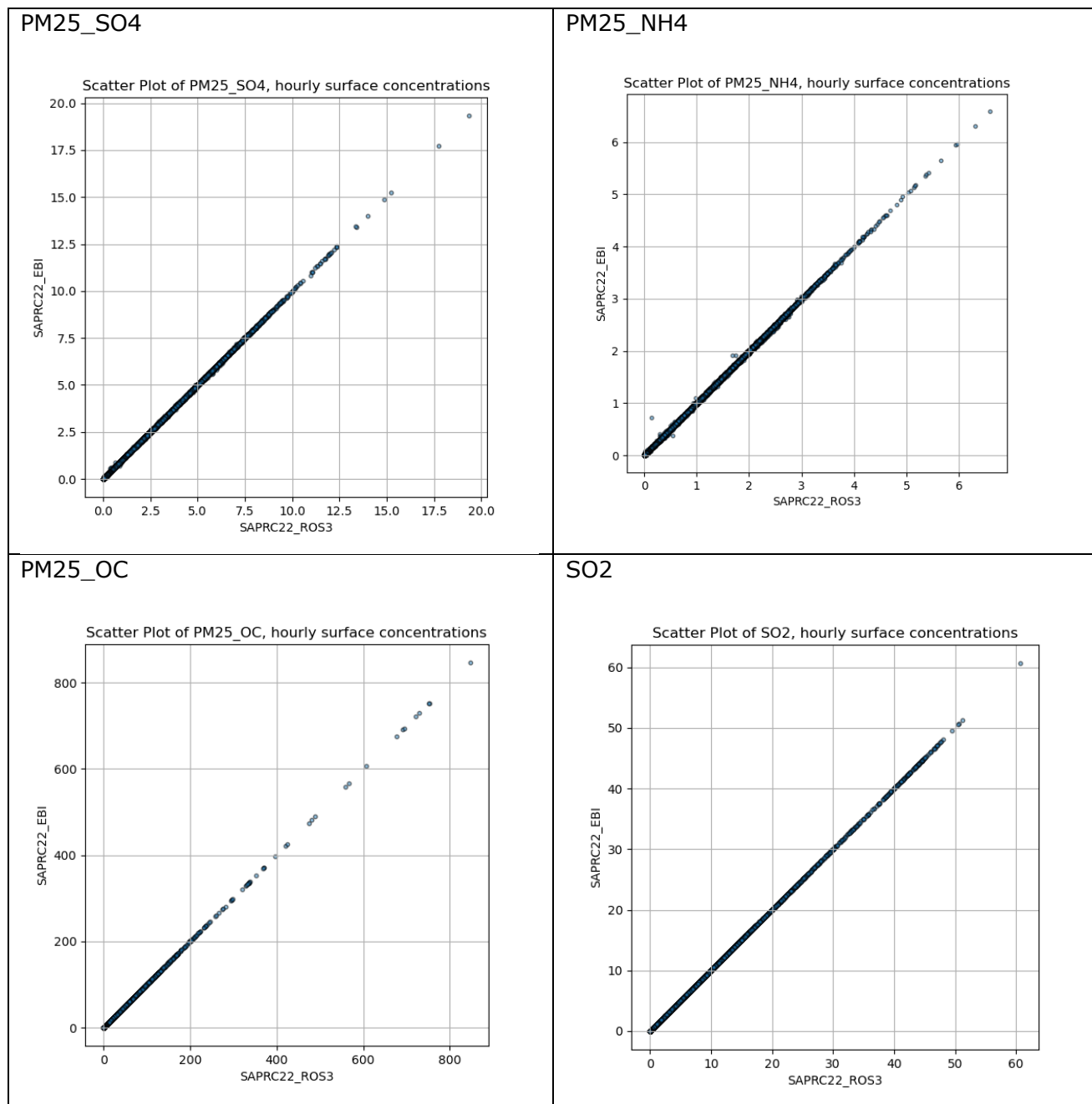


Figure 3-13. Comparison of hourly surface concentrations (all grid cells and 10 simulation days) of PM_{2.5}-Sulfate, PM_{2.5}-Ammonium, PM_{2.5}-Organic Carbon and SO₂ from CMAQ simulations with the SAPRC-22 mechanism using the EBI solver with those using the Rosenbrock solver.

3.3.3 Model Runtimes

The SAPRC-22 mechanism has more species and reactions than the SAPRC-07T mechanism, and this is expected to impact model runtimes. Table 3-2 below compares the runtimes for the three simulations that were conducted for this project. CMAQ with the SAPRC-22 mechanism and the EBI solver is about 21% slower than CMAQ with the SAPRC-07T mechanism and the EBI solver. The generalized Rosenbrock solver was used in this work mainly to serve as a benchmark to assess the accuracy of the EBI solver created for SAPRC-22 for this project. Using the Rosenbrock solver results in more than a factor of 2 increase in total runtime. Considering that both solvers produced similar concentration results for key species, we see no compelling reason to use the Rosenbrock solver for SAPRC-22.

Table 3-2. Model runtimes for SAPRC-07T/EBI, SAPRC-22/EBI and SAPRC-22/Rosenbrock.

Simulation	Runtime (Hours per Day of Simulation)
SAPRC-07T/EBI	4.24
SAPRC-22/EBI	5.12
SAPRC-22/ROS3	13.6

4.0 GUIDANCE FOR SPECIATING TOTAL ORGANIC GAS (TOG) EMISSIONS FOR THE SAPRC-22 CHEMICAL MECHANISM IN CMAQ

The SAPRC website provides mappings from TOG compounds to model species (referred to as a mechanism mapping) for SAPRC-22 and SAPRC-07T in the spreadsheet “SpecDB.xls” on tab “compounds”.¹ Table 4-1 compares how TOG compounds are mapped to SAPRC-22 or SAPRC-07T model species with comments that highlight differences. Explicit model species represent a single compound (e.g., ACET represents acetone only) and are identified in Table 4-1, whereas lumped model species can represent many compounds. The CMAQ implementation of SAPRC07T altered some species names (e.g., ACET changed to ACETONE) as noted in Table 4-1 comments. Our CMAQ implementation of SAPRC-22 uses the standard species names as given in column 2 of Table 4-1. The differences between the two mechanisms can be summarized as follows:

- Most explicit organic species present in SAPRC-07T retain the same name in SAPRC-22; some notable exceptions are ethylene (ethene), propylene (propene) and α -pinene with name changes from ETHE to ETHEN, PRPE to PROPE and APIN to APINE, respectively.
- Several explicit organic species are added in SAPRC-22; Notably, SAPRC-22 has explicit species for ethane (ETHAN), propane (PROP), n-butane (NC4) and β -pinene (BPINE), which should be used because these species can account for large fractions of TOG emissions.
- Many lumped species names carry over from SAPRC-07T to SAPRC-22 (ALK1, ALK2, ALK3, ALK4, ALK5, ARO1, ARO2, OLE1, OLE2) but with important changes to their definitions. For example, ethane and propane are explicit species in SAPRC22 but are lumped into ALK1 and ALK2, respectively, in SAPRC07T. Consequently, we do not recommend using model ready emissions prepared for SAPRC-07T with SAPRC-22.

The mechanism mappings provided within the SpecDB spreadsheet¹ are considered authoritative for SAPRC mechanisms. These mappings identify compounds by name and a unique ID used to prepare California Air Resources Board (CARB) emission inventories for modeling. Many of the mapped compounds are also identified by SMILES string² which can provide a convenient and authoritative basis for translating SpecDB mappings to work with other compound numbering schemes, e.g., from EPA’s SPECIATE database³. Alternatively, SAPRC mechanisms mappings can be developed from scratch by applying compound descriptions hierarchically (e.g., by compound class with explicit species mapped first) although estimated OH rate constants (kOH) will be needed for many compounds to complete this process.

The SpecDB mappings lack information needed to fully support the AERO6 scheme for secondary organic aerosols that we are linking with SAPRC-22 in CMAQv5.4. The following SAPRC-22 model species from Table 3-1 are linked to the AERO6 SOA scheme: BENZ, TOLU, ARO1, ARO2, trimethyl benzenes (BZ123, BZ124, BZ135), APINE, BPINE, STYRS, ISOP, MXYL, OXYL, PXYL, NAPS, SESQ, and TERP.

Table 4-1. Assignments of TOG compounds to SAPRC-22 and SAPRC-07T model species.

¹ <https://intra.engr.ucr.edu/~carter/emitdb/>

² https://en.wikipedia.org/wiki/Simplified_Molecular_Input_Line_Entry_System

³ <https://www.epa.gov/air-emissions-modeling/speciate>

Compound Description for SAPRC-22	SAPRC-22 Species	Explicit in SAPRC-22	SAPRC-07T Species ^a	Comment ^{b, c}
Acetone	ACET	yes	ACET	ACETONE in CMAQ version of S07T
Acetylene	ACETL	yes	ACYE	Name change from S07T; ACETYLENE in CMAQ S07T
Acrylonitrile	ACRLNT	yes	n/a	Assigned to ALK5 in CMAQ S07T
Acrolein	ACRO	yes	ACRO	ACROLEIN in CMAQ version of S07T
Saturated compounds that react only with OH and have kOH between 1.4×10^{-13} and 3.5×10^{-13}	ALK1	no	ALK1	Definition change from S07T; S07T ALK1 includes ethane; S07T kOH range is $< 3.4 \times 10^{-13}$
Saturated compounds that have kOH between 3.5×10^{-13} and 1.7×10^{-12}	ALK2	no	ALK2	Definition change from S07T; S07T ALK2 includes propane; S07T kOH range is 3.4×10^{-13} to 1.7×10^{-12}
Saturated compounds that have kOH between 1.7×10^{-12} and 3.5×10^{-12}	ALK3	no	ALK3	Definition change from S07T; S07T ALK3 includes n-butane and acetylene; S07T kOH range is 1.7×10^{-12} to 3.4×10^{-12}
Saturated compounds that have kOH between 3.5×10^{-12} and 6.0×10^{-12}	ALK4	no	ALK4	Definition change from S07T; S07T kOH range is 3.4×10^{-12} to 6.8×10^{-12}
Saturated compounds that have kOH greater than 6.0×10^{-12} and have fewer than 12 C + O atoms.	ALK5	no	ALK5	Definition change from S07T; S07T kOH range is $> 6.8 \times 10^{-12}$
Saturated compounds that have kOH greater than 6.0×10^{-12} and have 12 or more C + O atoms.	ALK6	no	n/a	Compounds assigned to ALK6 tend to contain O, e.g., glycol ethers
Amines	AMINS	no	n/a	Except for tertiary amines (see TAMNS)
α -Pinene	APINE	yes	APIN	Name change from S07T
Alkylbenzenes that have kOH between 5×10^{-12} and 1×10^{-11} . Mostly alkylbenzenes with only one substituent	ARO1	no	ARO1	Definition change from S07T; S07T kOH range is $< 1.4 \times 10^{-11}$
Alkylbenzenes that have kOH higher than 1×10^{-11} . Mostly alkylbenzenes with more than one substituent	ARO2	no	ARO2	Definition change from S07T; S07T kOH range is $> 1.4 \times 10^{-11}$; ARO2MN in CMAQ S07T (ARO2 minus Naphthalenes)
Biacetyl and other α -diketones	BACL	no	BACL	
Aromatic aldehydes (e.g., benzaldehyde)	BALD	no	BALD	

Compound Description for SAPRC-22	SAPRC-22 Species	Explicit in SAPRC-22	SAPRC-07T Species ^a	Comment ^{b, c}
Benzene and substituted benzenes that have kOH between 5×10^{-13} and 5×10^{-12}	BENX	no	n/a	E.g., chlorobenzene
Benzene	BENZ	yes	BENZ	BENZENE in CMAQ S07T
β-Pinene	BPINE	yes	n/a	β-pinene lumped with TERP in S07T
1,3-Butadiene	BUT13	yes	13BDE	Name change from S07T; BUTADIENE13 in CMAQ S07T
1,2,3-Trimethyl Benzene	BZ123	yes	n/a	Assigned to ARO2 in CMAQ S07T
1,2,4-Trimethyl Benzene	BZ124	yes	B124	Name change from S07T; TMBENZ124 in CMAQ S07T
1,3,5-Trimethyl Benzene	BZ135	yes	n/a	Assigned to ARO2 in S07T
Ethyl Benzene	C2BEN	yes	n/a	Assigned to ARO1 in S07T
Dihydroxy phenols	CATL	no	n/a	Assigned to CRES in S07T
Carbon tetrachloride	CCL4	yes	n/a	
Methane	CH4	yes	CH4	
Chloroform	CHCL3	yes	n/a	
Vinyl chloride	CLETHE	yes	n/a	Assigned to ALK5 in S07T
Cresols	CRES	no	CRES	
Ethylene dibromide	ETBR2	yes	n/a	Assigned to ALK1 in S07T
Propionaldehyde	ETCHO	yes	n/a	Assigned to RCHO in S07T
Ethylene dichloride	ETCL2	yes	n/a	Assigned to ALK1 in S07T
Ethane	ETHAN	yes	n/a	Assigned to ALK1 in S07T
Ethylene	ETHEN	yes	ETHE	Name change from S07T; ETHENE in CMAQ S07T
Ethanol	ETOH	yes	ETOH	
Ethylene oxide	ETOX	yes	n/a	Assigned to NROG in S07T
Furans	FURNS	no	n/a	Assigned to ARO2 in S07T
Glycolaldehyde	GLCHO	yes	n/a	Assigned to CCHO in S07T
Glyoxal	GLY	yes	GLY	
Formaldehyde	HCHO	yes	HCHO	
Formic Acid	HCOOH	yes	FACD	Name change from S07T; HCOOH in CMAQ S07T
Any compound with C=N bond. Assumed to rapidly hydrolyze.	IMINE	no	n/a	
Isoprene	ISOP	yes	ISOP	ISOPRENE in CMAQ S07T
Photoreactive saturated ketones	KET2	no	n/a	E.g., methyl isopropyl ketone, hydroxy acetone; PRD2 is the nearest equivalent in S07T
Ketones with at least 1 C=C double bond.	LVKS	no	n/a	E.g., 4-hepten-2-one; IPRD is the nearest equivalent in S07T
Methacrolein	MACR	yes	MACR	

Compound Description for SAPRC-22	SAPRC-22 Species	Explicit in SAPRC-22	SAPRC-07T Species ^a	Comment ^{b, c}
Acetaldehyde	MECHO	yes	<i>CCHO</i>	Name change from S07T
Methylene chloride	MECL2	yes	n/a	Assigned to ALK1 in S07T
Methyl ethyl ketone	MEK	yes	MEK	
Methanol	MEOH	yes	MEOH	
Methyl Glyoxal and other alpha-keto aldehydes	MGLY	no	MGLY	
Methyl t-Butyl Ether	MTBE	yes	n/a	Assigned to ALK3 in S07T
Methyl Vinyl Ketone	MVK	yes	MVK	
m-Xylene	MXYL	yes	MXYL	
Naphthalenes, tetralins, and indans	NAPS	no	n/a	NAPHTHAL In CMAQ S07T; see ARO2 comment
n-Butane	NC4	yes	n/a	Assigned to ALK3 in S07T
Unreactive mass	NROG	no	NROG	
Non-volatile	NVOL	no	NVOL	
Organic acids	OACID	no	<i>AACD</i>	E.g., acetic acid; Name change from S07T; CMAQ S07T uses CCOOH for acetic acid and RCOOH for higher organic acids
Alkenes other than ethene with k_{OH} between 8×10^{-12} and 6×10^{-11}	OLE1	no	OLE1	Definition change from S07T; S07T k_{OH} range is $< 4.7 \times 10^{-11}$
Alkenes other than terpenes or 1,3-butadiene that have k_{OH} rate constants greater than 6×10^{-12}	OLE2	no	OLE2	Definition change from S07T; S07T k_{OH} range is $> 4.7 \times 10^{-11}$
Alkenes other than terpenes or sesquiterpenes with only $-CR=CH_2$ groups.	OLE3	no	n/a	E.g., 2-methylpropene (isobutene)
Alkenes other than terpenes with non-conjugated $-CH=CR-$ or $-CR=CR-$ groups and possibly other double bonds and no double bonds in rings	OLE4	no	n/a	E.g., 2-methyl-2-butene
Unsaturated aldehydes with $C=C$ next to $-CHO$ (mostly from aromatics)	OLEA1	no	n/a	E.g., crotonaldehyde, trans-2-hexenal
Unsaturated aldehydes with $C=C$ not next to the $-CHO$ (mostly from aromatics)	OLEA2	no	n/a	E.g., cis-3-hexenal
Non-photoreactive, non-hydrocarbon compounds with $C=C$ double bonds	OLEP	no	n/a	E.g., 1,2-dichloroethene isomers

Compound Description for SAPRC-22	SAPRC-22 Species	Explicit in SAPRC-22	SAPRC-07T Species ^a	Comment ^{b, c}
Other organic products with vapor pressures less than 1 ppb	OTHN	no	n/a	E.g., octanedioic acid
o-Xylene	OXYL	yes	OXYL	
Organic peroxyacids	PACID	no	<i>PACD</i>	Name change from S07T; CMAQ S07T uses CCOOOH for peroxyacetic acid and RCOOOH for higher organic peroxyacids
Perchloroethylene	PCE	yes	n/a	Assigned to ALK1 in S07T
p-dichlorobenzene	PCLBEN	yes	n/a	Assigned to ARO1 in S07T
Phenol	PHEN	yes	n/a	Assigned to CRES in S07T
Unspecified photoreactive compounds such as nitrites or chloropicrin	PHOT	yes	n/a	Products omit NO _y from N atom in example precursors
Propane	PROP	yes	n/a	Assigned to ALK2 in S07T
Propene	PROPE	yes	<i>PRPE</i>	Name change from S07T; PROPENE in CMAQ S07T
p-Xylene	PXYL	yes	PXYL	
Other volatile organic nitrates that react with OH radicals slower than 5 x 10 ⁻¹²	R1NO3	no	n/a	E.g., isopropyl nitrate; assigned to NROG in S07T
Saturated aldehydes with 4 to 9 C atoms	RCHO	no	RCHO	
Methyl Hydroperoxide	MEOOH	yes	<i>COOH</i>	Name change from S07T
Other hydroperoxides with 3+ C atoms and vapor pressure greater than 1 ppb	ROOH	no	ROOH	E.g., cumene hydroperoxide
Sesquiterpenes	SESQ	no	SESQ	
Styrenes	STYRS	no	n/a	Assigned to OLE1 and OLE2 in S07T
Amines bonded to a tertiary carbon	TAMNS	no	n/a	E.g., 2-amino-2-methyl-1-propanol (AMP)
Lumped terpenes (C ₁₀ H ₁₆) with at least one ring and alkene group	TERP	no	TERP	
Toluene	TOLU	yes	TOLU	TOLUENE in CMAQ S07T
Xylenols and higher alkylphenols	XYNL	no	n/a	E.g., dimethyl phenols, 2-ethylphenol; assigned to CRES in S07T

Notes:

(a) Italicized SAPRC-07T names highlight a difference in species name or definition as explained by the comment.

(b) kOH values have units cm⁻³ molecule⁻¹ s⁻¹.

(c) SAPRC-07T is abbreviated as S07T in comments.

AERO6 uses model species SOAALK to account for SOA production from large alkanes and model species PCVOC to account for SOA production from combustion related IVOCs and other missing emissions/processes (Murphy et al. 2017). Emissions of linear and branched alkanes with more than 8 carbons and cyclic alkanes with more than 6 carbons should be mapped⁴ to SOAALK in addition to ALK4 or ALK5 (i.e., double counted) to account for their impacts to oxidants (via ALK4 or ALK5) and SOA (via SOAALK). Emissions of compounds with volatility (C^*) in the range $3 \times 10^2 \mu\text{g m}^{-3} < C^* < 3 \times 10^6 \mu\text{g m}^{-3}$ can be mapped (Seltzer et al., 2022) to PCVOC to account for their SOA formation potential although this may have the following limitations:

- (1) PCVOC emissions have no impact on oxidants within CMAQ, so mapping emitted IVOCs to PCVOC will implicitly assume that they have no oxidant formation potential.
- (2) CMAQ internally estimates PCVOC emissions for many source categories, by scaling PCVOC to primary organic aerosol (POA) emissions, so double counting of emissions mass may occur depending upon how IVOC and POA emissions were estimated outside CMAQ.

⁴ https://www.epa.gov/system/files/documents/2022-09/2019_EmisMod_TSD_August2022_1.pdf

5.0 SUMMARY

The SAPRC-22 gas-phase chemistry was successfully implemented in CMAQv5.4. The model was tested using an available CMAQ test case for a two-week period in July 2016. Comparisons of results from the CMAQ simulation using SAPRC-22 chemistry with those from a simulation using SAPRC-07T chemistry showed that the spatial patterns of species concentrations were similar and differences in values were reasonable. A CMAQ/SAPRC-22 simulation with the generalized mechanism-independent Rosenbrock solver was also conducted to check the accuracy of the faster but mechanism-dependent EBI solver generated for SAPRC-22. This simulation showed that the results with the two solvers were almost identical, providing confidence in using the EBI solver for simulations with SAPRC-22.

6.0 REFERENCES

- Carter, W.P.L., 2010. Development of the SAPRC-07 chemical mechanism, *Atmos. Environ.*, **44**(40), 5324-5335, <https://doi.org/10.1016/j.atmosenv.2010.01.026>.
- Carter, W.P.L., 2023. *Documentation of the SAPRC-22 Mechanism*. Report to the California Air Resources Board, Contract No. 21AQP011, <https://intra.engr.ucr.edu/~carter/SAPRC/22/S22doc.pdf>.
- EPA, 2023. *Technical Support Document (TSD): Preparation of Emissions Inventories for the 2016v3 North American Emissions Modeling Platform*. U.S. Environmental Protection Agency, Office of Air Quality Planning and Standards, Air Quality Assessment Division, Emissions Inventory and Analysis Group. Research Triangle Park, NC. EPA-454/B-23-002, January 2023, https://www.epa.gov/system/files/documents/2023-03/2016v3_EmisMod_TSD_January2023_2.pdf.
- Hutzell, W.T., Luecken, D.J., Appel, K.W. and Carter, W.P.L., 2012. Interpreting predictions from the SAPRC07 mechanism based on regional and continental simulations. *Atmospheric Environment*, **46**, 417-429, <https://doi.org/10.1016/j.atmosenv.2011.09.030>.
- Koo, B., Metzger, S., Vennam, P., Emery, C., Wilson, G., and Yarwood, G., (2020). Comparing the ISORROPIA and EQSAM Aerosol Thermodynamic Options in CAMx. In: *Mensink, C., Gong, W., Hakami, A. (eds) Air Pollution Modeling and its Application XXVI*, ITM 2018. Springer Proceedings in Complexity. Springer, Cham, https://doi.org/10.1007/978-3-030-22055-6_16.
- Murphy, B.N., Woody, M.C., Jimenez, J.L., Carlton, A.M.G., Hayes, P.L., Liu, S., Ng, N.L., Russell, L.M., Setyan, A., Xu, L., Young, J., Zaveri, R.A., Zhang, Q., and Pye, H.O.T., 2017. Semivolatile POA and parameterized total combustion SOA in CMAQv5.2: impacts on source strength and partitioning, *Atmos. Chem. Phys.*, **17**, 11107–11133, <https://doi.org/10.5194/acp-17-11107-2017>.
- Seltzer, K.M., Murphy, B.N. Murphy, Pennington, E.A., Allen, C., Talgo, K., and Pye, H.O.T., 2022. Volatile Chemical Product Enhancements to Criteria Pollutants in the United States, *Environ. Sci. Technol.*, **56**(11), 6905-6913, <https://pubs.acs.org/doi/10.1021/acs.est.1c04298>.

Appendix A: SAPRC-22 Reactions and Species

Table A-1. CMAQv5.4 SAPRC-22 Mechanism Definition

! SAPRC22 Standard 6/22/24 (vers R, Lumpmodel=B)

! Created from SAPRC22STD.xls 22-Jun-2024 12:20

SAPRC22_AE65_AQ

REACTIONS[CM] =

```
<1> NO2 = NO + O3P # 1.0/<NO2_06>;
<2> O3P + O2 + M = O3 # 6.00e-34^2.40;
<3> O3P + O3 = # 8.00e-12@2060;
<4> O3P + NO = NO2 # 9.00e-32^-1.50&3.00e-11&0.60&1.00;
<5> O3P + NO2 = NO # 5.10e-12@-210;
<6> O3P + NO2 = NO3 # 2.50e-31^-1.80&2.20e-11^-0.70&0.60&1.00;
<7> O3 + NO = NO2 # 3.00e-12@1500;
<8> O3 + NO2 = NO3 # 1.20e-13@2450;
<9> NO + NO3 = 2*NO2 # 1.50e-11@-170;
<10> NO + NO + O2 = 2*NO2 # 3.30e-39@-530;
<11> NO2 + NO3 = N2O5 # 3.60e-30^-4.10&1.90e-12^0.20&0.35&1.33;
<12> N2O5 = NO2 + NO3 # 1.30e-03^-3.50@11000&9.70e+14^0.10@11080&0.35&1.33;
<13> N2O5 + H2O = 2*HNO3 # 0.00e+00;
<14> N2O5 + H2O + H2O = 2*HNO3 # 0.00e+00;
<15> NO2 + NO3 = NO + NO2 # 4.50e-14@1260;
<16> NO3 = NO # 1.0/<NO3NO_06>;
<17> NO3 = NO2 + O3P # 1.0/<NO3NO2_6>;
<18> O3 = O1D # 1.0/<O3O1D_06>;
<19> O3 = O3P # 1.0/<O3O3P_06>;
<20> O1D + H2O = 2*OH # 1.63e-10@-60;
<21> O1D + M = O3P # 2.65e-11@-98;
<22> OH + NO = HONO # 7.00e-31^-2.60&3.60e-11^-0.10&0.60&1.00;
<23> HONO = OH + NO # 1.0/<HONO_06>;
<24> OH + HONO = NO2 # 1.80e-11@390;
<25> OH + NO2 = HNO3 # 3.20e-30^-4.50&3.00e-11&0.41&1.24;
<26> OH + NO3 = HO2 + NO2 # 2.20e-11;
<27> OH + HNO3 = NO3 %2 # 2.40e-14@-460&2.70e-17@-2199&6.50e-34@-1335;
<28> HNO3 = OH + NO2 # 1.0/<HNO3>;
<29> OH + O3 = HO2 # 1.70e-12@940;
<30> HO2 + NO = OH + NO2 # 3.30e-12@-270;
<31> HO2 + NO = HNO3 %3 # 2.39e-12^-13.77@1711&1.83e-32^-4.85@772;
<32> HO2 + NO + H2O = HNO3 # 1.20e-35@-2944;
<33> HO2 + NO2 = HNO4 # 1.40e-31^-3.10&4.00e-12&0.40&1.26;
<34> HNO4 = HO2 + NO2 # 4.10e-05@10650&6.00e+15@11170&0.40&1.26;
<35> HNO4 = 0.8*HO2 + 0.8*NO2 + 0.2*OH + 0.2*NO3 # 1.0/<HNO4_06>;
<36> HNO4 + OH = NO2 # 1.30e-12@-380;
<37> HO2 + O3 = OH # 1.00e-14@490;
<38> HO2 + HO2 = HO2H %3 # 3.00e-13@-460&2.10e-33@-920;
<39> HO2 + HO2 + H2O = HO2H %3 # 4.20e-34@-2660&2.94e-54@-3120;
<40> NO3 + HO2 = OH + NO2 # 3.50e-12;
<41> NO3 + NO3 = 2*NO2 # 8.50e-13@2450;
<42> HO2H = 2*OH # 1.0/<H2O2>;
<43> HO2H + OH = HO2 # 1.80e-12;
<44> OH + HO2 = # 4.80e-11@-250;
<45> CO + OH = HO2 + CO2 %3 # 1.44e-13@0&3.43e-33@0;
<46> H2 + OH = HO2 # 2.80e-12@1800;
<R2NO> SumRO2 + NO = NO # 2.55e-12@-380;
<R2H2> SumRO2 + HO2 = HO2 # 1.49e-11;
<R2N3> SumRO2 + NO3 = NO3 # 2.30e-12;
<R2R2> SumRO2 + SumRO2 = # 1.60e-14;
```

```

<R3N2> SumRCO3 + NO2 = NO2 # 7.70e-12^-0.20;
<R3NO> SumRCO3 + NO = NO # 6.70e-12@-340;
<R3H2> SumRCO3 + HO2 = HO2 # 3.14e-12@-580;
<R3N3> SumRCO3 + NO3 = NO3 # 4.00e-12;
<R3R2> SumRCO3 + SumRO2 = # 4.40e-13@-1070;
<R3R3> SumRCO3 + SumRCO3 = # 1.70e-11;
<57> RO2C + NO = NO2 # 1.0*K<R2NO>;
<58> RO2C + HO2 = # 1.0*K<R2H2>;
<59> RO2C + NO3 = NO2 # 1.0*K<R2N3>;
<60> RO2C + SumRO2 = SumRO2 # 1.0*K<R2R2>;
<61> RO2C + SumRCO3 = SumRCO3 # 1.0*K<R3R2>;
<62> RO2XC + NO = # 1.0*K<R2NO>;
<63> RO2XC + HO2 = # 1.0*K<R2H2>;
<64> RO2XC + NO3 = NO2 # 1.0*K<R2N3>;
<65> RO2XC + SumRO2 = SumRO2 # 1.0*K<R2R2>;
<66> RO2XC + SumRCO3 = SumRCO3 # 1.0*K<R3R2>;
<67> MEO2 + NO = NO2 + HCHO + HO2 # 2.80e-12@-300;
<68> MEO2 + HO2 = 0.9*MEOOH + 0.1*HCHO # 3.80e-13@-780;
<69> MEO2 + NO3 = HCHO + HO2 + NO2 # 1.20e-12;
<70> MEO2 + SumRO2 = 0.3*HO2 + 0.65*HCHO + 0.35*MEOH # 2.16e-13;
<71> MEO2 + SumRCO3 = 0.9*HO2 + HCHO # 2.00e-12@-500;
<72> ETO2 + NO = NO2 + HO2 + MECHO # 2.55e-12@-380;
<73> ETO2 + HO2 = ROOH # 7.44e-12;
<74> ETO2 + NO3 = NO2 + HO2 + MECHO # 2.30e-12;
<75> ETO2 + SumRO2 = 0.5*HO2 + 0.25*ETOH + 0.75*MECHO # 2.90e-14;
<76> ETO2 + SumRCO3 = 0.8*HO2 + MECHO # 1.60e-11;
<77> BZO2 + NO = NO2 + BZO # 1.0*K<R2NO>;
<78> BZO2 + HO2 = ROOH # 1.0*K<R2H2>;
<79> BZO2 + NO3 = BZO + NO2 # 1.0*K<R2N3>;
<80> BZO2 + SumRO2 = SumRO2 + BZO # 1.0*K<R2R2>;
<81> BZO2 + SumRCO3 = SumRCO3 + BZO # 1.0*K<R3R2>;
<Q1NO> MECO3 + NO = NO2 + MEO2 + CO2 + SumRO2 # 8.10e-12@-270;
<Q1N2> MECO3 + NO2 = PAN # 9.70e-29^-5.60&9.30e-12^-1.50&0.60&1.00;
<Q1N3> MECO3 + NO3 = NO2 + MEO2 + CO2 + SumRO2 # 4.00e-12;
<Q1H2> MECO3 + HO2 = 0.13*O3 + 0.5*OH + 0.5*MEO2 + 0.13*OACID + 0.37*PACID +
    0.5*CO2 + 0.5*SumRO2 # 2.20e-11;
<Q1R2> MECO3 + SumRO2 = 0.9*MEO2 + 0.1*OACID + 0.9*CO2 + 0.9*SumRO2 # 1.60e-11;
<Q1R3> MECO3 + SumRCO3 = MEO2 + CO2 + SumRO2 # 1.40e-11;
<Q6N2> BZCO3 + NO2 = PBZN # 1.11e-11;
<Q6NO> BZCO3 + NO = NO2 + CO2 + BZO2 + SumRO2 # 1.60e-11;
<Q6H2> BZCO3 + HO2 = 0.5*ALK5 + 0.13*O3 + 0.5*OH + 0.5*BZO2 + 0.5*CO2 +
    0.5*SumRO2 # 1.0*K<R3H2>;
<Q6N3> BZCO3 + NO3 = NO2 + CO2 + BZO2 + SumRO2 # 1.0*K<R3N3>;
<Q6R2> BZCO3 + SumRO2 = 2*SumRO2 + BZO2 + CO2 # 1.0*K<R3R2>;
<Q6R3> BZCO3 + SumRCO3 = SumRCO3 + CO2 + BZO2 + SumRO2 # 1.0*K<R3R3>;
<94> TBUO + NO2 = R1NO3 # 3.50e-12@-553;
<95> TBUO = ACET + MEO2 + SumRO2 # 1.40e+13@6856;
<96> BZO + NO2 = NPHE # 2.08e-12;
<97> BZO + HO2 = CRES # 1.0*K<R2H2>;
<98> BZO + O3 = BZO2 + SumRO2 # 2.86e-13;
<99> BZO + BZO = # 1.49e-11;
<100> NPRAD + NO2 = NPHE # 1.0*K<R3N2>;
<101> NPRAD + HO2 = NAPPRD # 1.0*K<R3H2>;
<102> NPRAD = NAPPRD # 1.00e-03;
<103> PNAMIN + NO2 = NAMIN # 1.0*K<R3N2>;
<104> PNAMIN + HO2 = AMINS # 1.0*K<R3H2>;
<105> PNAMIN = AMINS # 1.00e-03;
<G1N2> HCHO2 + NO2 = HCHO + NO3 # 7.00e-12;

```

<G1WA> HCHO2 + H2O = HCOOH # 2.40e-15;
 <G1S2> HCHO2 + SO2 = SULF + HCHO + SULRXN # 3.80e-11;
 <G2N2> MECHO2 + NO2 = MECHO + NO3 # 7.00e-12;
 <G2WA> MECHO2 + H2O = OACID # 2.40e-15;
 <G2S2> MECHO2 + SO2 = SULF + MECHO + SULRXN # 3.80e-11;
 <G3N2> RCHO2 + NO2 = RCHO + NO3 # 7.00e-12;
 <G3WA> RCHO2 + H2O = OACID # 2.40e-15;
 <G3S2> RCHO2 + SO2 = SULF + RCHO + SULRXN # 3.80e-11;
 <S2OH> SO2 + OH = HO2 + SULF + SULRXN # 3.30e-31^-4.30&1.60e-12&0.60&1.00;
 <C1OH> CH4 + OH = MEO2 + SumRO2 # 2.45e-12@1775;
 <117> HCHO = 2*HO2 + CO # 1.0/<HCHOR_13>;
 <118> HCHO = H2 + CO # 1.0/<HCHOM_13>;
 <119> HCHO + OH = HO2 + CO # 5.50e-12@-125;
 <120> HCHO + NO3 = HNO3 + HO2 + CO # 5.80e-16;
 <P1UI> PAN = NO2 + MECO3 + SumRCO3
 # 1.08e+00^-5.60@14000&1.03e+17^-1.50@14000&0.60&1.00;
 <P1HV> PAN = 0.6*NO2 + 0.4*NO3 + 0.4*MEO2 + 0.6*MECO3 + 0.4*CO2 + 0.4*SumRO2 +
 0.6*SumRCO3 # 1.0/<PAN_11>;
 <123> GLY = 2*CO + 2*HO2 # 1.0/<GLY_I13R>;
 <124> GLY = HCHO + CO # 1.0/<GLY_I13M>;
 <125> GLY + OH = 1.7*CO + 0.7*HO2 + 0.3*OH + 0.3*CO2 # 1.15e-11;
 <126> GLY + NO3 = HNO3 + 1.7*CO + 0.7*HO2 + 0.3*OH + 0.3*CO2 # 4.00e-16;
 <BLOH> BALD + OH = BZCO3 + SumRCO3 # 1.20e-11;
 <BLHV> BALD = # 9.00e-2/<BALD_11>;
 <BLN3> BALD + NO3 = HNO3 + BZCO3 + SumRCO3 # 4.00e-15;
 <PBUI> PBZN = BZCO3 + NO2 + SumRCO3 # 2.10e+16@13600;
 <PBHV> PBZN = 0.6*BZCO3 + 0.6*NO2 + 0.4*CO2 + 0.4*BZO2 + 0.4*NO3 + 0.4*SumRO2 +
 0.6*SumRCO3 # 1.0/<PPN_11>;
 <NPOH> NPHE + OH = BZO + NO2 # 3.50e-12;
 <NPHV> NPHE = HONO + PHEN # 1.50e-3/<NO2_06>;
 <NAOH> NAPS + OH = 0.741*HO2 + 0.707*CATL + 0.034*RO2C + 0.017*AFG2A +
 0.017*AFG2B + 0.034*GLY + 0.33*NPRAD + 0.25*MACO3 + 0.034*SumRO2 +
 0.25*SumRCO3 + PAHRO2 # 1.55e-11@-117;
 <CTOH> CATL3 + OH = HO2 + OTHN # 5.97e-10;
 <CTN3> CATL3 + NO3 = HNO3 + OTHN # 4.86e-10;
 <PNOH> NAPPRD + OH = HO2 + OTHN # 2.00e-10;
 <PNN3> NAPPRD + NO3 = HNO3 + OTHN # 1.70e-10;
 <139> PHOT = 2*HO2 + 2*RO2C + 2*SumRO2 + ALK3 # 1.0/<BACL_11>;
 <140> IMINE = MECHO # 2.78e-04;
 <141> CLETHE + OH = xHO2 + RO2C + xHCHO + yROOH + SumRO2 # 2.54e-12@-325;
 <142> ACRLNT + OH = xHO2 + RO2C + xHCHO + yROOH + SumRO2 # 4.13e-12;
 <143> PCE + OH = xHO2 + RO2C + yROOH + SumRO2 # 3.50e-12@920;
 <144> PCLBEN + OH = xHO2 + RO2C + yROOH + SumRO2 # 4.03e-13;
 <145> MECL2 + OH = xHO2 + RO2C + yROOH + SumRO2 # 1.80e-12@860;
 <146> ETBR2 + OH = xHO2 + RO2C + yROOH + SumRO2 # 7.69e-12@1056;
 <147> ETCL2 + OH = xHO2 + RO2C + yROOH + SumRO2 # 8.69e-12@1070;
 <148> ETOX + OH = xHO2 + RO2C + yROOH + SumRO2 # 1.63e-12@856;
 <149> CHCL3 + OH = xHO2 + RO2C + yROOH + SumRO2 # 1.80e-12@850;
 <150> xHO2 + NO = NO + HO2 # 1.0*K<R2NO>;
 <151> xHO2 + HO2 = HO2 # 1.0*K<R2H2>;
 <152> xHO2 + NO3 = NO3 + HO2 # 1.0*K<R2N3>;
 <153> xHO2 + SumRO2 = SumRO2 + 0.5*HO2 # 1.0*K<R2R2>;
 <154> xHO2 + SumRCO3 = SumRCO3 + HO2 # 1.0*K<R3R2>;
 <155> xOH + NO = NO + OH # 1.0*K<R2NO>;
 <156> xOH + HO2 = HO2 # 1.0*K<R2H2>;
 <157> xOH + NO3 = NO3 + OH # 1.0*K<R2N3>;
 <158> xOH + SumRO2 = SumRO2 + 0.5*OH # 1.0*K<R2R2>;
 <159> xOH + SumRCO3 = SumRCO3 + OH # 1.0*K<R3R2>;

<160> xNO2 + NO = NO + NO2 # 1.0*K<R2NO>;
<161> xNO2 + HO2 = HO2 # 1.0*K<R2H2>;
<162> xNO2 + NO3 = NO3 + NO2 # 1.0*K<R2N3>;
<163> xNO2 + SumRO2 = SumRO2 + 0.5*NO2 # 1.0*K<R2R2>;
<164> xNO2 + SumRCO3 = SumRCO3 + NO2 # 1.0*K<R3R2>;
<165> xNO3 + NO = NO + NO3 # 1.0*K<R2NO>;
<166> xNO3 + HO2 = HO2 # 1.0*K<R2H2>;
<167> xNO3 + NO3 = 2*NO3 # 1.0*K<R2N3>;
<168> xNO3 + SumRO2 = SumRO2 + 0.5*NO3 # 1.0*K<R2R2>;
<169> xNO3 + SumRCO3 = SumRCO3 + NO3 # 1.0*K<R3R2>;
<170> xHCHO + NO = NO + HCHO # 1.0*K<R2NO>;
<171> xHCHO + HO2 = HO2 # 1.0*K<R2H2>;
<172> xHCHO + NO3 = NO3 + HCHO # 1.0*K<R2N3>;
<173> xHCHO + SumRO2 = SumRO2 + 0.5*HCHO # 1.0*K<R2R2>;
<174> xHCHO + SumRCO3 = SumRCO3 + HCHO # 1.0*K<R3R2>;
<175> xGLY + NO = NO + GLY # 1.0*K<R2NO>;
<176> xGLY + HO2 = HO2 # 1.0*K<R2H2>;
<177> xGLY + NO3 = NO3 + GLY # 1.0*K<R2N3>;
<178> xGLY + SumRO2 = SumRO2 + 0.5*GLY # 1.0*K<R2R2>;
<179> xGLY + SumRCO3 = SumRCO3 + GLY # 1.0*K<R3R2>;
<180> xHCOOH + NO = NO + HCOOH # 1.0*K<R2NO>;
<181> xHCOOH + HO2 = HO2 # 1.0*K<R2H2>;
<182> xHCOOH + NO3 = NO3 + HCOOH # 1.0*K<R2N3>;
<183> xHCOOH + SumRO2 = SumRO2 + 0.5*HCOOH # 1.0*K<R2R2>;
<184> xHCOOH + SumRCO3 = SumRCO3 + HCOOH # 1.0*K<R3R2>;
<185> xMECHO + NO = NO + MECHO # 1.0*K<R2NO>;
<186> xMECHO + HO2 = HO2 # 1.0*K<R2H2>;
<187> xMECHO + NO3 = NO3 + MECHO # 1.0*K<R2N3>;
<188> xMECHO + SumRO2 = SumRO2 + 0.5*MECHO # 1.0*K<R2R2>;
<189> xMECHO + SumRCO3 = SumRCO3 + MECHO # 1.0*K<R3R2>;
<190> xETCHO + NO = NO + ETCHO # 1.0*K<R2NO>;
<191> xETCHO + HO2 = HO2 # 1.0*K<R2H2>;
<192> xETCHO + NO3 = NO3 + ETCHO # 1.0*K<R2N3>;
<193> xETCHO + SumRO2 = SumRO2 + 0.5*ETCHO # 1.0*K<R2R2>;
<194> xETCHO + SumRCO3 = SumRCO3 + ETCHO # 1.0*K<R3R2>;
<195> xGLCHO + NO = NO + GLCHO # 1.0*K<R2NO>;
<196> xGLCHO + HO2 = HO2 # 1.0*K<R2H2>;
<197> xGLCHO + NO3 = NO3 + GLCHO # 1.0*K<R2N3>;
<198> xGLCHO + SumRO2 = SumRO2 + 0.5*GLCHO # 1.0*K<R2R2>;
<199> xGLCHO + SumRCO3 = SumRCO3 + GLCHO # 1.0*K<R3R2>;
<200> xMEK + NO = NO + MEK # 1.0*K<R2NO>;
<201> xMEK + HO2 = HO2 # 1.0*K<R2H2>;
<202> xMEK + NO3 = NO3 + MEK # 1.0*K<R2N3>;
<203> xMEK + SumRO2 = SumRO2 + 0.5*MEK # 1.0*K<R2R2>;
<204> xMEK + SumRCO3 = SumRCO3 + MEK # 1.0*K<R3R2>;
<205> xACRO + NO = NO + ACRO # 1.0*K<R2NO>;
<206> xACRO + HO2 = HO2 # 1.0*K<R2H2>;
<207> xACRO + NO3 = NO3 + ACRO # 1.0*K<R2N3>;
<208> xACRO + SumRO2 = SumRO2 + 0.5*ACRO # 1.0*K<R2R2>;
<209> xACRO + SumRCO3 = SumRCO3 + ACRO # 1.0*K<R3R2>;
<210> xACET + NO = NO + ACET # 1.0*K<R2NO>;
<211> xACET + HO2 = HO2 # 1.0*K<R2H2>;
<212> xACET + NO3 = NO3 + ACET # 1.0*K<R2N3>;
<213> xACET + SumRO2 = SumRO2 + 0.5*ACET # 1.0*K<R2R2>;
<214> xACET + SumRCO3 = SumRCO3 + ACET # 1.0*K<R3R2>;
<215> xMACR + NO = NO + MACR # 1.0*K<R2NO>;
<216> xMACR + HO2 = HO2 # 1.0*K<R2H2>;
<217> xMACR + NO3 = NO3 + MACR # 1.0*K<R2N3>;

<218> xMACR + SumRO2 = SumRO2 + 0.5*MACR # 1.0*K<R2R2>;
<219> xMACR + SumRCO3 = SumRCO3 + MACR # 1.0*K<R3R2>;
<220> xMVK + NO = NO + MVK # 1.0*K<R2NO>;
<221> xMVK + HO2 = HO2 # 1.0*K<R2H2>;
<222> xMVK + NO3 = NO3 + MVK # 1.0*K<R2N3>;
<223> xMVK + SumRO2 = SumRO2 + 0.5*MVK # 1.0*K<R2R2>;
<224> xMVK + SumRCO3 = SumRCO3 + MVK # 1.0*K<R3R2>;
<225> xBACL + NO = NO + BACL # 1.0*K<R2NO>;
<226> xBACL + HO2 = HO2 # 1.0*K<R2H2>;
<227> xBACL + NO3 = NO3 + BACL # 1.0*K<R2N3>;
<228> xBACL + SumRO2 = SumRO2 + 0.5*BACL # 1.0*K<R2R2>;
<229> xBACL + SumRCO3 = SumRCO3 + BACL # 1.0*K<R3R2>;
<230> xMGLY + NO = NO + MGLY # 1.0*K<R2NO>;
<231> xMGLY + HO2 = HO2 # 1.0*K<R2H2>;
<232> xMGLY + NO3 = NO3 + MGLY # 1.0*K<R2N3>;
<233> xMGLY + SumRO2 = SumRO2 + 0.5*MGLY # 1.0*K<R2R2>;
<234> xMGLY + SumRCO3 = SumRCO3 + MGLY # 1.0*K<R3R2>;
<235> xBUDAL + NO = NO + BUDAL # 1.0*K<R2NO>;
<236> xBUDAL + HO2 = HO2 # 1.0*K<R2H2>;
<237> xBUDAL + NO3 = NO3 + BUDAL # 1.0*K<R2N3>;
<238> xBUDAL + SumRO2 = SumRO2 + 0.5*BUDAL # 1.0*K<R2R2>;
<239> xBUDAL + SumRCO3 = SumRCO3 + BUDAL # 1.0*K<R3R2>;
<240> xFURNS + NO = NO + FURNS # 1.0*K<R2NO>;
<241> xFURNS + HO2 = HO2 # 1.0*K<R2H2>;
<242> xFURNS + NO3 = NO3 + FURNS # 1.0*K<R2N3>;
<243> xFURNS + SumRO2 = SumRO2 + 0.5*FURNS # 1.0*K<R2R2>;
<244> xFURNS + SumRCO3 = SumRCO3 + FURNS # 1.0*K<R3R2>;
<245> xBALD + NO = NO + BALD # 1.0*K<R2NO>;
<246> xBALD + HO2 = HO2 # 1.0*K<R2H2>;
<247> xBALD + NO3 = NO3 + BALD # 1.0*K<R2N3>;
<248> xBALD + SumRO2 = SumRO2 + 0.5*BALD # 1.0*K<R2R2>;
<249> xBALD + SumRCO3 = SumRCO3 + BALD # 1.0*K<R3R2>;
<250> xBENX + NO = NO + BENX # 1.0*K<R2NO>;
<251> xBENX + HO2 = HO2 # 1.0*K<R2H2>;
<252> xBENX + NO3 = NO3 + BENX # 1.0*K<R2N3>;
<253> xBENX + SumRO2 = SumRO2 + 0.5*BENX # 1.0*K<R2R2>;
<254> xBENX + SumRCO3 = SumRCO3 + BENX # 1.0*K<R3R2>;
<255> xRCHO + NO = NO + RCHO # 1.0*K<R2NO>;
<256> xRCHO + HO2 = HO2 # 1.0*K<R2H2>;
<257> xRCHO + NO3 = NO3 + RCHO # 1.0*K<R2N3>;
<258> xRCHO + SumRO2 = SumRO2 + 0.5*RCHO # 1.0*K<R2R2>;
<259> xRCHO + SumRCO3 = SumRCO3 + RCHO # 1.0*K<R3R2>;
<260> xKET2 + NO = NO + KET2 # 1.0*K<R2NO>;
<261> xKET2 + HO2 = HO2 # 1.0*K<R2H2>;
<262> xKET2 + NO3 = NO3 + KET2 # 1.0*K<R2N3>;
<263> xKET2 + SumRO2 = SumRO2 + 0.5*KET2 # 1.0*K<R2R2>;
<264> xKET2 + SumRCO3 = SumRCO3 + KET2 # 1.0*K<R3R2>;
<265> xLVKS + NO = NO + LVKS # 1.0*K<R2NO>;
<266> xLVKS + HO2 = HO2 # 1.0*K<R2H2>;
<267> xLVKS + NO3 = NO3 + LVKS # 1.0*K<R2N3>;
<268> xLVKS + SumRO2 = SumRO2 + 0.5*LVKS # 1.0*K<R2R2>;
<269> xLVKS + SumRCO3 = SumRCO3 + LVKS # 1.0*K<R3R2>;
<270> xOLEA1 + NO = NO + OLEA1 # 1.0*K<R2NO>;
<271> xOLEA1 + HO2 = HO2 # 1.0*K<R2H2>;
<272> xOLEA1 + NO3 = NO3 + OLEA1 # 1.0*K<R2N3>;
<273> xOLEA1 + SumRO2 = SumRO2 + 0.5*OLEA1 # 1.0*K<R2R2>;
<274> xOLEA1 + SumRCO3 = SumRCO3 + OLEA1 # 1.0*K<R3R2>;
<275> xOLEA2 + NO = NO + OLEA2 # 1.0*K<R2NO>;

<276> xOLEA2 + HO2 = HO2 # 1.0*K<R2H2>;
<277> xOLEA2 + NO3 = NO3 + OLEA2 # 1.0*K<R2N3>;
<278> xOLEA2 + SumRO2 = SumRO2 + 0.5*OLEA2 # 1.0*K<R2R2>;
<279> xOLEA2 + SumRCO3 = SumRCO3 + OLEA2 # 1.0*K<R3R2>;
<280> xOLEP + NO = NO + OLEP # 1.0*K<R2NO>;
<281> xOLEP + HO2 = HO2 # 1.0*K<R2H2>;
<282> xOLEP + NO3 = NO3 + OLEP # 1.0*K<R2N3>;
<283> xOLEP + SumRO2 = SumRO2 + 0.5*OLEP # 1.0*K<R2R2>;
<284> xOLEP + SumRCO3 = SumRCO3 + OLEP # 1.0*K<R3R2>;
<285> xOACID + NO = NO + OACID # 1.0*K<R2NO>;
<286> xOACID + HO2 = HO2 # 1.0*K<R2H2>;
<287> xOACID + NO3 = NO3 + OACID # 1.0*K<R2N3>;
<288> xOACID + SumRO2 = SumRO2 + 0.5*OACID # 1.0*K<R2R2>;
<289> xOACID + SumRCO3 = SumRCO3 + OACID # 1.0*K<R3R2>;
<290> xPACID + NO = NO + PACID # 1.0*K<R2NO>;
<291> xPACID + HO2 = HO2 # 1.0*K<R2H2>;
<292> xPACID + NO3 = NO3 + PACID # 1.0*K<R2N3>;
<293> xPACID + SumRO2 = SumRO2 + 0.5*PACID # 1.0*K<R2R2>;
<294> xPACID + SumRCO3 = SumRCO3 + PACID # 1.0*K<R3R2>;
<295> xAMINS + NO = NO + AMINS # 1.0*K<R2NO>;
<296> xAMINS + HO2 = HO2 # 1.0*K<R2H2>;
<297> xAMINS + NO3 = NO3 + AMINS # 1.0*K<R2N3>;
<298> xAMINS + SumRO2 = SumRO2 + 0.5*AMINS # 1.0*K<R2R2>;
<299> xAMINS + SumRCO3 = SumRCO3 + AMINS # 1.0*K<R3R2>;
<305> xRPNO3 + NO = NO + RPNO3 # 1.0*K<R2NO>;
<306> xRPNO3 + HO2 = HO2 # 1.0*K<R2H2>;
<307> xRPNO3 + NO3 = NO3 + RPNO3 # 1.0*K<R2N3>;
<308> xRPNO3 + SumRO2 = SumRO2 + 0.5*RPNO3 # 1.0*K<R2R2>;
<309> xRPNO3 + SumRCO3 = SumRCO3 + RPNO3 # 1.0*K<R3R2>;
<310> xRCNO3 + NO = NO + RCNO3 # 1.0*K<R2NO>;
<311> xRCNO3 + HO2 = HO2 # 1.0*K<R2H2>;
<312> xRCNO3 + NO3 = NO3 + RCNO3 # 1.0*K<R2N3>;
<313> xRCNO3 + SumRO2 = SumRO2 + 0.5*RCNO3 # 1.0*K<R2R2>;
<314> xRCNO3 + SumRCO3 = SumRCO3 + RCNO3 # 1.0*K<R3R2>;
<315> xRHNO3 + NO = NO + RHNO3 # 1.0*K<R2NO>;
<316> xRHNO3 + HO2 = HO2 # 1.0*K<R2H2>;
<317> xRHNO3 + NO3 = NO3 + RHNO3 # 1.0*K<R2N3>;
<318> xRHNO3 + SumRO2 = SumRO2 + 0.5*RHNO3 # 1.0*K<R2R2>;
<319> xRHNO3 + SumRCO3 = SumRCO3 + RHNO3 # 1.0*K<R3R2>;
<320> xRDNO3 + NO = NO + RDNO3 # 1.0*K<R2NO>;
<321> xRDNO3 + HO2 = HO2 # 1.0*K<R2H2>;
<322> xRDNO3 + NO3 = NO3 + RDNO3 # 1.0*K<R2N3>;
<323> xRDNO3 + SumRO2 = SumRO2 + 0.5*RDNO3 # 1.0*K<R2R2>;
<324> xRDNO3 + SumRCO3 = SumRCO3 + RDNO3 # 1.0*K<R3R2>;
<330> xHPCRB + NO = NO + HPCRB # 1.0*K<R2NO>;
<331> xHPCRB + HO2 = HO2 # 1.0*K<R2H2>;
<332> xHPCRB + NO3 = NO3 + HPCRB # 1.0*K<R2N3>;
<333> xHPCRB + SumRO2 = SumRO2 + 0.5*HPCRB # 1.0*K<R2R2>;
<334> xHPCRB + SumRCO3 = SumRCO3 + HPCRB # 1.0*K<R3R2>;
<335> xAFG1 + NO = NO + AFG1 # 1.0*K<R2NO>;
<336> xAFG1 + HO2 = HO2 # 1.0*K<R2H2>;
<337> xAFG1 + NO3 = NO3 + AFG1 # 1.0*K<R2N3>;
<338> xAFG1 + SumRO2 = SumRO2 + 0.5*AFG1 # 1.0*K<R2R2>;
<339> xAFG1 + SumRCO3 = SumRCO3 + AFG1 # 1.0*K<R3R2>;
<340> xAFG2A + NO = NO + AFG2A # 1.0*K<R2NO>;
<341> xAFG2A + HO2 = HO2 # 1.0*K<R2H2>;
<342> xAFG2A + NO3 = NO3 + AFG2A # 1.0*K<R2N3>;
<343> xAFG2A + SumRO2 = SumRO2 + 0.5*AFG2A # 1.0*K<R2R2>;

```

<344> xAFG2A + SumRCO3 = SumRCO3 + AFG2A # 1.0*K<R3R2>;
<345> xAFG2B + NO = NO + AFG2B # 1.0*K<R2NO>;
<346> xAFG2B + HO2 = HO2 # 1.0*K<R2H2>;
<347> xAFG2B + NO3 = NO3 + AFG2B # 1.0*K<R2N3>;
<348> xAFG2B + SumRO2 = SumRO2 + 0.5*AFG2B # 1.0*K<R2R2>;
<349> xAFG2B + SumRCO3 = SumRCO3 + AFG2B # 1.0*K<R3R2>;
<350> xAFG3 + NO = NO + AFG3 # 1.0*K<R2NO>;
<351> xAFG3 + HO2 = HO2 # 1.0*K<R2H2>;
<352> xAFG3 + NO3 = NO3 + AFG3 # 1.0*K<R2N3>;
<353> xAFG3 + SumRO2 = SumRO2 + 0.5*AFG3 # 1.0*K<R2R2>;
<354> xAFG3 + SumRCO3 = SumRCO3 + AFG3 # 1.0*K<R3R2>;
<355> xPAN2 + NO = NO + PAN2 # 1.0*K<R2NO>;
<356> xPAN2 + HO2 = HO2 # 1.0*K<R2H2>;
<357> xPAN2 + NO3 = NO3 + PAN2 # 1.0*K<R2N3>;
<358> xPAN2 + SumRO2 = SumRO2 + 0.5*PAN2 # 1.0*K<R2R2>;
<359> xPAN2 + SumRCO3 = SumRCO3 + PAN2 # 1.0*K<R3R2>;
<365> xMEO2 + NO = NO + MEO2 + SumRO2 # 1.0*K<R2NO>;
<366> xMEO2 + HO2 = HO2 # 1.0*K<R2H2>;
<367> xMEO2 + NO3 = NO3 + MEO2 + SumRO2 # 1.0*K<R2N3>;
<368> xMEO2 + SumRO2 = 1.5*SumRO2 + 0.5*MEO2 # 1.0*K<R2R2>;
<369> xMEO2 + SumRCO3 = SumRCO3 + MEO2 + SumRO2 # 1.0*K<R3R2>;
<370> xETO2 + NO = NO + ETO2 + SumRO2 # 1.0*K<R2NO>;
<371> xETO2 + HO2 = HO2 # 1.0*K<R2H2>;
<372> xETO2 + NO3 = NO3 + ETO2 + SumRO2 # 1.0*K<R2N3>;
<373> xETO2 + SumRO2 = 1.5*SumRO2 + 0.5*ETO2 # 1.0*K<R2R2>;
<374> xETO2 + SumRCO3 = SumRCO3 + ETO2 + SumRO2 # 1.0*K<R3R2>;
<375> xMECO3 + NO = NO + MECO3 + SumRCO3 # 1.0*K<R2NO>;
<376> xMECO3 + HO2 = HO2 # 1.0*K<R2H2>;
<377> xMECO3 + NO3 = NO3 + MECO3 + SumRCO3 # 1.0*K<R2N3>;
<378> xMECO3 + SumRO2 = SumRO2 + 0.5*MECO3 + 0.5*SumRCO3 # 1.0*K<R2R2>;
<379> xMECO3 + SumRCO3 = 2*SumRCO3 + MECO3 # 1.0*K<R3R2>;
<380> xR2CO3 + NO = NO + R2CO3 + SumRCO3 # 1.0*K<R2NO>;
<381> xR2CO3 + HO2 = HO2 # 1.0*K<R2H2>;
<382> xR2CO3 + NO3 = NO3 + R2CO3 + SumRCO3 # 1.0*K<R2N3>;
<383> xR2CO3 + SumRO2 = SumRO2 + 0.5*R2CO3 + 0.5*SumRCO3 # 1.0*K<R2R2>;
<384> xR2CO3 + SumRCO3 = 2*SumRCO3 + R2CO3 # 1.0*K<R3R2>;
<385> xMACO3 + NO = NO + MACO3 + SumRCO3 # 1.0*K<R2NO>;
<386> xMACO3 + HO2 = HO2 # 1.0*K<R2H2>;
<387> xMACO3 + NO3 = NO3 + MACO3 + SumRCO3 # 1.0*K<R2N3>;
<388> xMACO3 + SumRO2 = SumRO2 + 0.5*MACO3 + 0.5*SumRCO3 # 1.0*K<R2R2>;
<389> xMACO3 + SumRCO3 = 2*SumRCO3 + MACO3 # 1.0*K<R3R2>;
<390> xTBUO + NO = NO + TBUO # 1.0*K<R2NO>;
<391> xTBUO + HO2 = HO2 # 1.0*K<R2H2>;
<392> xTBUO + NO3 = NO3 + TBUO # 1.0*K<R2N3>;
<393> xTBUO + SumRO2 = SumRO2 + 0.5*TBUO # 1.0*K<R2R2>;
<394> xTBUO + SumRCO3 = SumRCO3 + TBUO # 1.0*K<R3R2>;
<395> xBZO + NO = NO + BZO # 1.0*K<R2NO>;
<396> xBZO + HO2 = HO2 # 1.0*K<R2H2>;
<397> xBZO + NO3 = NO3 + BZO # 1.0*K<R2N3>;
<398> xBZO + SumRO2 = SumRO2 + 0.5*BZO # 1.0*K<R2R2>;
<399> xBZO + SumRCO3 = SumRCO3 + BZO # 1.0*K<R3R2>;
<400> yROOH + NO = NO # 1.0*K<R2NO>;
<401> yROOH + HO2 = HO2 + ROOH # 1.0*K<R2H2>;
<402> yROOH + NO3 = NO3 # 1.0*K<R2N3>;
<403> yROOH + SumRO2 = SumRO2 + 0.5*KET2 # 1.0*K<R2R2>;
<404> yROOH + SumRCO3 = SumRCO3 # 1.0*K<R3R2>;
<405> yRUOOH + NO = NO # 1.0*K<R2NO>;
<406> yRUOOH + HO2 = HO2 + RUOOH # 1.0*K<R2H2>;

```


<407> yRUOOH + NO3 = NO3 # 1.0*K<R2N3>;
 <408> yRUOOH + SumRO2 = SumRO2 + 0.5*OLEP # 1.0*K<R2R2>;
 <409> yRUOOH + SumRCO3 = SumRCO3 # 1.0*K<R3R2>;
 <410> yRAOOH + NO = NO # 1.0*K<R2NO>;
 <411> yRAOOH + HO2 = HO2 + RAOOH # 1.0*K<R2H2>;
 <412> yRAOOH + NO3 = NO3 # 1.0*K<R2N3>;
 <413> yRAOOH + SumRO2 = SumRO2 + 0.5*OLEP # 1.0*K<R2R2>;
 <414> yRAOOH + SumRCO3 = SumRCO3 # 1.0*K<R3R2>;
 <415> yHPCRB + NO = NO # 1.0*K<R2NO>;
 <416> yHPCRB + HO2 = HO2 + HPCRB # 1.0*K<R2H2>;
 <417> yHPCRB + NO3 = NO3 # 1.0*K<R2N3>;
 <418> yHPCRB + SumRO2 = SumRO2 + 0.5*KET2 # 1.0*K<R2R2>;
 <419> yHPCRB + SumRCO3 = SumRCO3 # 1.0*K<R3R2>;
 <420> yRPNO3 + NO = NO # 1.0*K<R2NO>;
 <421> yRPNO3 + HO2 = HO2 + RPNO3 # 1.0*K<R2H2>;
 <422> yRPNO3 + NO3 = NO3 # 1.0*K<R2N3>;
 <423> yRPNO3 + SumRO2 = SumRO2 + 0.5*R1NO3 # 1.0*K<R2R2>;
 <424> yRPNO3 + SumRCO3 = SumRCO3 # 1.0*K<R3R2>;
 <425> zR1NO3 + NO = NO + R1NO3 # 1.0*K<R2NO>;
 <426> zR1NO3 + HO2 = HO2 # 1.0*K<R2H2>;
 <427> zR1NO3 + NO3 = NO3 + KET2 + HO2 # 1.0*K<R2N3>;
 <428> zR1NO3 + SumRO2 = SumRO2 + 0.5*KET2 + 0.5*HO2 # 1.0*K<R2R2>;
 <429> zR1NO3 + SumRCO3 = SumRCO3 + KET2 + HO2 # 1.0*K<R3R2>;
 <430> zR2NO3 + NO = NO + R2NO3 # 1.0*K<R2NO>;
 <431> zR2NO3 + HO2 = HO2 # 1.0*K<R2H2>;
 <432> zR2NO3 + NO3 = NO3 + KET2 + HO2 # 1.0*K<R2N3>;
 <433> zR2NO3 + SumRO2 = SumRO2 + 0.5*KET2 + 0.5*HO2 # 1.0*K<R2R2>;
 <434> zR2NO3 + SumRCO3 = SumRCO3 + KET2 + HO2 # 1.0*K<R3R2>;
 <435> zRHNO3 + NO = NO + RHNO3 # 1.0*K<R2NO>;
 <436> zRHNO3 + HO2 = HO2 # 1.0*K<R2H2>;
 <437> zRHNO3 + NO3 = NO3 + KET2 + HO2 # 1.0*K<R2N3>;
 <438> zRHNO3 + SumRO2 = SumRO2 + 0.5*KET2 + 0.5*HO2 # 1.0*K<R2R2>;
 <439> zRHNO3 + SumRCO3 = SumRCO3 + KET2 + HO2 # 1.0*K<R3R2>;
 <440> zRCNO3 + NO = NO + RCNO3 # 1.0*K<R2NO>;
 <441> zRCNO3 + HO2 = HO2 # 1.0*K<R2H2>;
 <442> zRCNO3 + NO3 = NO3 + KET2 + HO2 # 1.0*K<R2N3>;
 <443> zRCNO3 + SumRO2 = SumRO2 + 0.5*KET2 + 0.5*HO2 # 1.0*K<R2R2>;
 <444> zRCNO3 + SumRCO3 = SumRCO3 + KET2 + HO2 # 1.0*K<R3R2>;
 <445> zRANO3 + NO = NO + RANO3 # 1.0*K<R2NO>;
 <446> zRANO3 + HO2 = HO2 # 1.0*K<R2H2>;
 <447> zRANO3 + NO3 = NO3 + RUOOH + HO2 # 1.0*K<R2N3>;
 <448> zRANO3 + SumRO2 = SumRO2 + 0.5*RUOOH + 0.5*HO2 # 1.0*K<R2R2>;
 <449> zRANO3 + SumRCO3 = SumRCO3 + RUOOH + HO2 # 1.0*K<R3R2>;
 <450> zRPNO3 + NO = NO + RPNO3 # 1.0*K<R2NO>;
 <451> zRPNO3 + HO2 = HO2 # 1.0*K<R2H2>;
 <452> zRPNO3 + NO3 = NO3 + RUOOH + HO2 # 1.0*K<R2N3>;
 <453> zRPNO3 + SumRO2 = SumRO2 + 0.5*RUOOH + 0.5*HO2 # 1.0*K<R2R2>;
 <454> zRPNO3 + SumRCO3 = SumRCO3 + RUOOH + HO2 # 1.0*K<R3R2>;
 <455> zRDNO3 + NO = NO + RDNO3 # 1.0*K<R2NO>;
 <456> zRDNO3 + HO2 = HO2 # 1.0*K<R2H2>;
 <457> zRDNO3 + NO3 = NO3 + R1NO3 + HO2 # 1.0*K<R2N3>;
 <458> zRDNO3 + SumRO2 = SumRO2 + 0.5*R1NO3 + 0.5*HO2 # 1.0*K<R2R2>;
 <459> zRDNO3 + SumRCO3 = SumRCO3 + R1NO3 + HO2 # 1.0*K<R3R2>;
 <460> zRNNNO3 + NO = NO + RNNNO3 # 1.0*K<R2NO>;
 <461> zRNNNO3 + HO2 = HO2 # 1.0*K<R2H2>;
 <462> zRNNNO3 + NO3 = NO3 + OTHN + HO2 # 1.0*K<R2N3>;
 <463> zRNNNO3 + SumRO2 = SumRO2 + 0.5*OTHN + 0.5*HO2 # 1.0*K<R2R2>;
 <464> zRNNNO3 + SumRCO3 = SumRCO3 + OTHN + HO2 # 1.0*K<R3R2>;

```

<465> zPAN2 + NO = NO + PAN2 # 1.0*K<R2NO>;
<466> zPAN2 + HO2 = HO2 # 1.0*K<R2H2>;
<467> zPAN2 + NO3 = NO3 + RCHO + HO2 # 1.0*K<R2N3>;
<468> zPAN2 + SumRO2 = SumRO2 + 0.5*RCHO + 0.5*HO2 # 1.0*K<R2R2>;
<469> zPAN2 + SumRCO3 = SumRCO3 + RCHO + HO2 # 1.0*K<R3R2>;
<Q2NO> R2CO3 + NO = NO2 + 0.95*xHO2 + 0.96*RO2C + 0.04*RO2XC + 0.95*xETCHO +
    0.04*zR1NO3 + yROOH + CO2 + SumRO2 # 6.70e-12@-340;
<Q2N2> R2CO3 + NO2 = PAN2 # 7.70e-12;
<Q2N3> R2CO3 + NO3 = NO2 + 0.95*xHO2 + 0.96*RO2C + 0.04*RO2XC + 0.95*xETCHO +
    0.04*zR1NO3 + yROOH + CO2 + SumRO2 # 4.00e-12;
<Q2H2> R2CO3 + HO2 = 0.13*O3 + 0.5*OH + 0.48*xHO2 + 0.48*RO2C + 0.02*RO2XC +
    0.48*xETCHO + 0.13*OACID + 0.37*PACID + 0.02*zR1NO3 + 0.5*yROOH +
    0.5*CO2 + 0.5*SumRO2 # 2.20e-11;
<Q2R2> R2CO3 + SumRO2 = 0.86*xHO2 + 0.86*RO2C + 0.04*RO2XC + 0.86*xETCHO +
    0.1*OACID + 0.04*zR1NO3 + 0.9*yROOH + 0.9*CO2 + 0.9*SumRO2 # 1.60e-11;
<Q2R3> R2CO3 + SumRCO3 = 0.95*xHO2 + 0.96*RO2C + 0.04*RO2XC + 0.95*xETCHO +
    0.04*zR1NO3 + yROOH + CO2 + SumRO2 # 1.40e-11;
<Q5UI> MACO3 = 0.93*xHO2 + 0.93*RO2C + 0.07*RO2XC + 0.93*xPACID + 0.07*zRCNO3 +
    SumRO2 # 7.79e+08@5003;
<Q5NO> MACO3 + NO = NO2 + MEO2 + HCHO + CO2 + CO + SumRO2 # 6.70e-12@-340;
<Q5N2> MACO3 + NO2 = APANS # 7.70e-12;
<Q5N3> MACO3 + NO3 = NO2 + MEO2 + HCHO + CO2 + CO + SumRO2 # 4.00e-12;
<Q5H2> MACO3 + HO2 = 0.13*O3 + 0.5*OH + 0.5*MEO2 + 0.5*HCHO + 0.37*PACID +
    0.13*OLEP + 0.5*CO2 + 0.5*CO + 0.5*SumRO2 # 2.20e-11;
<Q5R2> MACO3 + SumRO2 = 0.9*MEO2 + 0.9*HCHO + 0.1*OLEP + 0.9*CO2 + 0.9*CO +
    0.9*SumRO2 # 1.60e-11;
<Q5R3> MACO3 + SumRCO3 = MEO2 + HCHO + CO2 + CO + SumRO2 # 1.40e-11;
<C2OH> ETHAN + OH = ETO2 + SumRO2 # 1.51e-12^1.92@533;
<C3OH> PROP + OH = 0.95*xHO2 + 0.96*RO2C + 0.04*RO2XC + 0.01*xMEO2 +
    0.01*xMECHO + 0.27*xETCHO + 0.68*xACET + 0.04*zR1NO3 + yROOH + SumRO2
    # 2.00e-12^1.76@172;
<C4OH> NC4 + OH = 0.59*xHO2 + 1.02*RO2C + 0.08*RO2XC + 0.33*xETO2 +
    0.33*xMECHO + 0.12*xRCHO + 0.48*xMEK + 0.07*zR1NO3 + 0.01*zRHNO3 +
    1.1*yROOH + 1.1*SumRO2 # 2.09e-12^1.82@-42;
<E1OH> ETHEN + OH = xHO2 + RO2C + 1.48*xHCHO + 0.26*xGLCHO + yROOH + SumRO2
    # 1.10e-28^-3.50&8.40e-12^-1.75&0.60&1.00;
<E1O3> ETHEN + O3 = 0.18*H2 + 0.17*OH + 0.27*HO2 + 0.42*HCHO2 + HCHO +
    0.23*CO2 + 0.35*CO # 6.82e-15@2500;
<E1N3> ETHEN + NO3 = 0.01*xNO2 + 0.99*xHO2 + RO2C + 0.01*xHCHO + 0.99*xRCNO3 +
    yRPNO3 + SumRO2 # 3.30e-12@2880;
<E1OP> ETHEN + O3P = ETHEN_OP + 0.8*HO2 + 0.22*xHO2 + 0.29*RO2C + 0.51*MEO2 +
    0.07*xHCHO + 0.1*MECHO + 4.41*NROG + 0.51*CO + 0.8*SumRO2
    # 1.07e-11@800;
<495> ETHEN_OP = 0.07*xOH + 0.22*xPACID + 0.07*CO2 # 2.57e+00;
<496> ETHEN_OP + NO = NO + 0.07*xHO2 + 0.2*xHCHO + 0.02*xGLY + 0.24*yHPCRB +
    0.27*CO # 2.55e-12@-380;
<E2OH> PROPE + OH = 0.97*xHO2 + 0.97*RO2C + 0.03*RO2XC + 0.97*xHCHO +
    0.97*xMECHO + 0.03*zRHNO3 + yROOH + SumRO2 # 1.20e-11^-0.62@-210;
<E2O3> PROPE + O3 = PROPE_O3 + 0.09*H2 + 0.3*OH + 0.17*HO2 + 0.16*xHO2 +
    0.22*RO2C + 0.03*MEO2 + 0.21*HCHO2 + 0.12*MECHO2 + 0.5*HCHO +
    0.05*xHCHO + 0.5*MECHO + 0.05*MEOH + 0.09*CH4 + 0.24*CO2 + 0.22*CO +
    0.25*SumRO2 # 5.77e-15@1880;
<499> PROPE_O3 = 0.05*xOH + 0.16*xPACID + 0.05*CO2 # 2.45e+00;
<500> PROPE_O3 + NO = NO + 0.06*xHO2 + 0.15*xHCHO + 0.18*yHPCRB + 0.2*CO
    # 2.55e-12@-380;
<E2N3> PROPE + NO3 = 0.29*xNO2 + 0.68*xHO2 + 0.97*RO2C + 0.03*RO2XC +
    0.29*xHCHO + 0.29*xMECHO + 0.68*xRCNO3 + yRPNO3 + 0.03*zRDNO3 + SumRO2
    # 4.60e-13@1155;

```

<E2OP> PROPE + O3P = 0.25*ETCHO + 0.25*ACET + 0.5*ALK2 # 1.02e-11@280;
 <IPOH> ISOP + OH = ISOP_OH + 0.55*xHO2 + 0.55*RO2C + 0.05*RO2XC + 0.5*xHCHO +
 0.23*xMACR + 0.27*xMVK + 0.05*zRHNO3 + 0.6*yRUOOH + 0.05*xFURNS +
 0.6*SumRO2 # 2.70e-11@-390;
 <504> ISOP_OH = 0.06*OH + 0.34*HO2 + 0.39*HPCRB # 1.29e+00;
 <505> ISOP_OH + NO = NO + 0.35*xHO2 + 0.38*RO2C + 0.05*RO2XC + 0.03*xHCHO +
 0.34*xOLEA1 + 0.05*zRHNO3 + 0.42*yRUOOH + 0.43*SumRO2 # 2.55e-12@-380;
 <IPO3> ISOP + O3 = ISOP_O3 + 0.1*H2 + 0.13*OH + 0.29*HO2 + 0.12*xHO2 +
 0.33*RO2C + 0.01*RO2XC + 0.16*MECO3 + 0.13*xMECO3 + 0.04*xMACO3 +
 0.23*HCHO2 + 0.06*RCHO2 + 0.4*HCHO + 0.21*xHCHO + 0.39*MACR +
 0.16*MVK + 0.05*OLEP + 0.01*zRCNO3 + 0.15*yHPCRB + 0.13*CO2 + 0.33*CO +
 0.34*SumRO2 + 0.16*SumRCO3 + ISOPRXN # 1.05e-14@2000;
 <507> ISOP_O3 = 0.04*xOH + 0.12*xPACID + 0.04*CO2 # 2.81e+00;
 <508> ISOP_O3 + NO = NO + 0.04*xHO2 + 0.11*xHCHO + 0.13*yHPCRB + 0.14*CO
 # 2.55e-12@-380;
 <IPN3> ISOP + NO3 = ISOP_N3 + 0.7*xNO2 + 0.03*xHO2 + 0.9*RO2C + 0.1*RO2XC +
 0.48*xHCHO + 0.22*xOLEA1 + 0.48*xMVK + 0.03*xRCNO3 + yRPNO3 +
 0.1*zRDNO3 + SumRO2 + ISOPRXN # 2.95e-12@450;
 <510> ISOP_N3 = 0.12*NO2 + 0.04*HO2 + 0.04*RPNO3 + 0.12*HPCRB # 1.03e+00;
 <511> ISOP_N3 + NO = NO + 0.15*xHO2 + 0.15*RO2C + 0.01*RO2XC + 0.1*xHCHO +
 0.1*xRCNO3 + 0.05*xRHNO3 + 0.17*yRPNO3 + 0.01*zRDNO3 + 0.16*SumRO2
 # 2.55e-12@-380;
 <IPOP> ISOP + O3P = 0.23*RO2C + 0.02*RO2XC + 0.25*MEO2 + 0.23*xMACO3 +
 0.23*xHCHO + 0.75*OLEP + 0.02*zRCNO3 + 0.21*yHPCRB + 0.5*SumRO2
 # 3.50e-11;
 <E3OH> BUT13 + OH = BUT13_OH + 0.63*xHO2 + 0.63*RO2C + 0.04*RO2XC + 0.58*xHCHO +
 0.58*xACRO + 0.04*zRHNO3 + 0.67*yRUOOH + 0.05*xFURNS + 0.67*SumRO2
 # 1.12e-11@-530;
 <514> BUT13_OH = 0.31*HO2 + 0.33*HPCRB # 1.07e+00;
 <515> BUT13_OH + NO = NO + 0.31*xHO2 + 0.33*RO2C + 0.02*RO2XC + 0.03*xHCHO +
 0.31*xOLEA1 + 0.02*zRHNO3 + 0.35*yRUOOH + 0.35*SumRO2 # 2.55e-12@-380;
 <E3O3> BUT13 + O3 = BUT13_O3 + 0.09*H2 + 0.08*OH + 0.5*HO2 + 0.27*xHO2 +
 0.36*RO2C + 0.21*HCHO2 + 0.14*RCHO2 + 0.5*HCHO + 0.09*xHCHO +
 0.5*ACRO + 0.12*CO2 + 0.54*CO + 0.36*SumRO2 # 1.34e-14@2283;
 <517> BUT13_O3 = 0.09*xOH + 0.27*xPACID + 0.09*CO2 # 2.55e+00;
 <518> BUT13_O3 + NO = NO + 0.09*xHO2 + 0.25*xHCHO + 0.02*xGLY + 0.3*yHPCRB +
 0.33*CO # 2.55e-12@-380;
 <E3N3> BUT13 + NO3 = 0.89*xNO2 + 0.06*xHO2 + 0.94*RO2C + 0.06*RO2XC +
 0.74*xHCHO + 0.14*xOLEA1 + 0.74*xACRO + 0.06*xRCNO3 + yRPNO3 +
 0.06*zRDNO3 + SumRO2 # 1.10e-13;
 <E3OP> BUT13 + O3P = 0.25*OLEA2 + 0.25*MVK + 0.5*OLEP # 2.26e-11@40;
 <APOH> APINE + OH = APINE_OH + 0.02*HO2 + 0.59*xHO2 + 1.07*RO2C + 0.3*RO2XC +
 0.08*xHCHO + 0.51*xRCHO + 0.08*xOLEA2 + 0.17*xACET + 0.06*xMVK +
 0.03*xLVKS + 0.01*zR2NO3 + 0.01*zRCNO3 + 0.28*zRHNO3 + 0.53*yROOH +
 0.71*yRUOOH + 0.02*HPCRB + 0.13*yHPCRB + 1.37*SumRO2 +
 TRPRXN # 1.34e-11@-410;
 <522> APINE_OH = 0.01*OH + 0.08*HO2 + 0.07*HPCRB # 7.49e+00;
 <523> APINE_OH + NO = NO + 0.06*xHO2 + 0.08*RO2C + 0.02*RO2XC + 0.01*xHCHO +
 0.06*xOLEA2 + 0.02*zRHNO3 + 0.09*yHPCRB + 0.1*SumRO2 # 2.55e-12@-380;
 <APO3> APINE + O3 = 0.69*OH + 0.01*xOH + 0.01*HO2 + 0.17*xHO2 + 0.79*RO2C +
 0.26*RO2XC + 0.03*xMECO3 + 0.2*xR2CO3 + 0.29*RCHO2 + 0.2*xHCHO +
 0.02*RCHO + 0.13*xRCHO + 0.03*xBACL + 0.09*xACET + 0.03*KET2 +
 0.07*xPACID + 0.26*zRCNO3 + 0.82*yHPCRB + 0.03*CO2 + 0.17*CO +
 1.05*SumRO2 + TRPRXN # 8.22e-16@640;
 <APN3> APINE + NO3 = 0.81*xNO2 + 0.81*RO2C + 0.19*RO2XC + 0.81*xRCHO + yRPNO3 +
 0.19*zRDNO3 + SumRO2 # 1.20e-12@-490;
 <APOP> APINE + O3P = 0.5*KET2 + 0.5*ALK4 + TRPRXN # 3.20e-11;
 <BPOH> BPINE + OH = BPINE_OH + 0.01*xOH + 0.01*HO2 + 0.37*xHO2 + 1.43*RO2C +

$0.42 \cdot \text{RO2XC} + 0.03 \cdot \text{xR2CO3} + 0.35 \cdot \text{xHCHO} + 0.03 \cdot \text{xRCHO} + 0.23 \cdot \text{xOLEA2} +$
 $0.29 \cdot \text{xACET} + 0.11 \cdot \text{xKET2} + 0.02 \cdot \text{xPACID} + 0.02 \cdot \text{zR2NO3} + 0.09 \cdot \text{zRCNO3} +$
 $0.31 \cdot \text{zRHNO3} + 0.32 \cdot \text{yROOH} + \text{yRUOOH} + 0.01 \cdot \text{HPCRB} + 0.45 \cdot \text{yHPCRB} +$
 $1.85 \cdot \text{SumRO2} + \text{TRPRXN} \# 1.62\text{e-11@-460};$
 <528> $\text{BPINE_OH} = 0.04 \cdot \text{OH} + 0.12 \cdot \text{HO2} + 0.03 \cdot \text{PACID} + 0.01 \cdot \text{xPACID} + 0.11 \cdot \text{HPCRB} +$
 $0.01 \cdot \text{CO2} \# 3.45\text{e+00};$
 <529> $\text{BPINE_OH} + \text{NO} = \text{NO} + 0.11 \cdot \text{xHO2} + 0.15 \cdot \text{RO2C} + 0.04 \cdot \text{RO2XC} + 0.02 \cdot \text{xHCHO} +$
 $0.01 \cdot \text{xRCHO} + 0.08 \cdot \text{xOLEA2} + 0.01 \cdot \text{zRCNO3} + 0.03 \cdot \text{zRHNO3} + 0.01 \cdot \text{yRUOOH} +$
 $0.21 \cdot \text{yHPCRB} + 0.19 \cdot \text{SumRO2} \# 2.55\text{e-12@-380};$
 <BPO3> $\text{BPINE} + \text{O3} = 0.09 \cdot \text{H2} + 0.39 \cdot \text{OH} + 0.14 \cdot \text{HO2} + 0.23 \cdot \text{RO2C} + 0.08 \cdot \text{RO2XC} +$
 $0.23 \cdot \text{xR2CO3} + 0.21 \cdot \text{HCHO2} + 0.2 \cdot \text{RCHO2} + 0.5 \cdot \text{HCHO} + 0.5 \cdot \text{KET2} +$
 $0.08 \cdot \text{zRCNO3} + 0.26 \cdot \text{yHPCRB} + 0.12 \cdot \text{CO2} + 0.17 \cdot \text{CO} + 0.31 \cdot \text{SumRO2} + \text{TRPRXN}$
 $\# 1.39\text{e-15@1280};$
 <BPN3> $\text{BPINE} + \text{NO3} = 0.02 \cdot \text{OH} + 0.04 \cdot \text{xOH} + 0.18 \cdot \text{xHO2} + 2.25 \cdot \text{RO2C} + 0.63 \cdot \text{RO2XC} +$
 $0.14 \cdot \text{xR2CO3} + 0.04 \cdot \text{xHCHO} + 0.02 \cdot \text{xRCHO} + 0.27 \cdot \text{xACET} + 0.14 \cdot \text{xPACID} +$
 $0.02 \cdot \text{RCNO3} + 0.22 \cdot \text{xRCNO3} + 0.43 \cdot \text{zRCNO3} + \text{yRPNO3} + 0.19 \cdot \text{zRDNO3} +$
 $0.04 \cdot \text{CO2} + 2.88 \cdot \text{SumRO2} + \text{TRPRXN} \# 2.50\text{e-12};$
 <BPPO> $\text{BPINE} + \text{O3P} = 0.5 \cdot \text{RCHO} + 0.5 \cdot \text{ALK5} + \text{TRPRXN} \# 2.70\text{e-11};$
 <ACOH> $\text{ACETL} + \text{OH} = 0.67 \cdot \text{OH} + 0.33 \cdot \text{HO2} + 0.67 \cdot \text{GLY} + 0.33 \cdot \text{HCOOH} + 0.33 \cdot \text{CO}$
 $\# 5.50\text{e-30}\&8.30\text{e-13}^{\wedge}2.00\&0.60\&1.00;$
 <ACO3> $\text{ACETL} + \text{O3} = 0.26 \cdot \text{HO2} + 0.34 \cdot \text{RCHO2} + 0.34 \cdot \text{HCHO} + 0.18 \cdot \text{HCOOH} + 0.47 \cdot \text{CO2} +$
 $0.31 \cdot \text{CO} \# 1.00\text{e-20};$
 <BZOH> $\text{BENZ} + \text{OH} = 0.69 \cdot \text{HO2} + 0.28 \cdot \text{xHO2} + 0.28 \cdot \text{RO2C} + 0.04 \cdot \text{RO2XC} + 0.12 \cdot \text{OLEA2} +$
 $0.28 \cdot \text{xGLY} + 0.28 \cdot \text{xBUDAL} + 0.57 \cdot \text{PHEN} + 0.04 \cdot \text{zRANO3} + 0.31 \cdot \text{yRAOOH} +$
 $0.32 \cdot \text{SumRO2} + \text{BENZRO2} \# 2.30\text{e-12@190};$
 <TLOH> $\text{TOLU} + \text{OH} = 0.41 \cdot \text{HO2} + 0.5 \cdot \text{xHO2} + 0.5 \cdot \text{RO2C} + 0.08 \cdot \text{RO2XC} + 0.01 \cdot \text{OLEA1} +$
 $0.21 \cdot \text{OLEA2} + 0.22 \cdot \text{xGLY} + 0.22 \cdot \text{xMGLY} + 0.22 \cdot \text{xBUDAL} + 0.02 \cdot \text{xAFG1} +$
 $0.19 \cdot \text{xAFG2A} + 0.06 \cdot \text{xBALD} + 0.19 \cdot \text{CRES} + 0.01 \cdot \text{zR1NO3} + 0.07 \cdot \text{zRANO3} +$
 $0.08 \cdot \text{yROOH} + 0.51 \cdot \text{yRAOOH} + 0.58 \cdot \text{SumRO2} + \text{TOLRO2} \# 1.80\text{e-12@-340};$
 <OXOH> $\text{OXYL} + \text{OH} = 0.35 \cdot \text{HO2} + 0.54 \cdot \text{xHO2} + 0.54 \cdot \text{RO2C} + 0.11 \cdot \text{RO2XC} + 0.1 \cdot \text{OLEA1} +$
 $0.1 \cdot \text{OLEA2} + 0.12 \cdot \text{xGLY} + 0.22 \cdot \text{xMGLY} + 0.15 \cdot \text{xBACL} + 0.15 \cdot \text{xBUDAL} +$
 $0.01 \cdot \text{xAFG1} + 0.22 \cdot \text{xAFG2A} + 0.11 \cdot \text{xAFG2B} + 0.06 \cdot \text{xBALD} + 0.06 \cdot \text{LVKS} +$
 $0.08 \cdot \text{XYNL} + 0.02 \cdot \text{zR2NO3} + 0.09 \cdot \text{zRANO3} + 0.08 \cdot \text{yROOH} + 0.57 \cdot \text{yRAOOH} +$
 $0.65 \cdot \text{SumRO2} + \text{XYLRO2} \# 1.36\text{e-11};$
 <MXOH> $\text{MYXL} + \text{OH} = 0.21 \cdot \text{HO2} + 0.66 \cdot \text{xHO2} + 0.66 \cdot \text{RO2C} + 0.13 \cdot \text{RO2XC} + 0.01 \cdot \text{OLEA1} +$
 $0.13 \cdot \text{OLEA2} + 0.04 \cdot \text{xGLY} + 0.59 \cdot \text{xMGLY} + 0.05 \cdot \text{xAFG1} + 0.58 \cdot \text{xAFG2A} +$
 $0.03 \cdot \text{xBALD} + 0.07 \cdot \text{XYNL} + 0.01 \cdot \text{zR2NO3} + 0.12 \cdot \text{zRANO3} + 0.03 \cdot \text{yROOH} +$
 $0.75 \cdot \text{yRAOOH} + 0.79 \cdot \text{SumRO2} + \text{XYLRO2} \# 2.31\text{e-11};$
 <PXOH> $\text{PXYL} + \text{OH} = 0.38 \cdot \text{HO2} + 0.52 \cdot \text{xHO2} + 0.52 \cdot \text{RO2C} + 0.11 \cdot \text{RO2XC} + 0.02 \cdot \text{OLEA1} +$
 $0.23 \cdot \text{OLEA2} + 0.16 \cdot \text{xGLY} + 0.29 \cdot \text{xMGLY} + 0.29 \cdot \text{xAFG1} + 0.07 \cdot \text{xBALD} +$
 $0.16 \cdot \text{xAFG3} + 0.14 \cdot \text{XYNL} + 0.02 \cdot \text{zR2NO3} + 0.09 \cdot \text{zRANO3} + 0.09 \cdot \text{yROOH} +$
 $0.53 \cdot \text{yRAOOH} + 0.63 \cdot \text{SumRO2} + \text{XYLRO2} \# 4.14\text{e-12@-319};$
 <X1OH> $\text{BZ123} + \text{OH} = 0.18 \cdot \text{HO2} + 0.67 \cdot \text{xHO2} + 0.67 \cdot \text{RO2C} + 0.15 \cdot \text{RO2XC} + 0.03 \cdot \text{OLEA1} +$
 $0.03 \cdot \text{OLEA2} + 0.03 \cdot \text{xGLY} + 0.07 \cdot \text{xMGLY} + 0.54 \cdot \text{xBACL} + 0.54 \cdot \text{xAFG2A} +$
 $0.1 \cdot \text{xAFG2B} + 0.02 \cdot \text{xBALD} + 0.1 \cdot \text{LVKS} + 0.02 \cdot \text{XYNL} + 0.01 \cdot \text{zR2NO3} +$
 $0.14 \cdot \text{zRANO3} + 0.03 \cdot \text{yROOH} + 0.78 \cdot \text{yRAOOH} + 0.82 \cdot \text{SumRO2} +$
 $\text{XYLRO2} \# 3.27\text{e-11};$
 <X2OH> $\text{BZ124} + \text{OH} = 0.23 \cdot \text{HO2} + 0.63 \cdot \text{xHO2} + 0.63 \cdot \text{RO2C} + 0.14 \cdot \text{RO2XC} + 0.04 \cdot \text{OLEA1} +$
 $0.11 \cdot \text{OLEA2} + 0.03 \cdot \text{xGLY} + 0.51 \cdot \text{xMGLY} + 0.06 \cdot \text{xBACL} + 0.08 \cdot \text{xAFG1} +$
 $0.04 \cdot \text{xAFG2A} + 0.27 \cdot \text{xAFG2B} + 0.03 \cdot \text{xBALD} + 0.02 \cdot \text{LVKS} + 0.21 \cdot \text{xAFG3} +$
 $0.05 \cdot \text{XYNL} + 0.01 \cdot \text{zR2NO3} + 0.13 \cdot \text{zRANO3} + 0.04 \cdot \text{yROOH} + 0.73 \cdot \text{yRAOOH} +$
 $0.77 \cdot \text{SumRO2} + \text{XYLRO2} \# 3.25\text{e-11};$
 <X3OH> $\text{BZ135} + \text{OH} = 0.17 \cdot \text{HO2} + 0.68 \cdot \text{xHO2} + 0.68 \cdot \text{RO2C} + 0.15 \cdot \text{RO2XC} + 0.11 \cdot \text{OLEA2} +$
 $0.67 \cdot \text{xMGLY} + 0.67 \cdot \text{xAFG2A} + 0.02 \cdot \text{xBALD} + 0.05 \cdot \text{XYNL} + 0.01 \cdot \text{zR2NO3} +$
 $0.15 \cdot \text{zRANO3} + 0.02 \cdot \text{yROOH} + 0.81 \cdot \text{yRAOOH} + 0.83 \cdot \text{SumRO2} +$
 $\text{XYLRO2} \# 5.86\text{e-11};$
 <EBOH> $\text{C2BEN} + \text{OH} = 0.36 \cdot \text{HO2} + 0.51 \cdot \text{xHO2} + 0.54 \cdot \text{RO2C} + 0.12 \cdot \text{RO2XC} + 0.01 \cdot \text{xMEO2} +$
 $0.02 \cdot \text{xHCHO} + 0.01 \cdot \text{OLEA1} + 0.18 \cdot \text{OLEA2} + 0.18 \cdot \text{xGLY} + 0.18 \cdot \text{xMGLY} +$

```

0.18*xBUDAL + 0.02*xAFG1 + 0.16*xAFG2A + 0.03*xBALD + 0.16*XYNL +
0.05*zR2NO3 + 0.07*zRANO3 + 0.23*yROOH + 0.44*yRAOOH + 0.13*xBENX +
0.66*SumRO2 # 7.00e-12;
<MTOH> MTBE + OH = 0.72*xHO2 + 1.12*RO2C + 0.09*RO2XC + 0.19*xMEO2 + 0.2*xHCHO +
0.09*zR1NO3 + 0.89*yROOH + 0.17*ALK1 + 0.72*ALK2 + 0.01*ALK3 +
1.21*SumRO2 # 1.87e-13^3.34@-843;
<MLOH> MEOH + OH = HO2 + HCHO # 2.32e-13^2.72@-402;
<FAOH> HCOOH + OH = HO2 + CO2 # 4.50e-13;
<H1OH> MEOOH + OH = 0.03*OH + 0.97*MEO2 + 0.03*HCHO + 0.97*SumRO2
# 5.30e-12@-190;
<H1HV> MEOOH = OH + HO2 + HCHO # 1.0/<COOH>;
<A2OH> MECHO + OH = MECHO_OH + 0.04*xHO2 + 0.05*RO2C + 0.95*MECO3 + 0.01*xHCHO +
0.05*SumRO2 + 0.95*SumRCO3 # 2.40e-12^0.77@-546;
<550> MECHO_OH = 0.01*xOH + 0.04*xPACID + 0.01*CO2 # 3.75e+00;
<551> MECHO_OH + NO = NO + 0.01*xHO2 + 0.04*xHCHO + 0.04*yHPCRB + 0.05*CO
# 2.55e-12@-380;
<A2N3> MECHO + NO3 = HNO3 + MECO3 + SumRCO3 # 1.40e-12@1860;
<A2HV> MECHO = HO2 + 0.9*MEO2 + 0.1*MECO3 + 0.9*CO + 0.9*SumRO2 + 0.1*SumRCO3
# 1.0/<CCHOR_13>;
<EAOH> ETOH + OH = 0.95*HO2 + 0.05*xHO2 + 0.05*RO2C + 0.07*xHCHO + 0.95*MECHO +
0.01*xGLCHO + 0.05*yROOH + 0.05*SumRO2 # 4.42e-13^2.29@-606;
<GAOH> GLCHO + OH = 0.2*HO2 + 0.8*R2CO3 + 0.2*GLY + 0.8*SumRCO3 # 1.10e-11;
<GAN3> GLCHO + NO3 = HNO3 + 0.1*HO2 + 0.9*R2CO3 + 0.1*GLY + 0.9*SumRCO3
# 1.84e-14;
<GAHV> GLCHO = GLCHO_HV + 0.07*OH + 1.66*HO2 + 0.05*xHO2 + 0.07*RO2C +
0.83*xHCHO + 0.02*xHCHO + 0.1*MEOH + 0.93*CO + 0.07*SumRO2
# 1.0/<GLALD_14>;
<558> GLCHO_HV = 0.02*xOH + 0.05*xPACID + 0.02*CO2 # 2.62e+00;
<559> GLCHO_HV + NO = NO + 0.02*xHO2 + 0.04*xHCHO + 0.06*yHPCRB + 0.06*CO
# 2.55e-12@-380;
<A3OH> ETCHO + OH = 0.04*xHO2 + 0.04*RO2C + 0.96*R2CO3 + 0.04*xMECHO +
0.01*xPACID + 0.03*yHPCRB + 0.04*CO + 0.04*SumRO2 + 0.96*SumRCO3
# 6.63e-13^1.99@-1018;
<A3N3> ETCHO + NO3 = HNO3 + R2CO3 + SumRCO3 # 6.30e-15;
<A3HV> ETCHO = HO2 + ETO2 + CO + SumRO2 # 1.0/<C2CHOabs>;
<AROH> ACRO + OH = ACRO_OH + 0.31*xHO2 + 0.31*RO2C + 0.01*RO2XC + 0.68*MACO3 +
0.07*xHCHO + 0.01*xGLY + 0.24*xGLCHO + 0.01*zRHNO3 + 0.22*yHPCRB +
0.24*CO + 0.32*SumRO2 + 0.68*SumRCO3 # 7.10e-12@-333;
<564> ACRO_OH = 0.06*xPACID # 1.69e+00;
<565> ACRO_OH + NO = NO + 0.06*xGLY + 0.06*yHPCRB # 2.55e-12@-380;
<ARO3> ACRO + O3 = 0.16*H2 + 0.15*OH + 0.27*HO2 + 0.38*HCHO2 + 0.03*RCHO2 +
0.13*HCHO + 0.9*GLY + 0.02*HCOOH + 0.26*CO2 + 0.34*CO # 2.80e-19;
<ARN3> ACRO + NO3 = 0.94*HNO3 + 0.06*xHO2 + 0.06*RO2C + 0.94*MACO3 +
0.06*xRCNO3 + 0.05*yRPNO3 + 0.06*CO + 0.06*SumRO2 + 0.94*SumRCO3
# 1.10e-15;
<ARHV> ACRO = ACRO_HV + 0.22*OH + 0.49*HO2 + 0.17*xHO2 + 0.22*RO2C + 0.05*MEO2 +
0.15*MACO3 + 0.15*HCHO + 0.06*xHCHO + 0.06*MEOH + 0.12*CH4 +
0.25*ETHEN + 0.16*CO2 + 1.06*CO + 0.27*SumRO2 + 0.15*SumRCO3
# 1.0/<ACROL_16>;
<569> ACRO_HV = 0.06*xOH + 0.17*xPACID + 0.06*CO2 # 2.40e+00;
<570> ACRO_HV + NO = NO + 0.05*xHO2 + 0.15*xHCHO + 0.19*yHPCRB + 0.21*CO
# 2.55e-12@-380;
<K3OH> ACET + OH = 0.96*RO2C + 0.04*RO2XC + 0.96*xMECO3 + 0.96*xHCHO +
0.04*zRCNO3 + 0.85*yHPCRB + SumRO2 # 1.97e-14^3.88@-678;
<K3HV> ACET = MEO2 + MECO3 + SumRO2 + SumRCO3 # 1.0/<ACET_06>;
<K4OH> MEK + OH = 0.29*xHO2 + 0.94*RO2C + 0.07*RO2XC + 0.55*xMECO3 +
0.08*xR2CO3 + 0.11*xHCHO + 0.54*xMECHO + 0.29*xRCHO + 0.07*zRCNO3 +
0.91*yHPCRB + 1.01*SumRO2 # 5.42e-14^3.57@-889;

```

<K4HV> MEK = 0.15*MEO2 + 0.85*ETO2 + 0.85*MECO3 + 0.15*R2CO3 + SumRO2 + SumRCO3
1.75e-1/<MEK_06>;

<MAOH> MACR + OH = MACR_OH + 0.05*xHO2 + 0.75*RO2C + 0.04*RO2XC + 0.21*MACO3 +
0.05*xHCHO + 0.61*xKET2 + 0.79*SumRO2 + 0.21*SumRCO3 # 8.00e-12@-380;

<576> MACR_OH = 0.69*xOH + 0.08*xKET2 + 0.05*xPACID + 0.04*zRCNO3 + 0.69*CO2
5.26e-01;

<577> MACR_OH + NO = NO + 0.69*xHO2 + 0.09*xHCHO + 0.14*xMGLY + 0.04*zRHNO3 +
0.68*yHPCRB + 0.61*CO # 2.55e-12@-380;

<MAO3> MACR + O3 = 0.16*H2 + 0.19*OH + 0.01*xOH + 0.25*HO2 + 0.03*xHO2 +
0.04*RO2C + 0.01*MECO3 + 0.38*HCHO2 + 0.03*RCHO2 + 0.1*HCHO +
0.02*xHCHO + 0.02*MECHO + 0.9*MGLY + 0.01*OACID + 0.02*xPACID +
0.01*yHPCRB + 0.24*CO2 + 0.38*CO + 0.04*SumRO2 + 0.01*SumRCO3
1.40e-15@2100;

<MAN3> MACR + NO3 = MACR_N3 + 0.3*HNO3 + 0.66*RO2C + 0.04*RO2XC + 0.3*MACO3 +
0.66*xRCNO3 + 0.7*SumRO2 + 0.3*SumRCO3 # 3.40e-15;

<580> MACR_N3 = 0.66*xOH + 0.04*zRCNO3 + 0.66*CO2 # 5.24e-01;

<581> MACR_N3 + NO = NO + 0.66*xHO2 + 0.59*yRPNO3 + 0.04*zRDNO3 + 0.66*CO
2.55e-12@-380;

<MAHV> MACR = 0.45*OH + 0.3*HO2 + 0.43*RO2C + 0.02*RO2XC + 0.15*MEO2 +
0.43*xMECO3 + 0.15*MACO3 + 0.15*HCHO + 0.43*xHCHO + 0.02*zRCNO3 +
0.38*yHPCRB + 0.25*PROPE + CO + 0.6*SumRO2 + 0.15*SumRCO3
1.0/<MACR_06>;

<MVOH> MVK + OH = 0.28*xHO2 + 0.95*RO2C + 0.05*RO2XC + 0.66*xMECO3 +
0.28*xHCHO + 0.66*xGLCHO + 0.28*xMGLY + 0.05*zRCNO3 + 0.9*yHPCRB +
SumRO2 # 2.60e-12@-610;

<MVO3> MVK + O3 = 0.17*H2 + 0.18*OH + 0.26*HO2 + 0.02*xHO2 + 0.02*RO2C +
0.4*HCHO2 + 0.01*RCHO2 + 0.05*HCHO + 0.01*xHCHO + 0.01*MECHO +
0.95*MGLY + 0.01*xPACID + 0.01*yHPCRB + 0.23*CO2 + 0.36*CO +
0.02*SumRO2 # 8.50e-16@1520;

<MVHV> MVK = 0.4*MEO2 + 0.4*MACO3 + 0.6*PROPE + 0.6*CO + 0.4*SumRO2 +
0.4*SumRCO3 # 1.0/<MVK_16>;

<F1OH> BUDAL + OH = BUDAL_OH + 0.54*OH + 0.44*xHO2 + 0.44*RO2C + 0.02*RO2XC +
0.41*xGLY + 0.54*MALAH + 0.05*xPACID + 0.02*CO + 0.46*SumRO2
5.29e-11;

<587> BUDAL_OH = 0.39*xPACID + 0.02*zRCNO3 # 2.02e+01;

<588> BUDAL_OH + NO = NO + 0.37*xGLY + 0.35*yHPCRB # 2.55e-12@-380;

<F1HV> BUDAL = OH + HO2 + MALAH # 2.50e-1/<AFGS>;

<PHOH> PHEN + OH = 0.85*HO2 + 0.07*xHO2 + 0.07*RO2C + 0.01*RO2XC + 0.07*BZO +
0.01*OLEA1 + 0.04*OLEA2 + 0.03*xGLY + 0.04*xMGLY + 0.04*xBUDAL +
0.01*xAFG1 + 0.02*xAFG2A + 0.03*OLEP + 0.77*CATL + 0.01*zRANO3 +
0.08*yRAOOH + 0.08*SumRO2 # 4.70e-13@-1220;

<PHN3> PHEN + NO3 = HNO3 + BZO # 4.50e-12;

<L1OH> ALK1 + OH = 0.96*xHO2 + 0.97*RO2C + 0.03*RO2XC + 0.01*xMECO3 +
0.01*xHCHO + 0.09*xMGLY + 0.43*xOACID + 0.03*zRCNO3 + 0.62*yHPCRB +
39.33*NROG + 0.43*CO + SumRO2 # 3.35e-13;

<L2OH> ALK2 + OH = 0.1*xHO2 + 0.95*RO2C + 0.05*RO2XC + 0.01*xMEO2 +
0.84*xMECO3 + 0.01*xHCHO + 0.08*xRCHO + 0.01*xMGLY + 0.84*xOACID +
0.05*zRCNO3 + 0.64*yHPCRB + 1.38*NROG + SumRO2 # 1.67e-12;

<L3OH> ALK3 + OH = 0.34*xHO2 + 1.2*RO2C + 0.12*RO2XC + 0.14*xETO2 +
0.07*xR2CO3 + 0.33*xTBUO + 0.1*xHCHO + 0.1*xMECHO + 0.09*xRCHO +
0.07*xACET + 0.01*xKET2 + 0.12*xOACID + 0.07*zR1NO3 + 0.03*zRCNO3 +
0.02*zRHNO3 + 0.81*yROOH + 0.33*yHPCRB + 0.12*ALK1 + 16.91*NROG +
0.05*CO + 1.32*SumRO2 # 2.85e-12;

<L4OH> ALK4 + OH = 0.01*xOH + 0.27*HO2 + 0.36*xHO2 + 0.96*RO2C + 0.12*RO2XC +
0.23*xETO2 + 0.06*xHCHO + 0.13*xMECHO + 0.05*xETCHO + 0.07*xRCHO +
0.26*ACET + 0.3*xACET + 0.03*xMEK + 0.15*xKET2 + 0.09*zR1NO3 +
0.03*zRHNO3 + 1.05*yROOH + 0.01*HPCRB + 0.01*yHPCRB + 0.01*CO2 +
1.08*SumRO2 # 4.54e-12;

```

<L5OH> ALK5 + OH = ALK5_OH + 0.3*HO2 + 0.33*xHO2 + RO2C + 0.22*RO2XC +
0.02*xETO2 + 0.01*xR2CO3 + 0.01*HCHO + 0.03*xHCHO + 0.04*xMECHO +
0.02*xETCHO + 0.11*RCHO + 0.04*xRCHO + 0.06*GLCHO + 0.01*xGLCHO +
0.05*xACET + 0.05*xMEK + 0.13*KET2 + 0.23*xKET2 + 0.01*xPACID +
0.08*zR1NO3 + 0.05*zR2NO3 + 0.02*zRCNO3 + 0.06*zRHNO3 + 1.05*yROOH +
0.12*yHPCRB + 0.01*ALK4 + 0.01*ALK5 + 1.22*SumRO2 # 1.14e-11;
<597> ALK5_OH = 0.11*HO2 + 0.03*PACID + 0.08*HPCRB # 2.87e-01;
<598> ALK5_OH + NO = NO + 0.06*xHO2 + 0.11*RO2C + 0.02*RO2XC + 0.02*xMECO3 +
0.01*xR2CO3 + 0.02*xETCHO + 0.06*xRCHO + 0.01*xKET2 + 0.01*zRCNO3 +
0.01*zRHNO3 + 0.05*yROOH + 0.08*yHPCRB + 0.01*ALK5 + 0.12*SumRO2
# 2.55e-12@-380;
<L6OH> ALK6 + OH = ALK6_OH + 0.16*HO2 + 0.4*xHO2 + 0.97*RO2C + 0.33*RO2XC +
0.01*xR2CO3 + 0.01*xTBUO + 0.08*xHCHO + 0.01*xMECHO + 0.01*xETCHO +
0.07*RCHO + 0.08*xRCHO + 0.15*xACET + 0.09*KET2 + 0.26*xKET2 +
0.01*zR1NO3 + 0.15*zR2NO3 + 0.05*zRCNO3 + 0.12*zRHNO3 + 1.05*yROOH +
0.24*yHPCRB + 0.06*ALK3 + 0.01*ALK4 + 0.01*ALK5 + 1.31*SumRO2
# 1.63e-11;
<600> ALK6_OH = 0.03*HO2 + 0.08*HPCRB # 2.70e-01;
<601> ALK6_OH + NO = NO + 0.06*xHO2 + 0.12*RO2C + 0.02*RO2XC + 0.01*xR2CO3 +
0.02*xHCHO + 0.01*xMECHO + 0.02*xRCHO + 0.04*xACET + 0.01*xKET2 +
0.01*zRCNO3 + 0.01*zRHNO3 + 0.06*yROOH + 0.04*yHPCRB + 0.04*ALK2 +
0.01*ALK5 + 0.14*SumRO2 # 2.55e-12@-380;
<O1OH> OLE1 + OH = 0.78*xHO2 + 1.12*RO2C + 0.1*RO2XC + 0.01*xMEO2 + 0.11*xTBUO +
0.69*xHCHO + 0.01*xMECHO + 0.35*xETCHO + 0.35*xRCHO + 0.15*xGLCHO +
0.02*xACRO + 0.03*xACET + 0.02*xKET2 + 0.01*xMVK + 0.01*zR1NO3 +
0.01*zR2NO3 + 0.08*zRHNO3 + 1.15*yROOH + 0.07*yRUOOH + 1.22*SumRO2
# 3.18e-11;
<O1O3> OLE1 + O3 = 0.09*H2 + 0.26*OH + 0.01*xOH + 0.17*HO2 + 0.16*xHO2 +
0.18*RO2C + 0.01*RO2XC + 0.01*ETO2 + 0.01*xTBUO + 0.21*HCHO2 +
0.17*RCHO2 + 0.5*HCHO + 0.08*xMECHO + 0.19*ETCHO + 0.04*xETCHO +
0.31*RCHO + 0.02*xRCHO + 0.04*xACET + 0.02*ETOH + 0.01*zRCNO3 +
0.02*yROOH + 0.14*yHPCRB + 0.03*ETHAN + 0.02*PROP + 0.01*NC4 +
0.01*ALK2 + 0.02*ALK3 + 0.01*ALK4 + 0.24*CO2 + 0.37*CO + 0.2*SumRO2
# 8.70e-18;
<O1N3> OLE1 + NO3 = 0.09*xNO2 + 0.55*xHO2 + 1.41*RO2C + 0.13*RO2XC +
0.09*xETO2 + 0.15*xTBUO + 0.09*xHCHO + 0.09*xETCHO + 0.01*xRCHO +
0.14*xACET + 0.02*zR1NO3 + 0.81*xRCNO3 + 1.22*yRPNO3 + 0.11*zRDNO3 +
0.31*yROOH + 1.53*SumRO2 # 1.44e-14;
<O1OP> OLE1 + O3P = 0.25*RCHO + 0.1*MEK + 0.15*KET2 + 0.09*ALK2 + 0.36*ALK3 +
0.05*ALK4 # 4.43e-12;
<O2OH> OLE2 + OH = 0.92*xHO2 + 0.94*RO2C + 0.07*RO2XC + 1.25*xMECHO +
0.4*xETCHO + 0.12*xRCHO + 0.01*xACRO + 0.07*zRHNO3 + 0.99*yROOH +
0.01*yRUOOH + 1.01*SumRO2 # 6.29e-11;
<O2O3> OLE2 + O3 = OLE2_O3 + 0.46*OH + 0.06*HO2 + 0.32*xHO2 + 0.4*RO2C +
0.01*RO2XC + 0.04*MEO2 + 0.01*ETO2 + 0.16*MECHO2 + 0.1*RCHO2 +
0.07*xHCHO + 0.67*MECHO + 0.09*xMECHO + 0.22*ETCHO + 0.01*xETCHO +
0.07*RCHO + 0.01*xACET + 0.06*MEOH + 0.02*ETOH + 0.01*zRCNO3 +
0.09*yHPCRB + 0.12*CH4 + 0.04*ETHAN + 0.23*CO2 + 0.18*CO + 0.46*SumRO2
# 1.90e-16;
<608> OLE2_O3 = 0.01*OH + 0.08*xOH + 0.01*RCHO + 0.22*xPACID + 0.08*CO2
# 2.45e+00;
<609> OLE2_O3 + NO = NO + 0.08*xHO2 + 0.01*RO2C + 0.01*xR2CO3 + 0.2*xHCHO +
0.02*xGLY + 0.26*yHPCRB + 0.29*CO + 0.01*SumRO2 # 2.55e-12@-380;
<O2N3> OLE2 + NO3 = 0.8*xNO2 + 0.11*xHO2 + 1.01*RO2C + 0.08*RO2XC +
1.11*xMECHO + 0.33*xETCHO + 0.09*xRCHO + 0.01*xACET + 0.11*xRCNO3 +
1.08*yRPNO3 + 0.08*zRDNO3 + 0.01*yROOH + 1.09*SumRO2 # 4.34e-13;
<O2OP> OLE2 + O3P = 0.21*MEK + 0.29*KET2 + 0.21*ALK1 + 0.22*ALK2 + 0.07*ALK3
# 1.95e-11;

```

<O3OH> $\text{OLE3} + \text{OH} = 0.94 \times \text{xHO2} + 0.94 \times \text{RO2C} + 0.05 \times \text{RO2XC} + 0.94 \times \text{xHCHO} + 0.82 \times \text{xACET} + 0.12 \times \text{xMEK} + 0.05 \times \text{zRHNO3} + \text{yROOH} + \text{SumRO2} \# 5.26\text{e-11};$

<O3O3> $\text{OLE3} + \text{O3} = 0.09 \times \text{H2} + 0.58 \times \text{OH} + 0.14 \times \text{HO2} + 0.48 \times \text{RO2C} + 0.02 \times \text{RO2XC} + 0.45 \times \text{xMECO3} + 0.03 \times \text{xR2CO3} + 0.21 \times \text{HCHO2} + 0.5 \times \text{HCHO} + 0.45 \times \text{xHCHO} + 0.03 \times \text{xMECHO} + 0.43 \times \text{ACET} + 0.07 \times \text{MEK} + 0.02 \times \text{zRCNO3} + 0.42 \times \text{yHPCRB} + 0.12 \times \text{CO2} + 0.17 \times \text{CO} + 0.5 \times \text{SumRO2} \# 1.18\text{e-17};$

<O3N3> $\text{OLE3} + \text{NO3} = 0.86 \times \text{xNO2} + 0.95 \times \text{RO2C} + 0.05 \times \text{RO2XC} + 0.01 \times \text{xMEO2} + 0.07 \times \text{xETO2} + 0.86 \times \text{xHCHO} + 0.8 \times \text{xACET} + 0.06 \times \text{xMEK} + 0.08 \times \text{xRCNO3} + \text{yRPNO3} + 0.05 \times \text{zRDNO3} + \text{SumRO2} \# 3.62\text{e-13};$

<O3OP> $\text{OLE3} + \text{O3P} = 0.5 \times \text{RCHO} + 0.5 \times \text{ALK2} \# 1.70\text{e-11};$

<O4OH> $\text{OLE4} + \text{OH} = 0.92 \times \text{xHO2} + 0.92 \times \text{RO2C} + 0.08 \times \text{RO2XC} + 0.83 \times \text{xMECHO} + 0.09 \times \text{xETCHO} + 0.92 \times \text{xACET} + 0.08 \times \text{zRHNO3} + \text{yROOH} + \text{SumRO2} \# 8.71\text{e-11};$

<O4O3> $\text{OLE4} + \text{O3} = \text{OLE4_O3} + 0.72 \times \text{OH} + 0.03 \times \text{HO2} + 0.16 \times \text{xHO2} + 0.7 \times \text{RO2C} + 0.02 \times \text{RO2XC} + 0.03 \times \text{xMEO2} + 0.48 \times \text{xMECO3} + 0.11 \times \text{MECHO2} + 0.01 \times \text{RCHO2} + 0.53 \times \text{xHCHO} + 0.45 \times \text{xMECHO} + 0.02 \times \text{xMECHO} + 0.05 \times \text{ETCHO} + 0.5 \times \text{ACET} + 0.05 \times \text{xMEOH} + 0.02 \times \text{zRCNO3} + 0.44 \times \text{yHPCRB} + 0.08 \times \text{CH4} + 0.01 \times \text{ETHAN} + 0.12 \times \text{CO2} + 0.07 \times \text{CO} + 0.75 \times \text{SumRO2} \# 4.05\text{e-16};$

<618> $\text{OLE4_O3} = 0.05 \times \text{xOH} + 0.14 \times \text{xPACID} + 0.05 \times \text{CO2} \# 2.45\text{e+00};$

<619> $\text{OLE4_O3} + \text{NO} = \text{NO} + 0.05 \times \text{xHO2} + 0.14 \times \text{xHCHO} + 0.17 \times \text{yHPCRB} + 0.18 \times \text{CO} \# 2.55\text{e-12@-380};$

<O4N3> $\text{OLE4} + \text{NO3} = 0.92 \times \text{xNO2} + 0.92 \times \text{RO2C} + 0.08 \times \text{RO2XC} + 0.83 \times \text{xMECHO} + 0.09 \times \text{xETCHO} + 0.92 \times \text{xACET} + \text{yRPNO3} + 0.08 \times \text{zRDNO3} + \text{SumRO2} \# 9.31\text{e-12};$

<O4OP> $\text{OLE4} + \text{O3P} = 0.5 \times \text{KET2} + 0.5 \times \text{ALK2} \# 5.11\text{e-11};$

<TPOH> $\text{TERP} + \text{OH} = \text{TERP_OH} + 0.02 \times \text{HO2} + 0.53 \times \text{xHO2} + 1.07 \times \text{RO2C} + 0.28 \times \text{RO2XC} + 0.01 \times \text{xMACO3} + 0.15 \times \text{xHCHO} + 0.29 \times \text{xRCHO} + 0.13 \times \text{xOLEA2} + 0.09 \times \text{xACET} + 0.05 \times \text{xKET2} + 0.01 \times \text{xMVK} + 0.04 \times \text{xLVKS} + 0.05 \times \text{xOLEP} + 0.02 \times \text{zR2NO3} + 0.06 \times \text{zRCNO3} + 0.2 \times \text{zRHNO3} + 0.51 \times \text{yROOH} + 0.55 \times \text{yRUOOH} + 0.01 \times \text{HPCRB} + 0.24 \times \text{yHPCRB} + 1.35 \times \text{SumRO2} + \text{TRPRXN} \# 1.10\text{e-10};$

<623> $\text{TERP_OH} = 0.06 \times \text{OH} + 0.01 \times \text{xOH} + 0.09 \times \text{HO2} + 0.02 \times \text{AFG2A} + 0.01 \times \text{xKET2} + 0.03 \times \text{PACID} + 0.01 \times \text{xPACID} + 0.06 \times \text{HPCRB} \# 1.48\text{e+00};$

<624> $\text{TERP_OH} + \text{NO} = \text{NO} + 0.07 \times \text{xHO2} + 0.16 \times \text{RO2C} + 0.05 \times \text{RO2XC} + 0.03 \times \text{xR2CO3} + 0.02 \times \text{xHCHO} + 0.03 \times \text{xRCHO} + 0.02 \times \text{xOLEA1} + 0.02 \times \text{xOLEA2} + 0.01 \times \text{xAFG2A} + 0.01 \times \text{xOLEP} + 0.04 \times \text{zRCNO3} + 0.01 \times \text{zRHNO3} + 0.01 \times \text{yRUOOH} + 0.22 \times \text{yHPCRB} + 0.21 \times \text{SumRO2} \# 2.55\text{e-12@-380};$

<TPO3> $\text{TERP} + \text{O3} = \text{TERP_O3} + 0.03 \times \text{H2} + 0.58 \times \text{OH} + 0.02 \times \text{xOH} + 0.07 \times \text{HO2} + 0.08 \times \text{xHO2} + 0.37 \times \text{RO2C} + 0.12 \times \text{RO2XC} + 0.08 \times \text{xMECO3} + 0.11 \times \text{xR2CO3} + 0.08 \times \text{HCHO2} + 0.25 \times \text{RCHO2} + 0.18 \times \text{HCHO} + 0.06 \times \text{xHCHO} + 0.13 \times \text{xRCHO} + 0.03 \times \text{xGLY} + 0.01 \times \text{xMACR} + 0.01 \times \text{ACET} + 0.01 \times \text{xACET} + 0.17 \times \text{KET2} + 0.01 \times \text{LVKS} + 0.01 \times \text{xAFG3} + 0.01 \times \text{xPACID} + 0.02 \times \text{OLEP} + 0.12 \times \text{zRCNO3} + 0.01 \times \text{yROOH} + 0.01 \times \text{yHPCRB} + 0.41 \times \text{yHPCRB} + 0.06 \times \text{CO2} + 0.12 \times \text{CO} + 0.49 \times \text{SumRO2} + \text{TRPRXN} \# 1.17\text{e-16};$

<626> $\text{TERP_O3} = 0.06 \times \text{OH} + 0.07 \times \text{HO2} + 0.01 \times \text{PACID} + 0.01 \times \text{OTHN} + 0.05 \times \text{HPCRB} \# 1.07\text{e+00};$

<627> $\text{TERP_O3} + \text{NO} = \text{NO} + 0.02 \times \text{xHO2} + 0.12 \times \text{RO2C} + 0.04 \times \text{RO2XC} + 0.04 \times \text{xMECO3} + 0.04 \times \text{xMACO3} + 0.04 \times \text{xHCHO} + 0.04 \times \text{xOLEA2} + 0.04 \times \text{zRCNO3} + 0.13 \times \text{yHPCRB} + 0.16 \times \text{SumRO2} \# 2.55\text{e-12@-380};$

<TPN3> $\text{TERP} + \text{NO3} = \text{TERP_N3} + 0.51 \times \text{xNO2} + 0.01 \times \text{OH} + 0.01 \times \text{xOH} + 0.05 \times \text{xHO2} + 1.12 \times \text{RO2C} + 0.28 \times \text{RO2XC} + 0.01 \times \text{xHCHO} + 0.29 \times \text{xRCHO} + 0.21 \times \text{xOLEA2} + 0.15 \times \text{xACET} + 0.01 \times \text{xMVK} + 0.01 \times \text{xOLEP} + 0.01 \times \text{RCNO3} + 0.07 \times \text{xRCNO3} + 0.08 \times \text{zRCNO3} + \text{yRPNO3} + 0.19 \times \text{zRDNO3} + 0.04 \times \text{yROOH} + 0.01 \times \text{yRUOOH} + 1.4 \times \text{SumRO2} + \text{TRPRXN} \# 1.11\text{e-11};$

<629> $\text{TERP_N3} = 0.1 \times \text{OH} + 0.04 \times \text{HO2} + 0.1 \times \text{RCNO3} + 0.01 \times \text{HPCRB} \# 1.27\text{e+00};$

<630> $\text{TERP_N3} + \text{NO} = \text{NO} + 0.06 \times \text{xHO2} + 0.17 \times \text{RO2C} + 0.05 \times \text{RO2XC} + 0.03 \times \text{xR2CO3} + 0.01 \times \text{xHCHO} + 0.01 \times \text{xACET} + 0.06 \times \text{xRCNO3} + 0.05 \times \text{zRCNO3} + 0.01 \times \text{yRPNO3} + 0.01 \times \text{yROOH} + 0.01 \times \text{yHPCRB} + 0.01 \times \text{CO} + 0.22 \times \text{SumRO2} \# 2.55\text{e-12@-380};$

<TPOP> $\text{TERP} + \text{O3P} = 0.16 \times \text{RCHO} + 0.04 \times \text{OLEA2} + 0.18 \times \text{KET2} + 0.01 \times \text{LVKS} + 0.27 \times \text{OLEP} + 0.18 \times \text{ALK3} + 0.08 \times \text{ALK4} + 0.08 \times \text{ALK5} + \text{TRPRXN} \# 4.24\text{e-11};$

<SQOH> $\text{SESQ} + \text{OH} = \text{SESQ_OH} + 0.05 \times \text{OH} + 0.05 \times \text{HO2} + 0.47 \times \text{xHO2} + 0.77 \times \text{RO2C} +$

$0.23*RO2XC + 0.02*xHCHO + 0.44*xOLEA2 + 0.05*OLEP + 0.02*xOLEP +$
 $0.03*zRCNO3 + 0.18*zRHNO3 + 0.84*yRUOOH + 0.02*HPCRB + 0.09*yHPCRB +$
 $SumRO2 + SESQRXN \# 2.00e-10;$
 <633> $SESQ_OH = 0.07*OH + 0.13*HO2 + 0.01*OLEP + 0.01*zRPNO3 + 0.01*OTHN +$
 $0.12*HPCRB \# 5.58e+00;$
 <634> $SESQ_OH + NO = NO + 0.08*xHO2 + 0.32*RO2C + 0.12*RO2XC + 0.1*xHCHO +$
 $0.08*xOLEA2 + 0.1*xACET + 0.01*zRCNO3 + 0.12*zRHNO3 + 0.01*yROOH +$
 $0.24*yRUOOH + 0.21*yHPCRB + 0.44*SumRO2 \# 2.55e-12@-380;$
 <SQO3> $SESQ + O3 = SESQ_O3 + 0.66*OH + 0.01*HO2 + 0.19*RO2C + 0.07*RO2XC +$
 $0.15*xMACO3 + 0.33*RCHO2 + 0.01*HCHO + 0.15*xHCHO + 0.01*OLEP +$
 $0.07*zRCNO3 + 0.01*HPCRB + 0.22*yHPCRB + 0.26*SumRO2 +$
 $SESQRXN \# 3.14e-16;$
 <636> $SESQ_O3 = 0.07*OH + 0.36*HO2 + 0.06*OTHN + 0.01*HPCRB \# 4.32e+00;$
 <637> $SESQ_O3 + NO = NO + 0.01*xOH + 0.43*RO2C + 0.17*RO2XC + 0.17*xMECO3 +$
 $0.02*xHCHO + 0.02*xRCHO + 0.22*xOLEA2 + 0.17*zRCNO3 + 0.01*xHPCRB +$
 $0.51*yHPCRB + 0.04*CO + 0.6*SumRO2 \# 2.55e-12@-380;$
 <SQN3> $SESQ + NO3 = SESQ_N3 + 0.74*xNO2 + 0.02*OH + 0.82*RO2C + 0.22*RO2XC +$
 $0.74*xOLEA2 + 0.02*RCNO3 + 0.01*zRCNO3 + yRPNO3 + 0.21*zRDNO3 +$
 $1.04*SumRO2 + SESQRXN \# 1.90e-11;$
 <639> $SESQ_N3 = 0.01*OH + 0.01*HO2 + 0.01*RCNO3 \# 2.53e+00;$
 <640> $SESQ_N3 + NO = NO + 0.04*RO2C + 0.01*RO2XC + 0.01*zRCNO3 + 0.05*SumRO2$
 $\# 2.55e-12@-380;$
 <SQOP> $SESQ + O3P = 0.13*OLEA2 + 0.87*OLEP + SESQRXN \# 6.85e-11;$
 <BXOH> $BENX + OH = 0.69*HO2 + 0.28*xHO2 + 0.28*RO2C + 0.04*RO2XC + 0.12*OLEA2 +$
 $0.28*xGLY + 0.28*xBUDAL + 0.57*PHEN + 0.04*zRANO3 + 0.31*yRAOOH +$
 $0.32*SumRO2 \# 1.21e-12;$
 <B1OH> $ARO1 + OH = 0.27*HO2 + 0.48*xHO2 + 0.75*RO2C + 0.21*RO2XC + 0.05*xETO2 +$
 $0.01*xHCHO + 0.06*xMECHO + 0.11*xETCHO + 0.01*xRCHO + 0.01*OLEA1 +$
 $0.13*OLEA2 + 0.13*xGLY + 0.13*xMGLY + 0.13*xBUDAL + 0.01*xAFG1 +$
 $0.12*xAFG2A + 0.2*xBALD + 0.12*XYNL + 0.03*zR1NO3 + 0.11*zR2NO3 +$
 $0.01*zRHNO3 + 0.06*zRANO3 + 0.64*yROOH + 0.32*yRAOOH + 0.04*ARO1 +$
 $3.11*NROG + 0.96*SumRO2 + TOLRO2 \# 7.69e-12;$
 <B2OH> $ARO2 + OH = ARO2_OH + 0.29*HO2 + 0.56*xHO2 + 0.6*RO2C + 0.14*RO2XC +$
 $0.03*OLEA1 + 0.13*OLEA2 + 0.07*xGLY + 0.39*xMGLY + 0.03*xBACL +$
 $0.03*xBUDAL + 0.07*xAFG1 + 0.3*xAFG2A + 0.04*xAFG2B + 0.04*BALD +$
 $0.03*xBALD + 0.01*xKET2 + 0.01*LVKS + 0.04*xAFG3 + 0.08*XYNL +$
 $0.02*zR2NO3 + 0.01*zRCNO3 + 0.11*zRANO3 + 0.1*yROOH + 0.59*yRAOOH +$
 $0.04*yHPCRB + 0.03*xBENX + 1.38*NROG + 0.74*SumRO2 + XYLRO2 \# 2.13e-11;$
 <645> $ARO2_OH = 0.01*HPCRB \# 1.76e-01;$
 <646> $ARO2_OH + NO = NO + 0.01*RO2C + 0.01*yHPCRB + 0.32*NROG + 0.01*SumRO2$
 $\# 2.55e-12@-380;$
 <FUOH> $FURNS + OH = 0.75*HO2 + 0.24*xHO2 + 0.24*RO2C + 0.01*RO2XC + 0.07*xRCHO +$
 $0.03*xOLEA1 + 0.75*BUDAL + 0.14*xOLEP + 0.01*zRHNO3 + 0.15*yRUOOH +$
 $0.01*yHPCRB + 0.08*CO + 0.25*SumRO2 \# 3.84e-11;$
 <FUO3> $FURNS + O3 = 0.39*HO2 + 0.38*xHO2 + 0.38*RO2C + 0.01*RO2XC + 0.4*RCHO2 +$
 $0.14*OLEA1 + 0.05*xGLY + 0.01*zRCNO3 + 0.07*HPCRB + 0.25*yHPCRB +$
 $0.33*ALK1 + 0.19*CO2 + 0.79*CO + 0.39*SumRO2 \# 2.40e-18;$
 <FUN3> $FURNS + NO3 = 0.08*xNO2 + 0.87*xHO2 + 0.95*RO2C + 0.05*RO2XC +$
 $0.07*xOLEA1 + 0.01*xOLEP + 0.87*xRCNO3 + 0.63*yRPNO3 + 0.05*zRDNO3 +$
 $0.28*CO + SumRO2 \# 1.20e-12;$
 <STOH> $STYRS + OH = 0.06*HO2 + 0.79*xHO2 + 0.79*RO2C + 0.15*RO2XC + 0.74*xHCHO +$
 $0.03*OLEA2 + 0.02*xGLY + 0.02*xMGLY + 0.03*xBUDAL + 0.02*xAFG2A +$
 $0.74*xBALD + 0.02*XYNL + 0.14*zRHNO3 + 0.01*zRANO3 + 0.88*yROOH +$
 $0.06*yRAOOH + 0.94*SumRO2 + XYLRO2 \# 5.80e-11;$
 <STO3> $STYRS + O3 = 0.09*H2 + 0.08*OH + 0.17*HO2 + 0.03*RO2C + 0.03*xBZO +$
 $0.21*HCHO2 + 0.34*RCHO2 + 0.5*HCHO + 0.5*BALD + 0.05*PHEN +$
 $0.03*yROOH + 0.09*BENZ + 0.23*CO2 + 0.22*CO + 0.03*SumRO2 \# 1.60e-17;$
 <AMOH> $AMINS + OH = 0.02*HO2 + 0.96*xHO2 + 0.97*RO2C + 0.02*RO2XC + 0.01*xMEO2 +$

0.08*xHCHO + 0.02*RCHO + 0.02*zR2NO3 + 0.99*yROOH + 0.97*xAMINS +
0.99*SumRO2 # 4.35e-11;

<AMO3> AMINS + O3 = 0.61*AMINS + 29.54*NROG # 3.09e-18;

<TAOH> TAMNS + OH = 0.06*RO2C + 0.03*xMEO2 + 0.03*xHCHO + 0.06*yROOH +
0.97*PNAMIN + 0.03*xAMINS + 0.06*SumRO2 # 1.01e-11;

<A4OH> RCHO + OH = 0.01*OH + 0.01*xOH + 0.1*HO2 + 0.07*xHO2 + 0.09*RO2C +
0.01*RO2XC + 0.8*R2CO3 + 0.04*xHCHO + 0.01*xMECHO + 0.06*RCHO +
0.02*xRCHO + 0.05*MGLY + 0.01*xACET + 0.03*xPACID + 0.01*zRCNO3 +
0.04*yHPCRB + 0.01*ALK4 + 0.14*NROG + 0.01*CO2 + 0.02*CO + 0.1*SumRO2 +
0.8*SumRCO3 # 3.29e-11;

<A4N3> RCHO + NO3 = HNO3 + 0.07*HO2 + 0.01*RO2C + 0.92*R2CO3 + 0.04*RCHO +
0.03*MGLY + 0.01*SumRO2 + 0.92*SumRCO3 # 2.23e-14;

<A4HV> RCHO = 1.4*HO2 + 0.51*xHO2 + 0.65*RO2C + 0.04*RO2XC + 0.01*xETO2 +
0.05*xMECO3 + 0.15*xHCHO + 0.17*MECHO + 0.05*xMECHO + 0.05*xETCHO +
0.01*RCHO + 0.23*xRCHO + 0.22*GLCHO + 0.01*xGLCHO + 0.01*xMGLY +
0.07*xACET + 0.01*xMEK + 0.06*xKET2 + 0.01*zR1NO3 + 0.03*zRHNO3 +
0.59*yROOH + 0.08*yHPCRB + 0.03*ALK4 + 0.08*NROG + CO + 0.69*SumRO2
1.0/<C2CHOabs>;

<A5OH> OLEA1 + OH = OLEA1_OH + 0.22*HO2 + 0.26*xHO2 + 0.59*RO2C + 0.05*RO2XC +
0.03*MACO3 + 0.01*xHCHO + 0.06*RCHO + 0.05*xGLY + 0.2*xGLCHO +
0.05*AFG1 + 0.03*AFG2A + 0.08*AFG2B + 0.17*xKET2 + 0.02*xPACID +
0.01*HPCRB + 0.05*yHPCRB + 0.65*SumRO2 + 0.03*SumRCO3 # 5.06e-11;

<659> OLEA1_OH = 0.33*xOH + 0.1*HO2 + 0.2*xKET2 + 0.19*xPACID + 0.05*zRCNO3 +
0.1*HPCRB + 0.33*CO2 # 7.52e-01;

<660> OLEA1_OH + NO = NO + 0.42*xHO2 + 0.09*RO2C + 0.01*RO2XC + 0.09*xGLY +
0.2*xGLCHO + 0.48*xMGLY + 0.09*xHCOOH + 0.05*zRHNO3 + 0.6*yHPCRB +
0.04*CO + 0.1*SumRO2 # 2.55e-12@-380;

<A5O3> OLEA1 + O3 = 0.62*OH + 0.01*xOH + 0.59*HO2 + 0.02*xHO2 + 0.12*RO2C +
0.01*MECO3 + 0.09*xR2CO3 + 0.16*RCHO2 + 0.04*HCHO + 0.1*xHCHO +
0.02*MECHO + 0.42*GLY + 0.06*GLCHO + 0.81*MGLY + 0.02*KET2 + 0.1*MEOH +
0.02*HCOOH + 0.01*OACID + 0.02*xPACID + 0.08*yHPCRB + 0.05*ALK4 +
0.33*CO2 + 0.11*CO + 0.12*SumRO2 + 0.01*SumRCO3 # 3.50e-18;

<A5N3> OLEA1 + NO3 = OLEA1_N3 + 0.56*xNO2 + 0.04*HNO3 + 0.22*HO2 + 0.16*xHO2 +
0.71*RO2C + 0.06*RO2XC + 0.01*xHCHO + 0.03*xGLY + 0.53*xGLCHO +
0.02*AFG1 + 0.01*AFG2B + 0.03*xKET2 + 0.22*xPACID + 0.19*RCNO3 +
0.01*xRCNO3 + 0.02*zRCNO3 + 0.15*xRHNO3 + 0.17*yRPNO3 + 0.02*zRDNO3 +
0.15*CO + 0.78*SumRO2 # 9.64e-14;

<663> OLEA1_N3 = 0.01*AFG2B + 0.31*xPACID + 0.03*zRCNO3 # 5.51e+01;

<664> OLEA1_N3 + NO = NO + 0.01*RO2C + 0.31*xMGLY + 0.34*yRPNO3 + 0.03*zRDNO3 +
0.01*SumRO2 # 2.55e-12@-380;

<A5HV> OLEA1 = 0.43*OH + 0.01*xOH + 0.49*HO2 + 0.11*xHO2 + 0.36*RO2C +
0.02*RO2XC + 0.1*MEO2 + 0.13*xMECO3 + 0.1*xR2CO3 + 0.03*MACO3 +
0.13*xHCHO + 0.01*xMECHO + 0.06*xRCHO + 0.09*GLCHO + 0.13*xGLCHO +
0.06*MGLY + 0.03*AFG2A + 0.08*AFG2B + 0.03*KET2 + 0.04*xKET2 +
0.01*HCOOH + 0.01*xPACID + 0.25*OLEP + 0.02*zRCNO3 + 0.01*yROOH +
0.29*yHPCRB + 0.02*ALK4 + 0.01*ALK5 + 0.05*CO2 + 1.02*CO +
0.48*SumRO2 + 0.03*SumRCO3 # 1.0/<MACR_06>;

<A6OH> OLEA2 + OH = OLEA2_OH + 0.06*OH + 0.09*xOH + 0.05*HO2 + 0.31*xHO2 +
0.65*RO2C + 0.13*RO2XC + 0.01*xMECO3 + 0.03*xR2CO3 + 0.18*MACO3 +
0.03*xMACO3 + 0.22*xHCHO + 0.31*xRCHO + 0.02*OLEA1 + 0.02*OLEA2 +
0.02*xGLY + 0.01*xGLCHO + 0.01*xMGLY + 0.08*xKET2 + 0.01*LVKS +
0.05*MALAH + 0.12*xPACID + 0.02*OLEP + 0.11*zRCNO3 + 0.01*zRHNO3 +
0.04*HPCRB + 0.47*yHPCRB + 0.04*CO2 + 0.09*CO + 0.78*SumRO2 +
0.18*SumRCO3 # 8.78e-11;

<667> OLEA2_OH = 0.04*OH + 0.03*xOH + 0.03*HO2 + 0.02*xKET2 + 0.05*xPACID +
0.05*HPCRB + 0.02*CO2 # 4.14e+00;

<668> OLEA2_OH + NO = NO + 0.06*xHO2 + 0.07*RO2C + 0.02*RO2XC + 0.02*xMECO3 +
0.01*xR2CO3 + 0.02*xHCHO + 0.05*xRCHO + 0.03*xMGLY + 0.01*xACET +

```

0.01*MALAH + 0.02*zRCNO3 + 0.18*yHPCRB + 0.02*CO + 0.09*SumRO2
# 2.55e-12@-380;
<A603> OLEA2 + O3 = OLEA2_O3 + 0.06*H2 + 0.52*OH + 0.12*HO2 + 0.08*xHO2 +
0.35*RO2C + 0.07*RO2XC + 0.12*xMECO3 + 0.13*HCHO2 + 0.15*RCHO2 +
0.29*HCHO + 0.08*xHCHO + 0.38*RCHO + 0.11*xRCHO + 0.16*GLY +
0.01*xGLY + 0.2*MGLY + 0.01*xPACID + 0.07*zRCNO3 + 0.14*yHPCRB +
0.14*CO2 + 0.29*CO + 0.43*SumRO2 # 1.65e-17;
<670> OLEA2_O3 = 0.15*xOH + 0.03*xBACL + 0.1*xKET2 + 0.08*xPACID + 0.05*CO2
# 1.53e+00;
<671> OLEA2_O3 + NO = NO + 0.02*OH + 0.05*xHO2 + 0.01*RO2C + 0.01*RO2XC +
0.07*xR2CO3 + 0.01*xRCHO + 0.03*xGLY + 0.07*xMGLY + 0.02*KET2 +
0.01*zRCNO3 + 0.24*yHPCRB + 0.08*CO + 0.02*SumRO2 # 2.55e-12@-380;
<A6N3> OLEA2 + NO3 = OLEA2_N3 + 0.09*xNO2 + 0.28*HNO3 + 0.13*xOH + 0.2*xHO2 +
1.17*RO2C + 0.22*RO2XC + 0.03*xMECO3 + 0.14*xR2CO3 + 0.07*MACO3 +
0.07*xMACO3 + 0.05*xHCHO + 0.28*xRCHO + 0.01*xMGLY + 0.01*xAFG3 +
0.05*MALAH + 0.26*xPACID + 0.32*xRCNO3 + 0.21*zRCNO3 + 0.02*xRHNO3 +
0.07*yRPNO3 + 0.01*zRDNO3 + 0.02*yHPCRB + 0.07*CO2 + 0.03*CO +
1.4*SumRO2 + 0.07*SumRCO3 # 1.19e-12;
<673> OLEA2_N3 = 0.01*NO2 + 0.02*OH + 0.03*xOH + 0.13*xPACID + 0.01*RCNO3 +
0.01*HPCRB + 0.03*CO2 # 3.17e+01;
<674> OLEA2_N3 + NO = NO + 0.02*xNO2 + 0.02*xHO2 + 0.03*RO2C + 0.01*xMECO3 +
0.07*xHCHO + 0.05*xRCHO + 0.02*xMGLY + 0.01*xRCNO3 + 0.03*yRPNO3 +
0.24*yHPCRB + 0.07*CO + 0.03*SumRO2 # 2.55e-12@-380;
<A6HV> OLEA2 = OLEA2_HV + 0.15*OH + 0.01*xOH + 1.04*HO2 + 0.35*xHO2 +
1.06*RO2C + 0.26*RO2XC + 0.04*xMECO3 + 0.45*xHCHO + 0.17*xRCHO +
0.01*OLEA2 + 0.06*xOLEA2 + 0.19*xMACR + 0.13*xAFG2A + 0.01*xAFG2B +
0.01*xLVKS + 0.13*OLEP + 0.24*zRCNO3 + 0.02*zRHNO3 + 0.07*yRUOOH +
0.04*HPCRB + 1.23*yHPCRB + 1.11*CO + 1.33*SumRO2 # 1.0/<C2CHOabs>;
<676> OLEA2_HV = 0.14*OH + 0.02*HO2 + 0.01*AFG2A + 0.11*HPCRB + 0.01*CO2
# 2.22e+00;
<677> OLEA2_HV + NO = NO + 0.09*xHO2 + 0.22*RO2C + 0.06*RO2XC + 0.01*xMECO3 +
0.04*xHCHO + 0.03*xOLEA2 + 0.03*xLVKS + 0.05*zRCNO3 + 0.01*zRHNO3 +
0.01*yRUOOH + 0.27*yHPCRB + 0.28*SumRO2 # 2.55e-12@-380;
<K5OH> KET2 + OH = 0.56*HO2 + 0.16*xHO2 + 0.45*RO2C + 0.07*RO2XC + 0.02*MECO3 +
0.05*xMECO3 + 0.01*R2CO3 + 0.12*xR2CO3 + 0.02*HCHO + 0.14*xHCHO +
0.09*xMECHO + 0.01*xETCHO + 0.13*RCHO + 0.16*xRCHO + 0.37*MGLY +
0.01*xACET + 0.07*KET2 + 0.03*xKET2 + 0.07*zRCNO3 + 0.5*yHPCRB +
0.52*SumRO2 + 0.02*SumRCO3 # 9.56e-12;
<K5HV> KET2 = 0.37*HO2 + 0.3*xHO2 + 0.37*RO2C + 0.02*RO2XC + 0.09*MEO2 +
0.13*ETO2 + 0.51*MECO3 + 0.4*R2CO3 + 0.36*HCHO + 0.03*xHCHO +
0.02*xMECHO + 0.04*xETCHO + 0.26*xRCHO + 0.02*zRHNO3 + 0.39*yROOH +
0.08*ALK4 + 0.01*ALK5 + 0.09*CO + 0.61*SumRO2 + 0.91*SumRCO3
# 7.53e-2/<MEK_06>;
<K6OH> LVKS + OH = LVKS_OH + 0.22*HO2 + 0.04*xHO2 + 0.39*RO2C + 0.03*RO2XC +
0.13*MECO3 + 0.09*xMECO3 + 0.25*xR2CO3 + 0.04*xHCHO + 0.2*RCHO +
0.09*xRCHO + 0.13*OLEA1 + 0.01*MGLY + 0.04*xBACL + 0.25*xKET2 +
0.03*zRCNO3 + 0.36*yHPCRB + 0.42*SumRO2 + 0.13*SumRCO3 # 6.09e-11;
<681> LVKS_OH = 0.23*OH + 0.01*HO2 + 0.23*HPCRB # 4.68e-01;
<682> LVKS_OH + NO = NO + 0.01*xOH + 0.09*xHO2 + 0.21*RO2C + 0.04*RO2XC +
0.1*xMECO3 + 0.09*xRCHO + 0.01*xBACL + 0.01*xAFG2B + 0.01*xPACID +
0.04*zRCNO3 + 0.09*xHPCRB + 0.23*yHPCRB + 0.01*CO + 0.25*SumRO2
# 2.55e-12@-380;
<K6O3> LVKS + O3 = LVKS_O3 + 0.05*H2 + 0.53*OH + 0.38*HO2 + 0.02*xHO2 +
0.18*RO2C + 0.01*RO2XC + 0.01*MECO3 + 0.09*xMECO3 + 0.01*xR2CO3 +
0.13*HCHO2 + 0.13*RCHO2 + 0.03*HCHO + 0.02*xHCHO + 0.01*MECHO +
0.03*RCHO + 0.01*xGLY + 0.59*MGLY + 0.01*xMGLY + 0.32*BACL +
0.01*xBACL + 0.06*KET2 + 0.03*HCOOH + 0.01*OACID + 0.02*xPACID +
0.01*zRCNO3 + 0.04*yHPCRB + 0.4*CO2 + 0.3*CO + 0.19*SumRO2 +

```

```

0.01*SumRCO3 # 3.04e-17;
<684> LVKS_O3 = 0.02*OH + 0.01*xOH + 0.04*xMECO3 + 0.12*xPACID + 0.01*CO2
# 5.13e+00;
<685> LVKS_O3 + NO = NO + 0.05*xHO2 + 0.02*RO2C + 0.01*RO2XC + 0.02*xHCHO +
0.1*xMGLY + 0.01*zRCNO3 + 0.13*yHPCRB + 0.02*CO + 0.03*SumRO2
# 2.55e-12@-380;
<K6HV> LVKS = 0.06*xOH + 0.14*HO2 + 0.12*RO2C + 0.03*RO2XC + 0.26*MEO2 +
0.06*xMECO3 + 0.25*MACO3 + 0.14*HCHO + 0.06*xMGLY + 0.06*MALAH +
0.06*xPACID + 0.6*OLEP + 0.03*zRCNO3 + 0.6*CO + 0.41*SumRO2 +
0.25*SumRCO3 # 1.0/<MVK_16>;
<O5OH> OLEP + OH = OLEP_OH + 0.01*OH + 0.13*HO2 + 0.55*xHO2 + 0.65*RO2C +
0.14*RO2XC + 0.02*xHCHO + 0.53*xRCHO + 0.11*OLEA1 + 0.01*xBACL +
0.02*xKET2 + 0.01*xMVK + 0.01*OLEP + 0.14*zRCNO3 + 0.03*yROOH +
0.01*yRUOOH + 0.01*HPCRB + 0.74*yHPCRB + 0.8*SumRO2 # 8.34e-11;
<688> OLEP_OH = 0.05*OH + 0.01*xOH + 0.1*HO2 + 0.12*HPCRB + 0.01*CO2
# 2.30e+00;
<689> OLEP_OH + NO = NO + 0.07*xHO2 + 0.25*RO2C + 0.03*RO2XC + 0.06*xMECO3 +
0.11*xHCHO + 0.06*xGLCHO + 0.02*xMGLY + 0.03*xKET2 + 0.02*zRCNO3 +
0.01*zRHNO3 + 0.07*yRUOOH + 0.14*yHPCRB + 0.04*CO + 0.28*SumRO2
# 2.55e-12@-380;
<O5O3> OLEP + O3 = OLEP_O3 + 0.03*H2 + 0.61*OH + 0.05*HO2 + 0.08*xHO2 +
0.33*RO2C + 0.09*RO2XC + 0.04*xMECO3 + 0.11*xR2CO3 + 0.06*HCHO2 +
0.23*RCHO2 + 0.14*HCHO + 0.14*xHCHO + 0.08*RCHO + 0.09*xRCHO +
0.02*xGLY + 0.13*KET2 + 0.09*zRCNO3 + 0.35*yHPCRB + 0.03*ALK2 +
0.09*CO2 + 0.18*CO + 0.41*SumRO2 # 1.50e-16;
<691> OLEP_O3 = 0.17*OH + 0.04*HO2 + 0.01*KET2 + 0.17*HPCRB # 9.36e-01;
<692> OLEP_O3 + NO = NO + 0.17*RO2C + 0.06*RO2XC + 0.15*xMECO3 + 0.01*xR2CO3 +
0.15*xRCHO + 0.06*zRCNO3 + 0.18*yHPCRB + 0.22*SumRO2 # 2.55e-12@-380;
<O5N3> OLEP + NO3 = 0.74*xNO2 + 0.08*xHO2 + 0.88*RO2C + 0.16*RO2XC +
0.01*xMECO3 + 0.18*xHCHO + 0.64*xRCHO + 0.11*xKET2 + 0.09*xRCNO3 +
0.14*zRCNO3 + 0.28*yRPNO3 + 0.02*zRDNO3 + 0.01*yHPCRB + 1.04*SumRO2
# 8.50e-12;
<OAOH> OACID + OH = 0.3*xHO2 + 0.3*RO2C + 0.7*MEO2 + 0.02*xHCHO + 0.28*xMGLY +
0.26*yHPCRB + 0.72*CO2 + SumRO2 # 7.47e-13;
<PAOH> PACID + OH = 0.19*xOH + 0.56*xHO2 + 0.74*RO2C + 0.26*MECO3 + 0.19*xHCHO +
0.56*xPACID + 0.19*CO2 + 0.74*SumRO2 + 0.26*SumRCO3 # 3.00e-14;
<PAHV> PACID = OH + MEO2 + CO2 + SumRO2 # 1.0/<PAA>;
<MGOH> MGLY + OH = 0.01*xHO2 + 0.01*RO2C + 0.99*MECO3 + 0.01*xHCHO +
0.01*xPACID + CO + 0.01*SumRO2 + 0.99*SumRCO3 # 1.19e-11;
<MGN3> MGLY + NO3 = HNO3 + MECO3 + CO + SumRCO3 # 5.00e-16;
<MGHV> MGLY = HO2 + MECO3 + CO + SumRCO3 # 1.0/<MGLY_13>;
<BAHV> BACL = 2*MECO3 + 2*SumRCO3 # 1.0/<BACL_11>;
<CROH> CRES + OH = 0.84*HO2 + 0.11*xHO2 + 0.11*RO2C + 0.02*RO2XC + 0.03*BZO +
0.02*OLEA1 + 0.08*OLEA2 + 0.02*xGLY + 0.05*xMGLY + 0.03*xBACL +
0.03*xBUDAL + 0.01*xAFG1 + 0.05*xAFG2A + 0.02*xAFG2B + 0.01*xBALD +
0.17*LVKS + 0.14*OLEP + 0.42*CATL + 0.02*zRANO3 + 0.02*yROOH +
0.12*yRAOOH + 0.13*SumRO2 # 4.65e-11;
<CRN3> CRES + NO3 = HNO3 + BZO # 1.27e-11;
<XLOH> XYNL + OH = 0.78*HO2 + 0.16*xHO2 + 0.17*RO2C + 0.04*RO2XC + 0.02*BZO +
0.02*OLEA1 + 0.06*OLEA2 + 0.01*xGLY + 0.04*xMGLY + 0.1*xBACL +
0.01*xBUDAL + 0.01*xAFG1 + 0.09*xAFG2A + 0.03*xAFG2B + 0.01*xBALD +
0.26*LVKS + 0.01*xAFG3 + 0.16*OLEP + 0.01*XYNL + 0.27*CATL +
0.03*zRANO3 + 0.02*yROOH + 0.18*yRAOOH + 0.2*SumRO2 # 6.73e-11;
<XLN3> XYNL + NO3 = HNO3 + BZO # 3.09e-11;
<CAOH> CATL + OH = 0.96*HO2 + 0.03*xHO2 + 0.03*RO2C + 0.01*RO2XC + 0.01*BZO +
0.01*OLEA1 + 0.02*OLEA2 + 0.02*xBACL + 0.02*xAFG2A + 0.73*LVKS +
0.06*OLEP + 0.12*CATL3 + 0.01*zRANO3 + 0.03*yRAOOH + 0.03*SumRO2
# 1.56e-10;

```

```

<CAN3> CATL + NO3 = HNO3 + BZO # 4.04e-11;
<N4OH> RCNO3 + OH = RCNO3_OH + 0.34*NO2 + 0.1*xNO2 + 0.02*OH + 0.01*xOH +
    0.05*HO2 + 0.13*xHO2 + 0.49*RO2C + 0.07*RO2XC + 0.05*xMECO3 +
    0.07*R2CO3 + 0.08*xR2CO3 + 0.03*MACO3 + 0.13*xHCHO + 0.05*xMECHO +
    0.05*RCHO + 0.05*xRCHO + 0.01*xGLY + 0.05*MGLY + 0.02*xACET +
    0.12*KET2 + 0.01*xKET2 + 0.01*PACID + 0.04*xPACID + 0.04*OLEP +
    0.07*RCNO3 + 0.21*xRCNO3 + 0.07*zRCNO3 + 0.03*yRPNO3 + 0.03*yHPCRB +
    0.06*ALK2 + 4.74*NROG + 0.01*CO2 + 0.01*CO + 0.56*SumRO2 +
    0.09*SumRCO3 # 2.06e-11;
<708> RCNO3_OH = 0.05*HO2 + 0.01*xPACID + 0.05*RCNO3 # 2.76e+00;
<709> RCNO3_OH + NO = NO + 0.01*xHO2 + 0.07*RO2C + 0.02*RO2XC + 0.03*xR2CO3 +
    0.03*xHCHO + 0.01*xGLY + 0.04*xRCNO3 + 0.02*zRCNO3 + 0.01*yRPNO3 +
    0.02*yHPCRB + 0.09*SumRO2 # 2.55e-12@-380;
<N4HV> RCNO3_HV = RCNO3_HV + 0.77*NO2 + 0.04*xNO2 + 0.01*xOH + 0.38*HO2 +
    0.19*xHO2 + 0.53*RO2C + 0.1*RO2XC + 0.07*ETO2 + 0.11*MECO3 +
    0.04*R2CO3 + 0.03*xR2CO3 + 0.11*HCHO + 0.02*xHCHO + 0.05*MECHO +
    0.02*RCHO + 0.07*xRCHO + 0.09*OLEA2 + 0.03*MACR + 0.04*AFG2A +
    0.07*xACET + 0.04*xKET2 + 0.04*OACID + 0.03*xOACID + 0.02*PACID +
    0.04*xPACID + 0.02*xRCNO3 + 0.08*zRCNO3 + 0.03*xRHNO3 + 0.17*yRPNO3 +
    0.02*zRDNO3 + 0.22*yHPCRB + 8.77*NROG + 0.01*CO2 + 0.34*CO +
    0.71*SumRO2 + 0.14*SumRCO3 # 1.0/<CRBNIT>;
<711> RCNO3_HV = 0.25*HO2 + 0.06*PACID + 0.02*xPACID + 0.06*RCNO3 + 0.02*RPNO3 +
    0.1*HPCRB # 3.15e-01;
<712> RCNO3_HV + NO = NO + 0.05*xOH + 0.06*xHO2 + 0.21*RO2C + 0.04*RO2XC +
    0.05*xMECO3 + 0.05*xR2CO3 + 0.01*xHCHO + 0.14*xRCHO + 0.01*xMGLY +
    0.04*zRCNO3 + 0.07*xRHNO3 + 0.02*yRPNO3 + 0.14*yHPCRB + 0.05*CO2 +
    0.02*CO + 0.25*SumRO2 # 2.55e-12@-380;
<N3OH> RHNO3 + OH = 0.6*NO2 + 0.08*HO2 + 0.26*xHO2 + 0.3*RO2C + 0.05*RO2XC +
    0.07*xHCHO + 0.03*xGLCHO + 0.01*KET2 + 0.07*xKET2 + 0.01*LVKS +
    0.08*RCNO3 + 0.21*xRCNO3 + 0.05*xRHNO3 + 0.34*yRPNO3 + 0.05*zRDNO3 +
    0.02*ALK4 + 0.56*ALK5 + 0.35*SumRO2 # 3.80e-11;
<N3HV> RHNO3 = NO2 + 0.94*HO2 + 0.03*xHO2 + 0.09*RO2C + 0.02*RO2XC + 0.74*HCHO +
    0.02*xHCHO + 0.02*MECHO + 0.06*OLEA1 + 0.06*OLEA2 + 0.02*xOLEA2 +
    0.32*MACR + 0.02*ACET + 0.03*KET2 + 0.37*MVK + 0.02*OLEP +
    0.01*zRCNO3 + 0.01*zRHNO3 + 0.05*yRUOOH + 0.01*HPCRB + 0.05*yHPCRB +
    0.07*FURNS + 0.11*SumRO2 # 1.0/<IC3ONO2>;
<N5OH> RANO3 + OH = 0.42*NO2 + 0.58*HO2 + 0.02*AFG2A + 0.01*OLEP + 0.47*RCNO3 +
    0.11*xRHNO3 + 0.01*ALK5 + 0.37*ALK6 # 4.49e-11;
<N5HV> RANO3 = 0.66*RCNO3 + 0.34*xRHNO3 # 1.0/<COOH>;
<N6OH> RPNO3 + OH = RPNO3_OH + 0.01*zRNNNO3 + 0.37*NO2 + 0.07*xNO2 + 0.19*OH +
    0.12*HO2 + 0.11*xHO2 + 0.23*RO2C + 0.05*RO2XC + 0.1*xHCHO +
    0.04*xOLEA1 + 0.01*xOLEA2 + 0.03*xKET2 + 0.02*xMVK + 0.03*RCNO3 +
    0.1*xRCNO3 + 0.16*xRHNO3 + 0.12*RPNO3 + 0.01*xRPNO3 + 0.14*yRPNO3 +
    0.03*zRDNO3 + 0.32*ROOH + 0.05*HPCRB + 0.01*xHPCRB + 0.28*SumRO2
    # 5.19e-11;
<718> RPNO3_OH = 0.03*NO2 + 0.05*OH + 0.02*HO2 + 0.05*RCNO3 + 0.01*RPNO3 +
    0.03*HPCRB # 4.77e-01;
<719> RPNO3_OH + NO = NO + 0.03*xNO2 + 0.05*xHO2 + 0.09*RO2C + 0.01*RO2XC +
    0.02*xHCHO + 0.03*xGLY + 0.02*xOLEP + 0.03*xRCNO3 + 0.02*xRPNO3 +
    0.1*yRPNO3 + 0.01*zRDNO3 + 0.02*xHPCRB + 0.11*SumRO2 # 2.55e-12@-380;
<N6HV> RPNO3 = RPNO3_HV + 0.81*NO2 + OH + 0.14*HO2 + 0.01*RO2C + 0.63*HCHO +
    0.43*OLEA1 + 0.1*OLEA2 + 0.12*MVK + 0.15*OLEP + 0.14*RCNO3 +
    0.01*yRPNO3 + 0.01*SumRO2 # 1.0/<COOH>;
<721> RPNO3_HV = 0.03*NO2 + 0.01*HO2 + 0.01*RPNO3 + 0.03*HPCRB # 1.11e+00;
<722> RPNO3_HV + NO = NO + 0.04*xHO2 + 0.05*RO2C + 0.01*RO2XC + 0.03*xHCHO +
    0.03*xRCNO3 + 0.01*xRHNO3 + 0.05*yRPNO3 + 0.05*SumRO2 # 2.55e-12@-380;
<NDOH> RDNO3 + OH = 0.53*NO2 + 0.04*xNO2 + 0.01*HO2 + 0.25*xHO2 + 0.6*RO2C +
    0.15*RO2XC + 0.2*xHCHO + 0.03*xACET + 0.1*xKET2 + 0.02*RCNO3 +

```

$0.25 \times \text{RCNO}_3 + 0.01 \times \text{zRCNO}_3 + 0.52 \times \text{RHNO}_3 + 0.05 \times \text{xRDNO}_3 + 0.13 \times \text{zRDNO}_3 + 0.01 \times \text{CO} + 0.75 \times \text{SumRO}_2 \# 3.60\text{e-}11;$
 <NDHV> $\text{RDNO}_3 = \text{RDNO}_3_{\text{HV}} + 1.86 \times \text{NO}_2 + 0.03 \times \text{HO}_2 + 0.01 \times \text{xHO}_2 + 0.08 \times \text{RO}_2\text{C} + 0.03 \times \text{RO}_2\text{XC} + 0.01 \times \text{xR}_2\text{CO}_3 + 0.3 \times \text{HCHO} + 0.21 \times \text{OLEA}_1 + 0.35 \times \text{OLEA}_2 + 0.02 \times \text{xACET} + 0.05 \times \text{KET}_2 + 0.25 \times \text{MVK} + 0.03 \times \text{RCNO}_3 + 0.01 \times \text{xRCNO}_3 + 0.03 \times \text{zRCNO}_3 + 0.11 \times \text{SumRO}_2 \# 1.0 / <\text{DIONO}_2>;$
 <725> $\text{RDNO}_3_{\text{HV}} = 0.04 \times \text{NO}_2 + 0.02 \times \text{HO}_2 + 0.02 \times \text{RPNO}_3 + 0.04 \times \text{HPCRB} \# 1.09\text{e+}00;$
 <726> $\text{RDNO}_3_{\text{HV}} + \text{NO} = \text{NO} + 0.06 \times \text{xHO}_2 + 0.06 \times \text{RO}_2\text{C} + 0.01 \times \text{RO}_2\text{XC} + 0.04 \times \text{xHCHO} + 0.04 \times \text{xRCNO}_3 + 0.02 \times \text{xRHNO}_3 + 0.06 \times \text{yRPNO}_3 + 0.01 \times \text{zRDNO}_3 + 0.07 \times \text{SumRO}_2 \# 2.55\text{e-}12 @ -380;$
 <N1OH> $\text{R1NO}_3 + \text{OH} = 0.17 \times \text{NO}_2 + 0.41 \times \text{xNO}_2 + 0.01 \times \text{HO}_2 + 0.29 \times \text{xHO}_2 + 0.99 \times \text{RO}_2\text{C} + 0.12 \times \text{RO}_2\text{XC} + 0.1 \times \text{xHCHO} + 0.29 \times \text{xMECHO} + 0.05 \times \text{xETCHO} + 0.02 \times \text{xRCHO} + 0.05 \times \text{ACET} + 0.25 \times \text{xACET} + 0.05 \times \text{MEK} + 0.04 \times \text{xMEK} + 0.07 \times \text{KET}_2 + 0.05 \times \text{xKET}_2 + 0.19 \times \text{xRCNO}_3 + 0.1 \times \text{xRHNO}_3 + 0.01 \times \text{RPNO}_3 + 1.12 \times \text{yRPNO}_3 + 0.12 \times \text{zRDNO}_3 + 1.12 \times \text{SumRO}_2 \# 1.47\text{e-}12;$
 <N1HV> $\text{R1NO}_3 = \text{NO}_2 + 0.19 \times \text{HO}_2 + 0.38 \times \text{xHO}_2 + 0.45 \times \text{RO}_2\text{C} + 0.06 \times \text{RO}_2\text{XC} + 0.32 \times \text{ETO}_2 + 0.04 \times \text{TBUEO} + 0.01 \times \text{xTBUEO} + 0.05 \times \text{xHCHO} + 0.12 \times \text{MECHO} + 0.03 \times \text{ETCHO} + 0.03 \times \text{xETCHO} + 0.02 \times \text{xRCHO} + 0.34 \times \text{ACET} + 0.12 \times \text{xACET} + 0.11 \times \text{MEK} + 0.05 \times \text{KET}_2 + 0.22 \times \text{xKET}_2 + 0.01 \times \text{zR1NO}_3 + 0.06 \times \text{zRHNO}_3 + 0.51 \times \text{yROOH} + 0.83 \times \text{SumRO}_2 \# 1.0 / <\text{IC3ONO}_2>;$
 <N2OH> $\text{R2NO}_3 + \text{OH} = 0.06 \times \text{NO}_2 + 0.12 \times \text{xNO}_2 + 0.01 \times \text{HO}_2 + 0.52 \times \text{xHO}_2 + 1.12 \times \text{RO}_2\text{C} + 0.29 \times \text{RO}_2\text{XC} + 0.06 \times \text{xRCHO} + 0.04 \times \text{KET}_2 + 0.07 \times \text{xKET}_2 + 0.01 \times \text{RCNO}_3 + 0.49 \times \text{xRCNO}_3 + 0.01 \times \text{zRCNO}_3 + 0.03 \times \text{xRHNO}_3 + 1.36 \times \text{yRPNO}_3 + 0.29 \times \text{zRDNO}_3 + 0.01 \times \text{HPCRB} + 0.01 \times \text{CO} + 1.42 \times \text{SumRO}_2 \# 2.51\text{e-}11;$
 <N2HV> $\text{R2NO}_3 = \text{R2NO}_3_{\text{HV}} + \text{NO}_2 + 0.1 \times \text{HO}_2 + 0.46 \times \text{xHO}_2 + 0.74 \times \text{RO}_2\text{C} + 0.21 \times \text{RO}_2\text{XC} + 0.05 \times \text{xMACO}_3 + 0.04 \times \text{xRCHO} + 0.08 \times \text{KET}_2 + 0.37 \times \text{xKET}_2 + 0.04 \times \text{xMVK} + 0.03 \times \text{xPACID} + 0.07 \times \text{zRCNO}_3 + 0.14 \times \text{zRHNO}_3 + 0.51 \times \text{yROOH} + 0.02 \times \text{HPCRB} + 0.36 \times \text{yHPCRB} + 0.01 \times \text{CO} + 0.95 \times \text{SumRO}_2 \# 1.0 / <\text{IC3ONO}_2>;$
 <731> $\text{R2NO}_3_{\text{HV}} = 0.09 \times \text{OH} + 0.09 \times \text{HO}_2 + 0.07 \times \text{PACID} + 0.03 \times \text{xPACID} + 0.1 \times \text{HPCRB} \# 9.69\text{e-}01;$
 <732> $\text{R2NO}_3_{\text{HV}} + \text{NO} = \text{NO} + 0.13 \times \text{xHO}_2 + 0.22 \times \text{RO}_2\text{C} + 0.05 \times \text{RO}_2\text{XC} + 0.03 \times \text{xHCHO} + 0.11 \times \text{xRCHO} + 0.04 \times \text{xOLEA}_2 + 0.03 \times \text{zRCNO}_3 + 0.02 \times \text{zRHNO}_3 + 0.29 \times \text{yHPCRB} + 0.01 \times \text{CO} + 0.27 \times \text{SumRO}_2 \# 2.55\text{e-}12 @ -380;$
 <H4OH> $\text{RAOOH} + \text{OH} = 0.76 \times \text{OH} + 0.17 \times \text{HO}_2 + 0.06 \times \text{xHO}_2 + 0.06 \times \text{RO}_2\text{C} + 0.01 \times \text{RO}_2\text{XC} + 0.31 \times \text{RCHO} + 0.02 \times \text{xGLY} + 0.04 \times \text{xMGLY} + 0.02 \times \text{xBUDAL} + 0.02 \times \text{AFG}_2\text{A} + 0.03 \times \text{xAFG}_2\text{A} + 0.01 \times \text{KET}_2 + 0.03 \times \text{OLEP} + 0.01 \times \text{zRANO}_3 + 0.07 \times \text{yRAOOH} + 0.17 \times \text{HPCRB} + 0.01 \times \text{ALK}_5 + 0.37 \times \text{ALK}_6 + 0.07 \times \text{SumRO}_2 \# 8.27\text{e-}11;$
 <H4HV> $\text{RAOOH} = \text{HPCRB} \# 1.0 / <\text{COOH}>;$
 <H3OH> $\text{RUOOH} + \text{OH} = \text{RUOOH}_{\text{OH}} + 0.63 \times \text{OH} + 0.08 \times \text{HO}_2 + 0.1 \times \text{xHO}_2 + 0.11 \times \text{RO}_2\text{C} + 0.01 \times \text{RO}_2\text{XC} + 0.08 \times \text{xHCHO} + 0.03 \times \text{xMACR} + 0.05 \times \text{xMVK} + 0.02 \times \text{LVKS} + 0.01 \times \text{zRHNO}_3 + 0.01 \times \text{yROOH} + 0.1 \times \text{yRUOOH} + 0.08 \times \text{HPCRB} + 0.01 \times \text{xHPCRB} + 0.02 \times \text{ALK}_4 + 0.59 \times \text{ALK}_5 + 0.01 \times \text{xFURNS} + 0.12 \times \text{SumRO}_2 \# 5.97\text{e-}11;$
 <736> $\text{RUOOH}_{\text{OH}} = 0.06 \times \text{OH} + 0.12 \times \text{HO}_2 + 0.17 \times \text{HPCRB} \# 1.28\text{e-}02;$
 <737> $\text{RUOOH}_{\text{OH}} + \text{NO} = \text{NO} + 0.16 \times \text{xHO}_2 + 0.16 \times \text{RO}_2\text{C} + 0.01 \times \text{RO}_2\text{XC} + 0.03 \times \text{xHCHO} + 0.03 \times \text{xGLCHO} + 0.05 \times \text{xKET}_2 + 0.01 \times \text{zRPNO}_3 + 0.17 \times \text{yROOH} + 0.16 \times \text{xHPCRB} + 0.17 \times \text{SumRO}_2 \# 2.55\text{e-}12 @ -380;$
 <H3HV> $\text{RUOOH} = \text{OH} + 0.99 \times \text{HO}_2 + 0.02 \times \text{RO}_2\text{C} + 0.86 \times \text{HCHO} + 0.04 \times \text{OLEA}_1 + 0.39 \times \text{MACR} + 0.01 \times \text{KET}_2 + 0.46 \times \text{MVK} + 0.01 \times \text{OLEP} + 0.01 \times \text{yRUOOH} + 0.01 \times \text{yHPCRB} + 0.09 \times \text{FURNS} + 0.02 \times \text{SumRO}_2 \# 1.0 / <\text{COOH}>;$
 <H5OH> $\text{HPCRB} + \text{OH} = \text{HPCRB}_{\text{OH}} + 0.5 \times \text{OH} + 0.01 \times \text{HO}_2 + 0.03 \times \text{xHO}_2 + 0.03 \times \text{RO}_2\text{C} + 0.03 \times \text{xHCHO} + 0.35 \times \text{RCHO} + 0.01 \times \text{OLEA}_1 + 0.02 \times \text{AFG}_1 + 0.03 \times \text{xPACID} + 0.13 \times \text{OLEP} + 0.01 \times \text{HPCRB} + 0.03 \times \text{SumRO}_2 \# 5.40\text{e-}11;$
 <740> $\text{HPCRB}_{\text{OH}} = 0.44 \times \text{OH} + 0.02 \times \text{HO}_2 + 0.01 \times \text{AFG}_2\text{B} + 0.01 \times \text{PACID} + 0.08 \times \text{OLEP} + 0.36 \times \text{HPCRB} \# 6.37\text{e-}01;$
 <741> $\text{HPCRB}_{\text{OH}} + \text{NO} = \text{NO} + 0.42 \times \text{xHO}_2 + 0.43 \times \text{RO}_2\text{C} + 0.04 \times \text{RO}_2\text{XC} + 0.01 \times \text{xHCHO} + 0.05 \times \text{xGLY} + 0.01 \times \text{xGLCHO} + 0.24 \times \text{xMGLY} + 0.05 \times \text{xAFG}_2\text{B} + 0.01 \times \text{xPACID} + 0.01 \times \text{zRCNO}_3 + 0.03 \times \text{zRPNO}_3 + 0.34 \times \text{xHPCRB} + 0.45 \times \text{yHPCRB} + 0.04 \times \text{CO} + 0.47 \times \text{SumRO}_2 \# 2.55\text{e-}12 @ -380;$

<H5HV> HPCRB = OH + 0.01*xOH + 0.9*HO2 + 0.08*xHO2 + 0.09*RO2C + 0.01*RO2XC + 0.11*HCHO + 0.06*xHCHO + 0.11*OLEA1 + 0.05*AFG1 + 0.21*AFG2A + 0.53*AFG2B + 0.02*xAFG2B + 0.01*xLVKS + 0.06*xPACID + 0.01*zRCNO3 + 0.01*CO2 + 0.1*SumRO2 # 1.00e-1/<HPALDS>;

<H2OH> ROOH_OH + OH = ROOH_OH + 0.19*OH + 0.03*xOH + 0.22*HO2 + 0.37*xHO2 + 0.56*RO2C + 0.04*RO2XC + 0.07*xETO2 + 0.02*xTBUO + 0.01*HCHO + 0.35*xHCHO + 0.11*xMECHO + 0.01*xETCHO + 0.01*RCHO + 0.01*xRCHO + 0.06*GLCHO + 0.04*xGLCHO + 0.02*ACET + 0.15*xACET + 0.02*MEK + 0.02*xMEK + 0.08*KET2 + 0.04*xKET2 + 0.02*zR1NO3 + 0.01*zRHNO3 + 0.01*zRPNO3 + 0.59*yROOH + 0.22*HPCRB + 0.01*yHPCRB + 0.6*SumRO2 # 1.16e-11;

<744> ROOH_OH = 0.04*OH + 0.01*HO2 + 0.05*HPCRB # 6.78e-02;

<745> ROOH_OH + NO = NO + 0.03*xOH + 0.01*xHO2 + 0.05*RO2C + 0.01*RO2XC + 0.01*xHCHO + 0.03*xMECHO + 0.01*xETCHO + 0.01*xRCHO + 0.01*xACET + 0.01*zRPNO3 + 0.06*yROOH + 0.01*xHPCRB + 0.06*SumRO2 # 2.55e-12@-380;

<H2HV> ROOH = OH + 0.74*HO2 + 0.09*xHO2 + 0.13*RO2C + 0.02*RO2XC + 0.12*ETO2 + 0.02*TBUO + 0.77*HCHO + 0.25*MECHO + 0.01*ETCHO + 0.01*xETCHO + 0.01*xRCHO + 0.1*GLCHO + 0.18*ACET + 0.03*xACET + 0.04*MEK + 0.06*KET2 + 0.05*xKET2 + 0.01*zRCNO3 + 0.01*zRHNO3 + 0.1*yROOH + 0.03*yHPCRB + 0.27*SumRO2 # 1.0/<COOH>;

<F2OH> AFG1 + OH = AFG1_OH + 0.43*OH + 0.02*HO2 + 0.19*xHO2 + 0.19*RO2C + 0.02*RO2XC + 0.18*xGLY + 0.41*MALAH + 0.03*xPACID + 0.03*HPCRB + 0.01*CO + 0.21*SumRO2 # 3.39e-11;

<748> AFG1_OH = 0.02*OH + 0.33*HO2 + 0.02*PACID + 0.16*xPACID + 0.02*zRCNO3 + 0.33*HPCRB # 1.31e+01;

<749> AFG1_OH + NO = NO + 0.3*xHO2 + 0.32*RO2C + 0.03*RO2XC + 0.3*xGLY + 0.46*xMGLY + 0.02*MALAH + 0.43*yHPCRB + 0.35*SumRO2 # 2.55e-12@-380;

<F2HV> AFG1 = AFG1_HV + 0.3*OH + 0.94*HO2 + 0.03*xHO2 + 0.03*RO2C + 0.06*MEO2 + 0.01*AFG1 + 0.02*AFG2A + 0.26*PACID + 0.03*xPACID + 0.02*HPCRB + 0.03*CO + 0.09*SumRO2 # 2.20e-1/<AFGS>;

<751> AFG1_HV = 0.67*OH + 0.33*PACID + 0.33*HPCRB # 1.75e+01;

<752> AFG1_HV + NO = NO + 0.52*xOH + 0.07*xHO2 + 0.59*RO2C + 0.08*RO2XC + 0.52*MALAH + 0.08*zRCNO3 + 0.67*SumRO2 # 2.55e-12@-380;

<F3OH> AFG2A + OH = AFG2A_OH + 0.57*xHO2 + 0.73*RO2C + 0.07*RO2XC + 0.15*xMECO3 + 0.01*xR2CO3 + 0.2*MACO3 + 0.03*xRCHO + 0.47*xGLY + 0.54*xMGLY + 0.02*xPACID + 0.07*zRCNO3 + 0.46*yHPCRB + 0.03*CO + 0.8*SumRO2 + 0.2*SumRCO3 # 5.99e-11;

<754> AFG2A_OH = 0.22*xPACID # 2.34e+01;

<755> AFG2A_OH + NO = NO + 0.14*xRCHO + 0.06*xGLY + 0.2*yHPCRB # 2.55e-12@-380;

<F3HV> AFG2A = OH + 0.91*MEO2 + 0.09*ETO2 + MALAH + SumRO2 # 2.50e-1/<AFGS>;

<F4OH> AFG2B + OH = AFG2B_OH + 0.02*HO2 + 0.4*xHO2 + 0.4*RO2C + 0.05*RO2XC + 0.17*MACO3 + 0.39*xGLY + 0.39*xBACL + 0.05*zRCNO3 + 0.02*HPCRB + 0.38*yHPCRB + 0.45*SumRO2 + 0.17*SumRCO3 # 4.58e-11;

<758> AFG2B_OH = 0.36*HO2 + 0.36*HPCRB # 1.16e+01;

<759> AFG2B_OH + NO = NO + 0.2*xHO2 + 0.32*RO2C + 0.04*RO2XC + 0.12*xMECO3 + 0.13*xRCHO + 0.2*xGLY + 0.2*xBACL + 0.04*zRCNO3 + 0.31*yHPCRB + 0.36*SumRO2 # 2.55e-12@-380;

<F4HV> AFG2B = AFG2B_HV + 0.88*OH + 0.03*xHO2 + 0.03*RO2C + MEO2 + 0.87*MALAH + 0.03*xPACID + 0.03*CO + 1.03*SumRO2 # 2.20e-1/<AFGS>;

<761> AFG2B_HV = 0.09*OH + 0.05*PACID + 0.05*HPCRB # 8.62e+00;

<762> AFG2B_HV + NO = NO + 0.08*RO2C + 0.08*MALAH + 0.09*SumRO2 # 2.55e-12@-380;

<F5OH> AFG3 + OH = 0.27*xHO2 + 0.89*RO2C + 0.11*RO2XC + 0.62*xMECO3 + 0.62*xRCHO + 0.54*xMGLY + 0.11*zRCNO3 + 0.85*yHPCRB + SumRO2 # 7.20e-11;

<P2UI> PAN2 = NO2 + R2CO3 + SumRCO3 # 3.39e-04;

<P2OH> PAN2 + OH = 0.07*zPAN2 + 0.05*NO3 + 0.73*xNO3 + 1.15*xHO2 + 1.55*RO2C + 0.07*RO2XC + 0.17*xHCHO + 0.66*xMECHO + 0.07*xETCHO + 0.15*xPAN2 +

```

0.05*ALK3 + 0.73*CO2 + 0.5*CO + 1.62*SumRO2 # 3.42e-12;
<P2HV> PAN2 = NO2 + R2CO3 + SumRCO3 # 1.0/<PPN_11>;
<P4UI> APANS = NO2 + MACO3 + SumRCO3 # 3.39e-04;
<P4OH> APANS + OH = 0.01*zPAN2 + 0.74*NO3 + 0.18*xNO3 + 0.07*xHO2 + 0.25*RO2C +
0.01*RO2XC + 0.07*xHCHO + 0.18*xKET2 + 0.19*OACID + 0.07*xPAN2 +
0.56*ALK4 + 0.18*CO2 + 0.26*SumRO2 # 2.90e-11;
<P4O3> APANS + O3 = 0.16*H2 + 0.05*NO2 + 0.19*OH + 0.25*HO2 + 0.09*RO2C +
0.09*xR2CO3 + 0.38*HCHO2 + 0.01*RCHO2 + 0.1*HCHO + 0.04*xHCHO +
0.9*PAN2 + 0.21*CO2 + 0.31*CO + 0.09*SumRO2 # 8.20e-18;
<P4N3> APANS + NO3 = 0.05*zPAN2 + 0.95*xNO3 + 0.95*RO2C + 0.05*RO2XC +
0.95*xRCNO3 + 0.95*CO2 + SumRO2 # 1.60e-16;
<P4HV> APANS = 0.6*NO2 + 0.4*NO3 + 0.4*MEO2 + 0.6*MACO3 + 0.4*HCHO + 0.4*CO2 +
0.4*CO + 0.4*SumRO2 + 0.6*SumRCO3 # 1.0/<PPN_11>;

```

! Reactions for aerosol precursors from alkanes, benzene, xylenes, and toluene

```

<AALK> SOAALK + OH = OH + 0.006*SVAVB2 + 0.052*SVAVB3 +
0.081*SVAVB4 # 2.70e-12@-374;
<AE51> BENZRO2 + NO = NO + 0.034*SVAVB2 + 0.392*SVAVB4 # 1.0*K<R2NO>;
<AE52> BENZRO2 + HO2 = HO2 + 0.146*SVAVB1 # 1.0*K<R2H2>;
<AE53> XYLRO2 + NO = NO + 0.015*SVAVB2 + 0.023*SVAVB3 +
0.06*SVAVB4 # 1.0*K<R2NO>;
<AE54> XYLRO2 + HO2 = HO2 + 0.193*SVAVB1 # 1.0*K<R2H2>;
<AE55> TOLRO2 + NO = NO + 0.016*SVAVB2 + 0.051*SVAVB3 +
0.047*SVAVB4 # 1.0*K<R2NO>;
<AE56> TOLRO2 + HO2 = HO2 + 0.14*SVAVB1 # 1.0*K<R2H2>;
<AE57> PAHRO2 + NO = NO + 0.028*SVAVB2 + 0.225*SVAVB3 +
0.191*SVAVB4 # 1.0*K<R2NO>;
<AE58> PAHRO2 + HO2 = HO2 + 0.473*SVAVB1 # 1.0*K<R2H2>;

```

! iorganic heterogeneou reactions from hetchem version 5.0.2

```

<HET_NO2> NO2 = 0.5*HONO + 0.5*HNO3 # 1.0~<HETERO_NO2>;

```

! version 5.1 heterogenous N2O5 reactions leading to HNO3

```

<HET_N2O5IJ> N2O5 = HNO3 + H2NO3PIJ # 1.0~<HETERO_N2O5IJ>;
<HET_N2O5K> N2O5 = HNO3 + H2NO3PK # 1.0~<HETERO_N2O5K>;

<HET_H2NO3PIJA> H2NO3PIJ = HNO3 # 1.0~<HETERO_H2NO3PAIJ>;
<HET_H2NO3PKA> H2NO3PK = HNO3 # 1.0~<HETERO_H2NO3PAK>;

```

! Simplified ozone loss from oceanic halogens excluding chlorine

```

<HAL_Ozone> O3 = %H # 6.7006E-11@-10.7435 & 3.4153E-08@0.6713
& 2.0E-6;

```

! Uptake of NO3 (Mao et al. 2013 JGR)

```

<HET_NO3> NO3 = HNO3 # 1.0~<HETERO_NO3>;

```

! oligomerization reaction for secondary organic aerosols

```

<OLIG_ISOPRENE1> AISO1J = 0.50*AOLGBJ # 9.48816E-6;
<OLIG_ISOPRENE2> AISO2J = 0.50*AOLGBJ # 9.48816E-6;
<OLIG_SESQT1> ASQTJ = 1.50*AOLGBJ # 9.48816E-6;

```

! oligomerization reaction for aromatic secondary organic aerosols

```

<OLIG_AROMATIC1> AAVB2J = 0.907*AOLGAJ # 9.48816E-6;
<OLIG_AROMATIC2> AAVB3J = 0.925*AOLGAJ # 9.48816E-6;

```


<OLIG_AROMATIC3> AAVB4J = 0.943*AOLGAJ # 9.48816E-6;

!aging reactions for primary organic carbon

<RPOAGEPI> APOCI + OH = 1.25*APNCOMI + APOCI + OH # 2.5E-12;
 <RPOAGELI> APNCOMI + OH = OH # 1.0~<HETERO_PNCOMLI>;
 <RPOAGEPJ> APOCJ + OH = 1.25*APNCOMJ + APOCJ + OH # 2.5E-12;
 <RPOAGELJ> APNCOMJ + OH = OH # 1.0~<HETERO_PNCOMLJ>;

! Formation of Potential Combustion SOA (pcSOA)

<PCSOA> PCVOC + OH = OH + PCSOARXN # 1.25E-11;

! Aging with Func. and Frag. (Donahue et al. 2012)

<POA_AGE1> VLVPO1 + OH = OH
 + 0.4857 * VLVPO1 + 0.0062 * VSVPO1
 + 0.0025 * VSVPO2 + 0.0026 * VSVPO3
 + 0.0023 * VIVPO1 + 0.2944 * VLVOO1
 + 0.2021 * VLVOO2 + 0.0019 * VSVOO2
 + 0.0023 * VSVOO3 #4.0e-11;
 <POA_AGE2> VSVPO1 + OH = OH
 + 0.3003 * VLVPO1 + 0.2862 * VSVPO1
 + 0.0041 * VSVPO2 + 0.0035 * VSVPO3
 + 0.2239 * VLVOO1 + 0.1820 * VLVOO2 #4.0e-11;
 <POA_AGE3> VSVPO2 + OH = OH
 + 0.3856 * VLVPO1 + 0.0950 * VSVPO1
 + 0.1373 * VSVPO2 + 0.0005 * VSVPO3
 + 0.2051 * VLVOO1 + 0.1764 * VLVOO2 #4.0e-11;
 <POA_AGE4> VSVPO3 + OH = OH
 + 0.2181 * VLVPO1 + 0.3063 * VSVPO1
 + 0.0153 * VSVPO2 + 0.1043 * VSVPO3
 + 0.1893 * VLVOO1 + 0.1668 * VLVOO2 #4.0e-11;
 <POA_AGE5> VIVPO1 + OH = OH
 + 0.2412 * VLVPO1 + 0.2089 * VSVPO1
 + 0.3000 * VSVPO2 + 0.2028 * VLVOO1
 + 0.0471 * VLVOO2 #4.0e-11;
 <POA_AGE6> VLVOO1 + OH = OH
 + 0.6664 * VLVOO1 + 0.0143 * VLVOO2
 + 0.0123 * VSVOO1 + 0.1239 * VSVOO2
 + 0.1831 * VSVOO3 #4.0e-11;
 <POA_AGE7> VLVOO2 + OH = OH
 + 0.2858 * VLVOO1 + 0.3931 * VLVOO2
 + 0.0139 * VSVOO1 + 0.1027 * VSVOO2
 + 0.2045 * VSVOO3 #4.0e-11;
 <POA_AGE8> VSVOO1 + OH = OH
 + 0.3303 * VLVOO1 + 0.2272 * VLVOO2
 + 0.2607 * VSVOO1 + 0.0702 * VSVOO2
 + 0.1116 * VSVOO3 #4.0e-11;
 <POA_AGE9> VSVOO2 + OH = OH
 + 0.3444 * VLVOO1 + 0.2749 * VLVOO2
 + 0.0491 * VSVOO1 + 0.2577 * VSVOO2
 + 0.0739 * VSVOO3 #4.0e-11;
 <POA_AGE10> VSVOO3 + OH = OH
 + 0.3886 * VLVOO1 + 0.2421 * VLVOO2
 + 0.0640 * VSVOO1 + 0.0385 * VSVOO2
 + 0.2667 * VSVOO3 #4.0e-11;

! GLY and MGLY uptake

<HET_GLY> GLY = AGLYJ # 1.0~<HETERO_GLY>;

<HET_MGLY> MGLY = AGLYJ # 1.0~<HETERO_MGLY>;

! Reactive Tracers for formaldehyde, acetaldehyde and acrolein

<TR01> HCHO_PRIMARY = # 1.0/<HCHOR_13>;

<TR02> HCHO_PRIMARY = # 1.0/<HCHOM_13>;

<TR03> HCHO_PRIMARY + OH = OH # 5.50e-12@-125;

! Removed based on comment from peer reviews

!<TR04> HCHO_PRIMARY + HO2 = HO2 # 9.70e-15@-625;

!

<TR05> HCHO_PRIMARY + NO3 = NO3 # 5.80e-16;

!<TR06> HCHO_PRIMARY + CL = CL # 8.10e-11@30; ! This version does not have chlorine

<TR07> CCHO_PRIMARY + OH = OH # 2.40e-12^0.77@-546;

<TR08> CCHO_PRIMARY = # 1.0/<CCHOR_13>;

<TR09> CCHO_PRIMARY + NO3 = NO3 # 1.40e-12@1860;

!<TR10> CCHO_PRIMARY + CL = CL # 8.00e-11; ! This version does not have chlorine

<TR11> ACRO_PRIMARY + OH = OH # 7.10e-12@-333;

<TR12> ACRO_PRIMARY + O3 = O3 # 2.80e-19;

<TR13> ACRO_PRIMARY + NO3 = NO3 # 1.10e-15;

!<TR14> ACRO_PRIMARY + O3P = O3P # 2.37e-12; ! This version does not have the ACRO + O3P reaction

<TR15> ACRO_PRIMARY = # 1.0/<ACROL_16>;

!<TR16> ACRO_PRIMARY + CL = CL # 2.94e-10; ! This version does not have chlorine

endmech

constants

<C1> ATM_AIR = 1.0E+06

<C2> ATM_H2 = 0.56

<C3> ATM_N2 = 0.7808E+06

<C4> ATM_O2 = 0.2095E+06

<C5> ATM_CH4 = 1.85

end constants

Table A-2. CMAQv5.4 SAPRC-22 Gas-Phase Species

Model Species	Definition	Molecular Weight	Explicit or Lumped
ACET	Acetone	58.08	E
ACETL	Acetylene	26.04	E
ACRLNT	Acrylonitrile	53.06	E
ACRO	Acrolein	56.06	E
ACRO_HV	Intermediate from ACRO reaction	56.06	E
ACRO_OH	Intermediate from ACRO reaction	56.06	E
ACRO_PRIMARY	Acrolein emissions tracer	56.06	E
AFG1	Monounsaturated 1,4-dialdehydes formed from aromatics	99.49	L
AFG1_HV	Intermediate from AFG1 reaction	99.49	L
AFG1_OH	Intermediate from AFG1 reaction	99.49	L
AFG2A	Monounsaturated 1,4 aldehyde-ketones formed from aromatics, with at no substituents other than the aldehyde on the double bonds	101.7	L
AFG2A_OH	Intermediate from AFG2A reaction	101.7	L
AFG2B	Monounsaturated 1,4 aldehyde-ketones formed from aromatics, with at least one substituent other than the aldehyde on a double bond	112.13	L
AFG2B_HV	Intermediate from AFG2B reaction	112.13	L
AFG2B_OH	Intermediate from AFG2B reaction	112.13	L
AFG3	Monounsaturated 1,4-diketone aromatic products	112.13	L
ALK1	Saturated compounds that react only with OH and have kOH between 1.4e-13 and 3.5e-13 cm ³ molec ⁻¹ s ⁻¹	74.08	L
ALK2	Saturated compounds that have kOH between 3.5e-13 and 1.7e12 cm ³ molec ⁻¹ s ⁻¹	88.11	L
ALK3	Saturated compounds that and have kOH between 1.7e-12 and 3.5e-12 cm ³ molec ⁻¹ s ⁻¹	80.54	L
ALK4	Saturated compounds that and have kOH between 3.5e-12 and 6.0e-12 cm ³ molec ⁻¹ s ⁻¹	71.04	L
ALK5	Saturated compounds that and have kOH greater than 6.0e-12 cm ³ molec ⁻¹ s ⁻¹ and have fewer than 12 carbons+oxygens.	96.98	L
ALK5_OH	Intermediate from ALK5 reaction	96.98	L
ALK6	Saturated compounds that and have kOH greater than 6.0e-12 cm ³ molec ⁻¹ s ⁻¹ and have 12 or more carbons+oxygens.	200.16	L
ALK6_OH	Intermediate from ALK6 reaction	200.16	L
AMINS	Amines	52.06	L
APANS	PAN analogues formed from Methacrolein and other unsaturated PAN analogues.	133.06	L
APINE	a-pinene	136.23	E
APINE_OH	Intermediate from APINE reaction	136.23	E

Model Species	Definition	Molecular Weight	Explicit or Lumped
ARO1	Alkylbenzenes that react with KOH between 5e-12 and 1e-11 cm ³ molec ⁻¹ s ⁻¹ . Mostly alkylbenzenes with only one substituent	128.78	L
ARO2	Alkylbenzenes that react with KOH higher than 1e-11 cm ³ molec ⁻¹ s ⁻¹ . Mostly alkylbenzenes with more than one substituent	126.42	L
ARO2_OH	Intermediate from ARO2 reaction	126.42	L
BACL	Biacetyl and other alpha-diketones	86.09	L
BALD	Aromatic aldehydes (e.g., benzaldehyde)	72.06	L
BENX	Benzene and substituted benzenes that react with KOH between 5e-13 and 5e-12 cm ³ molec ⁻¹ s ⁻¹ .	78.11	L
BENZ	Benzene	78.11	E
BENZRO2	Peroxy radical tracer from Benzene and OH reaction used to produce aerosol material	159.11	L
BPINE	β-pinene	136.23	E
BPINE_OH	Intermediate from BPINE reaction	136.23	E
BUDAL	2-Butene-1,4-dial	84.07	E
BUDAL_OH	Intermediate from BUDAL reaction	84.07	E
BUT13	1,3-Butadiene	54.09	E
BUT13_O3	Intermediate from BUT13 reaction	54.09	E
BUT13_OH	Intermediate from BUT13 reaction	54.09	E
BZ123	1,2,3-Trimethyl Benzene	120.19	E
BZ124	1,2,4-Trimethyl Benzene	120.19	E
BZ135	1,3,5-Trimethyl Benzene	120.19	E
BZCO3	Peroxyacyl radical formed from Aromatic Aldehydes	137.11	L
BZO	Phenoxy or substituted phenoxy radicals	93.1	L
BZO2	Benzyl peroxy and substituted benzyl peroxy radicals	125.1	L
C2BEN	Ethyl Benzene	106.17	E
CATL	Dihydroxy Phenols	124.14	L
CATL3	Tri- and multi-hydroxy phenols	140.14	L
CCHO_PRIMARY	Acetaldehyde Emissions Tracer	44.05	E
CHCL3	Chloroform	119.38	E
CLETHE	Vinyl Chloride	62.5	E
CO	Carbon Monoxide	28.01	E
CO2	Carbon Dioxide	44.01	E
CRES	Cresols	108.14	L
ETBR2	Ethylene Dibromide	187.86	E
ETCHO	Propionaldehyde	58.08	E
ETCL2	Ethylene dichloride	98.96	E
ETHAN	Ethane	30.07	E
ETHEN	Ethene	28.05	E
ETHEN_OP	Intermediate from ETHEN reaction	28.05	E

Model Species	Definition	Molecular Weight	Explicit or Lumped
ETO2	Ethyl peroxy radicals	61.06	L
ETOH	Ethanol	46.07	E
ETOX	Ethylene oxide	44.05	E
FURNS	Furans	68.07	L
GLCHO	Glycolaldehyde	60.05	E
GLCHO_HV	Intermediate from GLCHO reaction	60.05	E
GLY	Glyoxal	58.04	E
H2NO3PIJ	Fine Mode Dissolved Nitric Acid	64	E
H2NO3PK	Coarse Mode Dissolved Nitric Acid	64	E
HCHO	Formaldehyde	30.03	E
HCHO_PRIMARY	Formaldehyde Emission Tracer	30.03	E
HCHO2	Unsubstituted stabilized Criegee biradical	46.03	E
HCOOH	Formic Acid	46.03	E
HNO3	Nitric Acid	63.02	E
HNO4	Peroxynitric Acid	79.02	E
HO2	Hydroperoxide Radicals	33.01	E
HO2H	Hydrogen Peroxide	34.01	E
HONO	Nitrous Acid	47.02	E
HPCRB	Unsaturated hydroperoxy carbonyls	128.44	L
HPCRB_OH	Intermediate from HPCRB reaction	128.44	L
IMINE	Any compound with C=N bond	34.07	L
ISOP	Isoprene	68.12	E
ISOP_N3	Intermediate from ISOP reaction	68.12	E
ISOP_O3	Intermediate from ISOP reaction	68.12	E
ISOP_OH	Intermediate from ISOP reaction	68.12	E
ISOPRXN	SOA precursor compounds from isoprene	68	E
KET2	Photoreactive saturated ketones	99.82	L
LVKS	Ketones with at least 1 C=C double bond.	154.23	L
LVKS_O3	Intermediate from LVKS reaction	154.23	L
LVKS_OH	Intermediate from LVKS reaction	154.23	L
MACO3	Peroxyacyl radicals formed from methacrolein and other acroleins.	101.08	L
MACR	Methacrolein	70.09	E
MACR_N3	Intermediate from MACR reaction	70.09	E
MACR_OH	Intermediate from MACR reaction	70.09	E
MALAH	Maleic Anhydride	98.06	E
MECHO	Acetaldehyde	44.05	E
MECHO_OH	Intermediate from MECHO reaction	44.05	E
MECHO2	Methyl substituted stabilized Criegee biradical	60.05	E
MECL2	Methylene Chloride	84.93	E
MECO3	Acetyl Peroxy Radicals	75.04	E
MEK	Methyl Ethyl Ketone	72.11	E

Model Species	Definition	Molecular Weight	Explicit or Lumped
MEO2	Methyl Peroxy Radicals	47.03	E
MEOH	Methanol	32.04	E
MEOOH	Methyl Hydroperoxide	48.04	E
MGLY	Methyl Glyoxal and other alpha-keto aldehydes	72.06	L
MTBE	Methyl t-Butyl Ether	88.15	E
MVK	Methyl Vinyl Ketone	70.09	E
MXYL	m-Xylene	106.17	E
N2O5	Nitrogen Pentoxide	108.02	E
NAMIN	Nitramines	89.14	L
NAPPRD	Phenolic and other products formed from naphthalenes.	172.22	L
NAPS	Naphthalenes, tetralins, and indans	142.2	L
NC4	n-Butane	58.12	E
NO	Nitric Oxide	30.01	E
NO2	Nitrogen Dioxide	46.01	E
NO3	Nitrate Radical	62.01	E
NPHE	Nitrophenols	86.09	L
NPRAD	Simplified representative of unknown naphthalene intermediates that react with NO2 forming products	175.21	L
NROG	Unreactive mass	1	L
O1D	Excited Oxygen Atoms	16	E
O3	Ozone	48	E
O3P	Ground State Oxygen Atoms	16	E
OACID	Organic acids	60.05	L
OH	Hydroxyl Radicals	17.01	E
OLE1	Alkenes other than ethene with kOH between 8e-12 and 6e-11 cm ³ molec ⁻¹ s ⁻¹ .	68.66	L
OLE2	Alkenes other than terpenes or 1,3-butadiene that have OH rate constants greater than 6e-12 cm ³ molec ⁻¹ s ⁻¹ .	64.92	L
OLE2_O3	Intermediate from OLE2 reaction	64.92	L
OLE3	Alkenes other than terpenes or sesquiterpenes with only -CR=CH2 groups.	58.05	L
OLE4	Alkenes other than terpenes with non-conjugated -CH=CR- or -CR=CR- groups and possibly other double bonds and no double bonds in rings	71.51	L
OLE4_O3	Intermediate from OLE4 reaction	71.51	L
OLEA1	Unsaturated aldehydes with C=C next to -CHO (mostly from aromatics)	143.12	L
OLEA1_N3	Intermediate from OLEA1 reaction	143.12	L
OLEA1_OH	Intermediate from OLEA1 reaction	143.12	L
OLEA2	Unsaturated aldehydes with C=C not next to the -CHO (mostly from aromatics)	151.84	L

Model Species	Definition	Molecular Weight	Explicit or Lumped
OLEA2_HV	Intermediate from OLEA2 reaction	151.84	L
OLEA2_N3	Intermediate from OLEA2 reaction	151.84	L
OLEA2_O3	Intermediate from OLEA2 reaction	151.84	L
OLEA2_OH	Intermediate from OLEA2 reaction	151.84	L
OLEP	Non-photoreactive, non-hydrocarbon compounds with C=C double bonds	160.86	L
OLEP_O3	Intermediate from OLEP reaction	160.86	L
OLEP_OH	Intermediate from OLEP reaction	160.86	L
OTHN	Other organic products with vapor pressures less than 1 ppb	240.28	L
OXYL	o-Xylene	106.17	E
PACID	Organic peroxyacids	76.05	L
PAHRO2	Naphthalene hydroxyperoxy radical tracker from naphthalene and OH reaction; an aerosol precursor	187.2	L
PAN	Peroxy Acetyl Nitrate	121.05	E
PAN2	Higher PAN analogues (saturated, no 2nd nitrate)	149.1	L
PBZN	PAN analogues formed from Aromatic Aldehydes	183.13	L
PCE	Perchloroethylene	165.83	E
PCLBEN	p-Dichlorobenzene	147	E
PCSOARXN	Tracer for potential combustion SOA VOCs	170	L
PCVOC	Tracer for potential combustion SOA VOCs	170	L
PHEN	Phenol	94.11	E
PHOT	Unspecified photoreactive compounds such as nitrites or chloropicrin	86.09	L
PNAMIN	Nitramine precursor formed from amines with no alpha hydrogens	72.13	L
PROP	Propane	44.1	E
PROPE	Propene	42.08	E
PROPE_O3	Intermediate from PROPE reaction	42.08	E
PXYL	p-Xylene	106.17	E
R1NO3	Other volatile organic nitrates that react with OH radicals slower than $5 \times 10^{-12} \text{ cm}^3 \text{ molec}^{-1} \text{ s}^{-1}$	146.57	L
R2CO3	Higher saturated peroxy acyl radicals (forms PAN2)	103.1	L
R2NO3	Other volatile organic nitrates that react with OH radicals faster than $5 \times 10^{-12} \text{ cm}^3 \text{ molec}^{-1} \text{ s}^{-1}$	190.37	L
R2NO3_HV	Intermediate from R2NO3 reaction	190.37	L
RANO3	Organic nitrates with peroxy groups (formed primarily from aromatics)	230.33	L
RAOOH	Hydroperoxides with peroxy groups and with C=C double bonds (mainly hydroperoxides formed from reaction with NO with OH-aromatic-O2 adducts after peroxy cyclization)	188.67	L
RCHO	Lumped C3+ Aldehydes (mechanism based on propionaldehyde)	100.45	L

Model Species	Definition	Molecular Weight	Explicit or Lumped
RCHO2	Other stabilized Criegee biradicals	74.08	L
RCNO3	Volatile organic carbonly nitrates	180.91	L
RCNO3_HV	Intermediate from RCNO3 reaction	180.91	L
RCNO3_OH	Intermediate from RCNO3 reaction	180.91	L
RDNO3	Volatile organic dinitrates	249.44	L
RDNO3_HV	Intermediate from RDNO3 reaction	249.44	L
RHNO3	Volatile organic hydroxy nitrates	176.59	L
RNNO3	Organic nitrates with vapor pressure less than 1 ppb	216.23	L
RO2C	Peroxy Radical Operator representing NO to NO2 and NO3 to NO2 conversions, and the effects of peroxy radical reactions on acyl peroxy and other peroxy radicals (used in some multi-step mechanisms).	14.01	L
RO2XC	Peroxy Radical Operator representing NO consumption (used in conjunction with organic nitrate formation), and the effects of peroxy radical reactions on NO3, acyl peroxy radicals, and other peroxy radicals. (used in some multi-step mechanisms)	14.01	L
ROOH	Other hydroperoxides with 3+ carbons and vapor pressure greater than 1 ppb	97	L
ROOH_OH	Intermediate from ROOH reaction	97	L
RPNO3	Organic nitrates with hydroperoxy groups	230.33	L
RPNO3_HV	Intermediate from RPNO3 reaction	230.33	L
RPNO3_OH	Intermediate from RPNO3 reaction	230.33	L
RUOOH	Hydroperoxides with C=C double bonds	145.11	L
RUOOH_OH	Intermediate from RUOOH reaction	145.11	L
SESQ	Sesquiterpenes	204.35	L
SESQ_N3	Intermediate from SESQ reaction	204.35	L
SESQ_O3	Intermediate from SESQ reaction	204.35	L
SESQ_OH	Intermediate from SESQ reaction	204.35	L
SESQRXN	Precursor of Sesquiterpenes Aerosol Material	204.35	L
SO2	Sulfur Dioxide	64.06	E
SOAALK	Alkanes that produce aerosol material	112	L
STYRS	Styrenes	104.15	L
SULF	Sulfates (SO3 or H2SO4)	98.08	L
SULRXN	Precursor of Aerosol Sulfate	98.08	E
SumRCO3	Total acyl peroxy radical concentration	48	L
SumRO2	Total peroxy radical concentration	32	L
SVAVB1	Low volatility organic gas from oxidation of anthropogenic VOCs	179	L
SVAVB2	Semivolatile organic gas from oxidation of anthropogenic VOCs	169	L

Model Species	Definition	Molecular Weight	Explicit or Lumped
SVAVB3	Semivolatile organic gas from oxidation of anthropogenic VOCs	158	L
SVAVB4	Semivolatile organic gas from oxidation of anthropogenic VOCs	198	L
TAMNS	Amines bonded to a tertiary carbon (inhibiting)	73.14	L
TBUO	t-Butoxy Radicals	73.11	E
TERP	Terpenes	136.23	L
TERP_N3	Intermediate from TERP reaction	136.23	L
TERP_O3	Intermediate from TERP reaction	136.23	L
TERP_OH	Intermediate from TERP reaction	136.23	L
TOLRO2	Peroxy Radical tracker from TOLU and ARO1 reactions with OH; an aerosol precursor	172.14	L
TOLU	Toluene	92.14	E
TRPRXN	Precursor of Terpene Aerosol Material	136	E
VIVPO1	Evaporated primary organic aerosol, intermediate-volatility	266	L
VLVOO1	Evaporated secondary organic aerosol, low-volatility, cat 1	136	L
VLVOO2	Evaporated secondary organic aerosol, low-volatility, cat 2	136	L
VLVPO1	Evaporated primary organic aerosol, intermediate-volatility	218	L
VSV001	Evaporated secondary organic aerosol, semi-volatile, cat 1	135	L
VSV002	Evaporated secondary organic aerosol, semi-volatile, cat 2	135	L
VSV003	Evaporated secondary organic aerosol, semi-volatile, cat 3	134	L
VSVPO1	Evaporated primary organic aerosol, semi-volatile, cat 1	230	L
VSVPO2	Evaporated primary organic aerosol, semi-volatile, cat 2	241	L
VSVPO3	Evaporated primary organic aerosol, semi-volatile, cat 3	253	L
xACET	Formation of product ACET in peroxy + NO reactions	58.08	E
xACRO	Formation of product ACRO in peroxy + NO reactions	56.06	E
xAFG1	Formation of product AFG1 in peroxy + NO reactions	99.49	L
xAFG2A	Formation of product AFG2A in peroxy + NO reactions	101.7	L
xAFG2B	Formation of product AFG2B in peroxy + NO reactions	112.13	L
xAFG3	Formation of product AFG3 in peroxy + NO reactions	112.13	L

Model Species	Definition	Molecular Weight	Explicit or Lumped
xAMINS	Formation of product AMINS in peroxy + NO reactions	52.06	L
xBACL	Formation of product BACL in peroxy + NO reactions	86.09	L
xBALD	Formation of product BALD in peroxy + NO reactions	72.06	L
xBENX	Formation of product BENX in peroxy + NO reactions	78.11	L
xBUDAL	Formation of product BUDAL in peroxy + NO reactions	84.07	E
xBZO	Formation of product BZO in peroxy + NO reactions	93.1	L
xETCHO	Formation of product ETCHO in peroxy + NO reactions	58.08	E
xETO2	Formation of product ETO2 in peroxy + NO reactions	61.06	L
xFURNS	Formation of product FURNS in peroxy + NO reactions	68.07	L
xGLCHO	Formation of product GLCHO in peroxy + NO reactions	60.05	E
xGLY	Formation of product GLY in peroxy + NO reactions	58.04	E
xHCHO	Formation of product HCHO in peroxy + NO reactions	30.03	E
xHCOOH	Formation of product HCOOH in peroxy + NO reactions	46.03	E
xHO2	Formation of product HO2 in peroxy + NO reactions	33.01	E
xHPCRB	Formation of product HPCRB in peroxy + NO reactions	128.44	L
xKET2	Formation of product KET2 in peroxy + NO reactions	99.82	L
xLVKS	Formation of product LVKS in peroxy + NO reactions	154.23	L
xMACO3	Formation of product MACO3 in peroxy + NO reactions	101.08	L
xMACR	Formation of product MACR in peroxy + NO reactions	70.09	E
xMECHO	Formation of product MECHO in peroxy + NO reactions	44.05	E
xMECO3	Formation of product MECO3 in peroxy + NO reactions	75.04	E
xMEK	Formation of product MEK in peroxy + NO reactions	72.11	E
xMEO2	Formation of product MEO2 in peroxy + NO reactions	47.03	E
xMGLY	Formation of product MGLY in peroxy + NO reactions	72.06	L

Model Species	Definition	Molecular Weight	Explicit or Lumped
xMVK	Formation of product MVK in peroxy + NO reactions	70.09	E
xNO2	Formation of product NO2 in peroxy + NO reactions	46.01	E
xNO3	Formation of product NO3 in peroxy + NO reactions	62.01	E
xOACID	Formation of product OACID in peroxy + NO reactions	60.05	L
xOH	Formation of product OH in peroxy + NO reactions	17.01	E
xOLEA1	Formation of product OLEA1 in peroxy + NO reactions	143.12	L
xOLEA2	Formation of product OLEA2 in peroxy + NO reactions	151.84	L
xOLEP	Formation of product OLEP in peroxy + NO reactions	160.86	L
xPACID	Formation of product PACID in peroxy + NO reactions	76.05	L
xPAN2	Formation of product PAN2 in peroxy + NO reactions	149.1	L
xR2CO3	Formation of product R2CO3 in peroxy + NO reactions	103.1	L
xRCHO	Formation of product RCHO in peroxy + NO reactions	100.45	E
xRCNO3	Formation of product RCNO3 in peroxy + NO reactions	180.91	L
xRDNO3	Formation of product RDNO3 in peroxy + NO reactions	249.44	L
xRHNO3	Formation of product RHNO3 in peroxy + NO reactions	176.59	L
xRPNO3	Formation of product RPNO3 in peroxy + NO reactions	230.33	L
xTBUO	Formation of product TBUO in peroxy + NO reactions	73.11	E
XYLRO2	Peroxy Radical tracker from xylene and ARO2 reactions with OH;an aerosol precursor	187.17	E
XYNL	Xylenols and higher alkylphenols	124.84	L
yHPCRB	Formation of adduct HPCRB following peroxy + HO2 reactions	128.44	L
yRAOOH	Formation of adduct RAOOH following peroxy + HO2 reactions	188.67	L
yROOH	Formation of adduct ROOH following peroxy + HO2 reactions	97	L
yRPNO3	Formation of adduct RPNO3 following peroxy + HO2 reactions	230.33	L
yRUOOH	Formation of adduct RUOOH following peroxy + HO2 reactions	145.11	L

Model Species	Definition	Molecular Weight	Explicit or Lumped
zPAN2	Formation of nitrate PAN2 in peroxy + NO reactions	149.1	L
zR1NO3	Formation of nitrate R1NO3 in peroxy + NO reactions	146.57	L
zR2NO3	Formation of nitrate R2NO3 in peroxy + NO reactions	190.37	L
zRANO3	Formation of nitrate RANO3 in peroxy + NO reactions	230.33	L
zRCNO3	Formation of nitrate RCNO3 in peroxy + NO reactions	180.91	L
zRDNO3	Formation of nitrate RDNO3 in peroxy + NO reactions	249.44	L
zRHNO3	Formation of nitrate RHNO3 in peroxy + NO reactions	176.59	L
zRNNO3	Formation of nitrate RNNO3 in peroxy + NO reactions	216.23	L
zRPNO3	Formation of nitrate RPNO3 in peroxy + NO reactions	230.33	L

Table A-3. CMAQv5.4 SAPRC-22 Aerosol-Phase Species

Model Species	Definition	Molecular Weight	Explicit or Lumped
AAL	Aluminum	27	E
AAVB1	Low Volatility SOA Product from SVAVB1	198	L
AAVB2	Semivolatile SOA Product from SVAVB2	179	L
AAVB3	Semivolatile SOA Product from SVAVB3	169	L
AAVB4	Semivolatile SOA Product from SVAVB4	158	L
ACA	Calcium	40.1	E
ACL	Chloride	35.5	E
ACORS	Anthropogenic Coarse-mode particle mass	100.0	L
AEC	Elemental Carbon	12	E
AFE	Iron	55.8	E
AGLY	Glyoxal	66.4	E
AH2O	Water	18	E
AH3OP	Hydronium Ion	19	E
AISO1	Semivolatile SOA Product from Isoprene	132.0	L
AISO2	High-Volatility SOA Product from Isoprene	133.0	L
AIVPO1	Intermediate Volatility Primary Organic Compounds	266	L
AK	Potassium	39.1	E
ALVOO1	Low Volatility Oxidized Combustion Organic Compounds	136	L
ALVOO2	Low Volatility Oxidized Combustion Organic Compounds	136	L
ALVPO1	Low Volatility Primary Organic Compounds	218	L
AMG	Magnesium	24.3	E
AMN	Manganese	54.9	E
ANA	Sodium	23	E
ANH4	Ammonium	18	E
ANO3	Nitrate	62	E
AOLGA	Oligomer products of anthropogenic SOA compounds	206.0	L
AOLGB	Oligomer products of biogenic SOA compounds	248.0	L
AORGC	Glyoxal and Methylglyoxal SOA	177.0	L
AORGH2O	Water associated with organic species of particulate matter	18	E
AOTHR	Other Particulate Mass	200	L
APCSO	Potential Combustion SOA	170	L
APNCOM	Non-Carbon Organic Matter associated with APOC	220	L
APOC	Primary Organic Carbon	220	L
ASEACAT	Sea spray cations	23.75	L
ASI	Silicon	28.1	E
ASO4	Sulfate	96	E
ASOIL	Lumped crustal species	100.0	L

Model Species	Definition	Molecular Weight	Explicit or Lumped
ASQT	Semivolatile SOA Product from Sesquiterpenes	273.0	L
ASVOO1	Semivolatile Oxidized Combustion Organic Compounds	135	L
ASVOO2	Semivolatile Oxidized Combustion Organic Compounds	135	L
ASVOO3	Semivolatile Oxidized Combustion Organic Compounds	134	L
ASVPO1	Semivolatile Primary Organic Compounds	230	L
ASVPO2	Semivolatile Primary Organic Compounds	241	L
ASVPO3	Semivolatile Primary Organic Compounds	253	L
ATI	Titanium	47.9	E
NUMACC	Accum. Mode Number Conc	NA	NA
NUMAIT	Aitken Mode Number Conc	NA	NA
NUMCOR	Coarse Mode Number Conc	NA	NA
SRFACC	Accum. Mode Surface Area Conc	NA	NA
SRFAIT	Aitken Mode Surface Area Conc	NA	NA
SRFCOR	Coarse Mode Surface Area Conc	NA	NA

Note that for each aerosol mass species, a letter will be appended to the names below in order to designate the size, or mode, of the aerosol being represented: I = Aitken mode, J = Accumulation mode, K = Coarse mode.